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[Intervention Review]

Perioperative immune checkpoint inhibitors with or without chemotherapy versus placebo with or without chemotherapy in elderly people with localised non-small cell lung cancer

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ABSTRACT

Rationale

Lung cancer is typically a cancer of the elderly, with a median age at diagnosis of 71, and more than one third of the people diagnosed with lung cancer are over 75 years old. Immune checkpoint inhibitors (ICIs) have revolutionised the treatment of cancers, including lung cancer.

ICIs targeting the programmed death-1/programmed death-ligand 1 (PD-1/PD-L1) axis, administered in the neoadjuvant setting, the adjuvant setting, or both, are currently the standard of care for resectable non-small-cell lung cancer (NSCLC) worldwide. These ICIs are commonly used in combination with platinum-based chemotherapy and have shown superior efficacy in patients eligible for curative surgery.

The concept of immunosenescence, which refers to age-related changes in the immune system — particularly a decline in the efficiency of T-cell mediated responses — raises concerns about the benefits of ICIs in the elderly population.

Objectives

To assess the benefits and harms of ICI with or without chemotherapy compared to no treatment or placebo with or without chemotherapy given before surgery, after, or both in older adults diagnosed with NSCLC at the early resectable stage.

Search methods

We searched for all eligible randomised controlled trials (RCTs) in electronic databases (CENTRAL, MEDLINE, and Embase), trial registries (clinicaltrials.gov and the World Health Organization ICTRP), references of eligible studies, meeting abstracts of the main world conferences, and the Food and Drug Administration (FDA) and European Medicines Agency (EMA) websites. The search was up to 3 July 2025.

Eligibility criteria

We included parallel designed RCTs comparing ICI with or without chemotherapy versus no treatment or placebo with or without chemotherapy administered before surgery, after, or both for early-stage resectable NSCLC. We excluded studies comparing differential sequencing of ICIs or combinations of ICIs with alternate interventions (e.g. radiotherapy).

Outcomes

Our critical outcomes were overall survival and grade ≥ 3 treatment-related adverse events.

Our important outcomes were disease-free survival, event-free survival, pathological complete response rate, major pathological response rate, and health-related quality of life.

Risk of bias

Two review authors independently used version 2 of the Cochrane risk of bias tool for randomised trials (RoB 2) to assess bias in the included studies.

Synthesis methods

We synthesised results for each outcome and pooled data where possible (using a random-effects model with DerSimonian and Laird methods for all outcomes; and the Mantel-Haenszel method for dichotomous outcomes). Where this was not possible due to the nature or the amount of data, we narratively summarised the results. Three authors independently assessed the certainty of the evidence, using the five GRADE considerations for each outcome.

Included studies

We included a total of 11 studies with 6788 participants, of whom 3152 were ≥ 65 years old (46.4% of all participants). We were also interested in those aged ≥ 75 years old but could not obtain the exact number of such participants in nine studies.

The inclusion criteria were similar across studies: adults presenting with a resectable NSCLC (stages II to IIIB according to the eighth edition of the Tumor, Node, Metastasis (TNM) classification), with an Eastern Cooperative Oncology Group performance status score of 0 or 1. One study excluded people ≥ 70 years old. Six studies excluded NSCLC with known alterations in epidermal growth factor receptor and anaplastic lymphoma kinase genes.

Synthesis of results

The following results relate to people ≥ 65 years old only. The term perioperative is used to label studies evaluating the administration of treatment before (neoadjuvant) and after (adjuvant) surgery.

Overall survival at any time point (in all studies)

Aggregated data from three studies (590 participants) showed that perioperative ICI probably results in little to no difference in overall survival compared to placebo or no treatment, with a hazard ratio (HR) of 0.88 (95% confidence interval (CI) 0.61 to 1.26) and moderate-certainty evidence (downgraded for risk of bias).

Treatment-related adverse events $\geq$ grade 3 at any time point

One study (39 participants) provided data indicating that perioperative ICI combined with neoadjuvant chemotherapy may result in little to no difference in treatment-related adverse events $\geq$ grade 3 compared to neoadjuvant chemotherapy alone, with a risk ratio (RR) of 2.75 (95% CI 0.38 to 19.83) and low-certainty evidence (downgraded for indirectness and imprecision).

Disease-free survival at any time point

Aggregated data from three studies (1403 participants) showed that adjuvant ICI probably slightly increases disease-free survival compared to placebo or no treatment (HR 0.85, 95% CI 0.73 to 0.99) with moderate-certainty evidence (downgraded for risk of bias).

Event-free survival at any time point

Aggregated data from seven studies (1531 participants) showed that neoadjuvant or perioperative ICI likely increases event-free survival compared to placebo or no treatment (HR 0.61, 95% CI 0.52 to 0.71) with moderate-certainty evidence (downgraded for risk of bias).

Pathological complete response

Aggregated data from six studies (1068 participants) showed that neoadjuvant or perioperative ICI may result in a large increase in pathological complete response rates compared to placebo or no treatment (RR 5.07, 95% CI 3.40 to 7.54) with low-certainty evidence (downgraded for risk of bias and imprecision).

Major pathological response

Aggregated data from six studies (1068 participants) showed that neoadjuvant or perioperative ICI probably results in a large increase in major pathological response rates compared to placebo or no treatment (RR 2.94, 95% CI 2.32 to 3.72) with moderate-certainty evidence (downgraded for risk of bias).

Authors' conclusions

In people aged ≥ 65 years old, the addition of ICIs probably results in little to no improvement in overall survival. Based on one study, treatment-related adverse events showed a similar profile, with low-certainty evidence. However, ICIs probably increase disease-free survival, event-free survival, and major pathological response rates by a clinically meaningful margin. ICI may also increase complete pathological response rates.

No study reported health-related quality of life assessments in older adults. Data were also insufficient to evaluate outcomes precisely in participants aged 65 to 75 years, those ≥ 75 years, or in PD-L1 stratified subgroups. We classified 12 studies as ongoing, as no results are yet available for elderly participants.

Funding

This Cochrane review had no dedicated funding.

Registration

Protocol available via DOI: 10.1002/14651858.CD014907.

PLAIN LANGUAGE SUMMARY

Do medicines that help the body's immune system recognise and attack cancer cells improve results for older adults having surgery for lung cancer?

Key messages

- Medicines that help the body's immune system recognise and attack cancer cells can shrink the cancer and likely slow its return in people aged 65 years and older having surgery for the most common type of lung cancer. However, it probably does not increase how long people live overall.
- We do not have enough evidence yet to know if its benefits outweigh the harms, and no study assessed well-being.

What is resectable non-small cell lung cancer?

About 85% of people with lung cancer have what is called non-small cell lung cancer (NSCLC). It is most often found in older adults, with an average age at diagnosis of 71. When the cancer has not spread out of the lung, and if the person is fit enough to have surgery, the tumour can be removed. However, the cancer can come back after treatment, and between 30% and 55% of people eventually die from the disease. To reduce the risk of cancer coming back, different treatments can be used together.

How is resectable non-small cell lung cancer treated?

In recent years, lung cancer treatment has been transformed with the discovery of medicines that help the body's immune system recognise and attack cancer cells (immunotherapy). Immunotherapy works together with drugs that kill cancer cells (chemotherapy). When given before surgery, after surgery, or both, immunotherapy improves the chances of curing the cancer and helps delay its return.

What did we want to find out?

The ageing process affects the immune system, reducing its capacity to fight cancer: this is called immunosenescence. The elderly population is a very diverse group, and each person needs to be carefully assessed because age alone is not enough to base the best treatment decisions on. We wanted to find out if, despite immunosenescence, immunotherapy is effective in people aged 65 years and older, to cure lung cancer and prolong life without unwanted treatment effects.

What did we do?

We searched for high-quality studies that tested immunotherapy, either alone or in combination with chemotherapy, given before surgery, after surgery, or both, in people aged 65 and older. We summarised their results and assessed how confident we can be in the evidence, based on criteria such as how the studies were done and how many people were included.

What did we find?

For people aged 65 and older, we found results from 11 studies involving 3152 people with NSCLC. The studies compared immunotherapy alone or in combination with chemotherapy to a placebo (a sham or 'dummy' treatment) or no treatment, with or without chemotherapy. Treatments were given either only before surgery, only after surgery, or both before and after surgery.

Immunotherapy reduces the amount of cancer found at surgery and probably delays the cancer from coming back. Immunotherapy probably makes little or no difference to overall survival. There was not enough evidence to determine its effect on potential harms, and no study assessed well-being.

What are the limitations of the evidence?

Our confidence in most of the evidence is moderate because of some concerns about how the studies were conducted. More complete and longer-term data are needed to better estimate how much a person's lifespan might be extended. We have very limited confidence in the assessment of unwanted effects, because only one study provided data about these.

How up to date is this evidence?

The evidence is up to date to 3 July 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Perioperative immune checkpoint inhibitors +/- chemotherapy versus placebo or no treatment +/- chemotherapy in people ≥ 65 years old with early-stage resectable non-small cell lung cancer

Population: people ≥ 65 years old with early stage resectable non-small cell lung cancer

Setting: randomised controlled phase II or III trials

Intervention: ICI +/- chemotherapy

Control: placebo or no treatment +/- chemotherapy

Outcomes	N° of participants (studies)	Certainty of evidence (GRADE)	Relative effects (95% CI)	Anticipated absolute effects* (95% CI)	
				Risk with placebo or no treatment +/- platinum-based chemotherapy	Risk with ICI +/- platinum-based chemotherapy
Overall survival (at any time point, in all studies)	590 (3 RCTs)	Moderate ⊕⊕⊕⊖ ^a	HR 0.88 (0.61 to 1.26)	254 deaths per 1000 (190 to 318 per 1000)	27 fewer deaths per 1000 (90 fewer to 55 more)
Treatment-related adverse events ≥ grade 3 (at any time point, in all studies)	39 (1 RCT)	Low ⊕⊕⊖⊖ ^b	RR 2.75 (0.38 to 19.38)	91 events per 1000 (73 to 109)	159 more events per 1000 (56 fewer to 1000 more)
Disease-free survival (at any time point, only in adjuvant studies)	1403 (3 RCTs)	Moderate ⊕⊕⊕⊖ ^c	HR 0.85 (0.73 to 0.99)	521 events per 1000 (424 to 538) per 1000	56 fewer events per 1000 (105 fewer to 4 fewer)
Event-free survival (at any time point, only in neoadjuvant and perioperative studies)	1531 (7 RCTs)	Moderate ⊕⊕⊕⊖ ^d	HR 0.61 (0.52 to 0.71)	521 events per 1000 (468 to 574 events)	159 fewer events per 1000 (114 fewer to 203 fewer)
Pathological complete response (pCR) (at surgery, only in neoadjuvant and perioperative studies)	1068 (6 RCTs)	Low ⊕⊕⊖⊖ ^e	RR 5.07 (3.40 to 7.54)	48 pCRs per 1000 (35 to 61 pCR)	195 more pCRs per 1000 (115 more to 314 more)
Major pathological response (MPR) (at surgery, only in neoadjuvant and perioperative studies)	1068 (6 RCTs)	Moderate ⊕⊕⊕⊖ ^f	RR 2.94 (2.32 to 3.72)	137 MPRs per 1000 (113 to 175 events)	265 more MPRs per 1000 (180 more to 371 more)

Health-related quality of life (at any time point, in all studies)	No available data
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***The risk in the intervention group** (and its 95% CI) is based on the assumed risk in the comparison group (based on partial data for time-to-event outcomes) and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **HR:** hazard ratio; **ICI:** immune checkpoint inhibitor; **MPR:** major pathological response; **pCR:** pathological complete response; **RCT:** randomised controlled trial; **RR:** risk ratio.

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

GRADE evaluation

^aDowngraded one point due to risk of bias (IMPOWER-010: risk of bias arising from deviations from intended interventions and selection of the reported results).

^bDowngraded by two points: one point for indirectness (only one study providing data), and one point for imprecision (confidence intervals include both large benefit and extreme harm).

^cDowngraded one point for risk of bias (IMPOWER-010: risk of bias arising from deviations from intended interventions, measurement of the outcome and selection of the reported results).

^dDowngraded one point due to risk of bias (CHECKMATE-816: risk of bias arising from deviations from intended interventions).

^eDowngraded by two points: one point due to risk of bias (CHECKMATE-816: risk of bias arising from deviations from intended interventions) and one for imprecision (wide 95% CI).

^fDowngraded one point due to risk of bias (CHECKMATE-816: risk of bias arising from deviations from intended interventions).

Calculation of anticipated absolute effects (95% CI)

- Overall survival: based on data from KEYNOTE-671 and NADIM-II: 75 died/129 lived from a total of 204 (ICI) versus 74 died/123 lived from a total of 197 (control).
- Treatment-related adverse events of grade ≥ 3 : based on data from NADIM-II only: 7 events/21 no-events from a total of 28 (ICI) versus 1 event/10 no-events from a total of 11 (control).
- Disease-free survival: based on data from CCTG BR.31 and IMPOWER-010: 315 events/395 no-events from a total of 710 (ICI) versus 251 events/253 no-events from a total of 504 (control).
- Event-free survival: based on data from KEYNOTE-671, NADIM-II, NEOTORCH and RATIONALE-315: 134 events/215 no-events from a total of 349 versus 198 events/161 no-events from a total of 359 (control).
- pCR: based on data from AEGEAN, CHECKMATE-77T, CHECKMATE-816, NADIM-II, RATIONALE-315 and TD-FOREKNOW: 136 events/390 no-events from a total of 526 (ICI) versus 26 events/516 no-events from a total of 542 (control).
- MPR: based on data from AEGEAN, CHECKMATE-77T, CHECKMATE-816, NADIM-II, RATIONALE-315 and TD-FOREKNOW: 213 events/313 no-events from a total of 526 (ICI) versus 74 events/468 no-events from a total of 542 (control).

BACKGROUND

Description of the condition

Epidemiology

Lung cancer is currently the most frequently diagnosed cancer worldwide, with an estimated 2.5 million new cases in 2022, which is 12.4% of all incident cancer cases [1]. It is the leading cause of cancer death with around 1.8 million related deaths, corresponding to 18.7% of all cancer deaths, according to a recent GLOBOCAN evaluation [1]. Heterogeneity across countries is mainly due to differences in tobacco smoking habits, which remain the greatest risk factor, with up to 90% of lung cancers attributed to smoking. Patterns in the incidence and mortality largely reflect the advance of the tobacco epidemic, explaining a rising incidence in women from high-income countries in the last decades. Also, environmental exposures and air pollution seem to partially explain the incidence of lung cancer not attributable to tobacco smoking [2]. Lung cancer is highly prevalent in the elderly population, with a median age at diagnosis of 71 years; approximately 72% of cases occur in individuals over 65 years of age and 37% in those aged 75 years and older [3].

Histology and molecular biology

Lung cancer is conventionally classified into two main histological types: small cell lung carcinoma (SCLC) and non-small cell lung carcinoma (NSCLC). This latter category accounts for 85% of the cases and is further subdivided into squamous cell carcinoma (SCC) and non-squamous subtypes, including large-cell carcinoma and adenocarcinoma. The most common histology is adenocarcinoma, with a rising incidence over the last few years, especially in high-income countries [2].

In addition to the histological definition, molecular characteristics are now taken into account, including the presence of oncogenic driver mutations which play major roles in both biological mechanisms and clinical sensitivity to treatments [4]. Molecular alterations are routinely investigated using new-generation sequencing with either a customised gene panel or comprehensive genomic profiling [5]. In the perioperative setting, testing for epidermal growth factor receptor (EGFR) and anaplastic lymphoma kinase (ALK) determines eligibility for appropriate adjuvant targeted therapies [6]. These actionable alterations tend to be more frequent in women and never-smokers presenting with a non-squamous cell carcinoma [2].

Programmed death-ligand 1 (PD-L1) expression, quantified by immunohistochemistry, is currently the main predictive biomarker for immunotherapy, routinely used in clinical practice and incorporated into most clinical trials [7]. Its clinical relevance is supported by several prospective clinical studies demonstrating a correlation between PD-L1 levels in tumour tissue and the clinical efficacy of immune checkpoint inhibitor (ICI) therapy [8], which has led to its endorsement by international guidelines [9]. Emerging biomarkers such as tumour mutational burden (TMB) also show potential in predicting response and survival outcomes with ICI [10].

Treatment principles of early-stage resectable NSCLC

Prognosis varies according to histological subtype and cancer stage: the five-year relative survival rates are 63.9% for localised

lung cancer, 34.4% for loco-regional disease, and 8.0% for metastatic disease [3].

For people with early-stage resectable NSCLC (stages I to IIIA/B), surgery is generally offered if the patient is fit enough and accepts the risks of surgery. A lobectomy, typically accompanied by systematic nodal dissection, is the preferred surgical approach. When surgery is refused or contraindicated, definitive radiotherapy is recommended, preferably stereotactic ablative radiotherapy (SABR) for smaller tumours, or in combination with chemotherapy for more advanced diseases [6, 11].

Multimodal protocols combining surgery, systemic treatment, and radiotherapy are used to maximise the chances of cure, prevent recurrence, and prolong survival. Systemic treatment may be given before surgery (neoadjuvant), after surgery (adjuvant), or both (referred to as perioperative in this review) for resectable or resected NSCLC, based on the Tumor, Node, Metastasis (TNM) staging system [12], histological type and molecular characteristics [6, 11].

Adjuvant chemotherapy with a platinum-based doublet is recommended for people with stage IIB to III resected NSCLC (eighth edition of the TNM) and may be considered for stage IIA [13]. Neoadjuvant chemotherapy provides a similar benefit, offering an approximate 5% absolute survival improvement in five-year survival, and can be considered to achieve tumour downstaging and allow for less extensive surgery [14]. Cisplatin is the most commonly used drug, administered at a cumulative dose of 300 mg/m² over three to four cycles. When cisplatin is not feasible, carboplatin is an accepted alternative. In tumours harbouring classical EGFR mutations (exon 19 deletions and L858R substitutions), adjuvant osimertinib is indicated for stages IB to IIIA. Immunotherapy may be combined with chemotherapy for a synergistic effect, but is generally not used in the presence of oncogenic driver mutations due to a lack of significant benefit in these cases [11].

Description of the intervention and how it might work

Antitumour immunity mechanisms

Immunosurveillance is the process by which immune cells detect and eliminate foreign pathogens such as bacteria and viruses, as well as cancerous cells. The relationship between the immune system and tumorigenesis was first theorised by Robert Schreiber, who defined the 'three Es' of cancer immune-editing: elimination (of tumour cells by the host immune system), equilibrium (between cancer and immunity), and escape (of tumour cells from the immune system). During the equilibrium phase, the immune system exerts a selective pressure on tumour cells, leading to the emergence of mutated variants that are able to evade immune detection [15, 16]. This immune evasion is facilitated by various mechanisms, including components of the innate immune system such as dendritic cells, natural killer cells, macrophages, and the complement system.

Dendritic cells play a central role in initiating antitumor adaptive immune responses by activating CD4 and CD8 T-cells [17]. Once primed by these antigen-presenting cells, lymphocytes target tumour-specific antigens, infiltrate tumours, and destroy cancer cells [18]. However, inhibitory signals — especially immune checkpoints — may oppose T-cell activation and proliferation, resulting in dysfunctional T-cells [19]. These exhausted T-cells

often present an intense and sustained co-expression of inhibitory receptors such as T-lymphocyte-associated protein 4 (CTLA-4), programmed death 1 (PD-1), lymphocyte-activation gene 3 (LAG-3), T-cell immunoglobulin mucin-3 (TIM-3), T-cell immunoglobulin and immunoreceptor tyrosine-based inhibitory motif (ITIM) domain (TIGIT), amongst others. Restoring T-cell function to enhance antitumor immunity is a primary goal of cancer immunotherapy [20].

Immune checkpoint inhibitors: mechanisms of action

ICIs are monoclonal antibodies that block interactions between immune checkpoints and their ligands (e.g. PD-1/PD-L1, CTLA-4/CD80-86), thereby restoring exhausted T-cell function and enhancing specific antitumor immunity. ICIs are the standard of care across most lung cancer stages and histological subtypes, used in various treatment combinations.

PD-1 inhibitors such as nivolumab (Opdivo, Bristol-Myers Squibb) [21], pembrolizumab (Keytruda, Merck Sharp & Dohme) [22], toripalimab (Tuoyi, Shanghai Junshi Bioscience) [23], tislelizumab (Tevimbra, BeiGene) and camrelizumab are human or humanised IgG4 monoclonal antibodies that prevent lymphocytes' PD-1 binding to PD-L1 on tumour cells. PD-L1 inhibitors like atezolizumab (Tecentriq, Roche) [24] and durvalumab (Imfizi, AstraZeneca) [25] are respectively humanised and human IgG1 antibodies targeting PD-L1 directly. Anti-CTLA-4 antibodies (e.g. ipilimumab) are often combined with anti-PD-(L)1 agents to promote de novo T-cell responses by blocking CTLA-4/CD80-86 interactions. Additional immunotherapies are currently under investigation, sometimes in combination with ICIs [18].

Immune checkpoint inhibitors in the elderly

Ageing is characterised by a progressive loss of physiological integrity, including the immune system decline known as immunosenescence [26]. This involves a reduction of peripheral naïve T-cells, a relative increase of memory T-cells, and an elevation of pro-inflammatory cytokine levels, contributing to 'inflamm-ageing'. These changes impair immune responses to infections and vaccines, and increase susceptibility to diseases such as cancer [27]. Senescent T-cells, identified by loss of CD28 and overexpression of CD57 and KLRG1, exhibit low proliferation capacity and IL-2 production [28]. In NSCLC patients, elevated senescent T-cell levels correlate with poorer outcomes following ICI therapy, suggesting their potential role as circulating biomarkers of resistance to single-agent ICIs [29]. These observations prompt questions regarding the effectiveness of ICIs in older patients.

Principles of perioperative immunotherapy in early-stage resectable NSCLC

The ICIs revolution is rapidly moving from metastatic to early-stage with many studies investigating the benefits of immunotherapy in the perioperative setting (neoadjuvant, adjuvant or both), alone or in combination with a platinum-based chemotherapy regimen. The primary goal of perioperative treatments is to cure people, which depends on factors such as the tumour's resectability, the quality of the resection, the person's operability, and the anticipated postoperative lung function. Stages I to IIIA (according to the eighth edition of the TNM) are usually resectable. Some of the stage IIIBs are potentially eligible for surgery and must be discussed in a multidisciplinary tumour board (MDT) [30, 31].

Neoadjuvant ICIs have many benefits: the presence of the tumour and its high neoantigen burden enables a broad T-cell response, including tumour-specific memory T-cells; the surgery specimen provides a useful opportunity to assess pathological response and biomarkers of efficacy; the association to neoadjuvant chemotherapy allows a synergistic action via neoantigens release and promotion of a robust immune response; the shrinkage of the tumour facilitates the surgical gesture and reduces the associated morbidity [32].

Adjuvant treatments are mainly directed against micrometastatic disease, which is amenable to eradication. Surgery affects the immune responses with increased levels of stress-related cytokines, growth hormones, regulatory T-cells, tumour-associated macrophages, and myeloid-derived suppressor cells, all contributing to persistent immunosuppression. Adjuvant ICIs administered shortly after surgery can reverse this immunosuppressive environment, prime anti-tumour immunity, and target circulating tumour cells. The combination with adjuvant chemotherapy has a synergising potential by inducing tumour cell death and affecting neoantigen exposure [32].

Perioperative immunotherapy is an emerging concept aiming to prime host immunity against existing tumour neoantigens and to eliminate local and micrometastatic disease. It combines the advantages of both adjuvant and neoadjuvant approaches [32].

Risks of perioperative immunotherapy

Immune-related adverse events (irAEs) can arise from the heightened immune system activity and loss of self-tolerance induced by immune checkpoint inhibition. Several mechanisms have been suggested to explain irAEs, including the induction of a new inflammatory syndrome or the exacerbation of an existing autoimmune condition [33]. While irAEs can affect any organ system, they most commonly involve the skin, gastrointestinal tract, endocrine glands, lungs, and liver. Less frequently, the nervous system, kidneys, blood, muscles, joints, heart, or eyes may also be affected [34]. The frequency of irAEs is similar for both anti-PD-1 and anti-PD-L1 therapies, with a prevalence ranging from 65% to 75%, including approximately 15% of grade 3 or 4 adverse events, and a low incidence of fatal toxic effects (0.36% to 0.38%) [35, 36]. Most irAEs occur within 10 weeks of initiating ICI therapy, though late events following treatment discontinuation have also been reported [37]. The timeline for the onset of irAEs often correlates with the affected organ system [38].

These adverse events must be carefully considered, as the goal is to cure patients while preserving their long-term quality of life. However, in comparison to the toxicity associated with chemotherapy, irAEs do not appear to be a significant barrier [31].

Current challenges

A recent systematic review and meta-analysis of ICIs as single agents or in combination with chemotherapy in early-stage NSCLC was conducted to discuss clinicopathological factors and treatment sequencing strategies [39]. It seems that perioperative regimens achieve better results than adjuvant treatments, but their superiority compared to neoadjuvant strategies alone is ambiguous. So far, no randomised controlled trial (RCT) has compared neoadjuvant versus adjuvant versus perioperative regimens for resectable NSCLC, and the optimal treatment sequencing remains unclear [31].

However, RCTs conducted in melanoma that directly compared perioperative with adjuvant strategies demonstrated superior event-free survival (EFS) and high complete pathological response rates with the perioperative ICIs [7, 40, 41]. A preclinical study on mice suggested that neoadjuvant ICI is superior to adjuvant ICI at prolonging survival, reducing distal recurrence and inducing anti-tumour immunity in resectable NSCLC [42].

Most trials in the neoadjuvant setting have adopted pathological response as a primary endpoint. ICIs are approved for use in the neoadjuvant and adjuvant settings on the basis of event-free survival and disease-free survival benefit, respectively. Still, the correlation of these endpoints with overall survival remains unclear in these settings. Another major challenge is the identification of clinicopathological predictive factors in order to improve participants' selection and maximise clinical outcomes [32].

Additionally, immunotherapy is not universally accessible, particularly in low-income countries or in settings without reimbursement, due to the high cost of the treatments [31]. Some studies are exploring whether reduced doses of ICI provide benefits comparable to the standard dosing (e.g. Biswal (ELoN) [43, 44]).

Why it is important to do this review

Even though the majority of lung cancer patients are over 70 years old, elderly people are significantly under-represented in clinical trials [45]. Older adults are at a higher risk for toxic effects when receiving high-risk treatments, because of their underlying comorbidities, fragilities and polymedication. A cluster-randomised trial evaluated the effect of tailored geriatric assessment and management versus the usual care provided by oncologists, showing a much lower proportion of grade ≥ 3 adverse events [46]. The International Society of Geriatric Oncology (SIOG) has made specific recommendations for NSCLC in the elderly, with the European Organisation for Research and Treatment of Cancer (EORTC) Elderly Task Force and Lung Cancer Group [47]. In general, treatment decisions should not be based on chronological age, but on a comprehensive geriatric assessment, including the person's life expectancy, their functional, cognitive, and emotional status, and the presence of comorbidities. Therefore, frail or comorbid people ≥ 70 years old should be carefully monitored to have the same benefit from gold standard treatments as their younger counterparts. A recent position paper by the Young International Society of Geriatric Oncology describes the need to generate better evidence on the use of immunotherapy in the older population as a whole [48].

No clinical study has specifically investigated perioperative (chemo-)immunotherapy in older adults with early NSCLC. This systematic review is written as part of a protocol comprising seven reviews of older adults with NSCLC at different stages and in different settings, under the Cochrane Lung Cancer Group [49]. The first review was published in 2024, evaluating ICIs and platinum-based chemotherapy +/- bevacizumab for people in the first-line setting. It showed a benefit in overall survival and progression-free survival for people with advanced NSCLC between 65 and 75 years old at the cost of an increased toxicity. It demonstrated no benefit for those aged 75 years and older [50]. Nevertheless, in the phase III IFCT-1805 ELDERLY trial involving participants aged 70 to 89 years, adding atezolizumab to first-line carboplatin and weekly paclitaxel significantly improved progression-free survival, objective response rate, and duration of response with a non-

significant trend toward prolonged overall survival [51]. Based on these combined results in the first-line setting, it seems that people over 75 years of age might have a poorer clinical response to ICIs [29, 50].

Today, optimising the management of elderly people in thoracic oncology is essential, as individuals aged 70 years or older account for more than half of all lung cancer diagnoses. ICIs have revolutionised the management of NSCLC by restoring antitumour immune responses and are now firmly established as a cornerstone therapy for advanced disease. The limited inclusion of older adults in clinical trials has resulted in a substantial gap in knowledge, with a poor understanding of their responses to anticancer therapies [52].

In the era of personalised medicine, it is essential to better characterise the benefits and harms of ICIs in older adults to guide appropriate treatment decisions and ensure that elderly people are not inadvertently denied therapies from which they could benefit due to poorly understood frailty or age-related biases. To our knowledge, this review and meta-analysis is the first to investigate the perioperative use of ICIs in resectable NSCLC among elderly people.

OBJECTIVES

To assess the benefits and harms of ICI with or without chemotherapy compared to no treatment or placebo with or without chemotherapy given before surgery, after, or both in older adults diagnosed with NSCLC at the early resectable stage.

METHODS

The review was conducted in accordance with the Methodological Expectations of Cochrane Intervention Reviews (MECIR) [53], and reported using PRISMA [54], as outlined in Chapter 3 of the *Cochrane Handbook for Systematic Reviews of Interventions* [55]. We followed a previously published protocol [49]; the amendments are summarised in [Supplementary material 8](#).

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs) reporting on the efficacy, safety, or both, of immune checkpoint inhibitors (ICIs) administered in the neoadjuvant setting, adjuvant setting, or both (i.e. perioperative settings) for people aged 65 or older with resectable NSCLC, with or without blinding.

Quasi-randomised trials, cluster-randomised, and cross-over RCTs were not eligible for this review.

Types of participants

We included treatment-naïve people aged 65 years and older with resectable NSCLC. We allowed different cut-off ages (65, 70, 75, or 80 years old), but the included RCTs only used the cut-off ages of 65 and 75 years to define older adults. Please see [Registration and protocol](#) about the heterogeneous meanings attached to the term 'elderly' in geriatric oncology and the use of age cut-offs in clinical trials.

We included studies conducted exclusively in older adults (i.e. 65 years and older) as well as those conducted in all adults (i.e. 18

years and older), if authors provided disaggregated data for our population of interest. We contacted the authors of all the fully published studies that included elderly participants but did not provide specific results for age subgroups. If we could not obtain data for this population, we classified these studies as ongoing.

Types of interventions

We included studies if researchers reported an intervention in the following setting: ICIs before or after surgery or both, versus placebo or no treatment (with or without standard platinum-based chemotherapy), for early-stage resectable NSCLC, in previously untreated people aged 65 years and older.

We accepted the following designs, grouped into three categories.

1. Neoadjuvant only

- Neoadjuvant ICI alone versus placebo or best supportive care, no adjuvant treatment
- Neoadjuvant combination of an ICI with chemotherapy versus chemotherapy alone or chemotherapy with placebo, no adjuvant treatment

2. Adjuvant only

- Adjuvant ICI alone versus placebo or best supportive care, no neoadjuvant treatment
- Adjuvant combination of an ICI with chemotherapy versus chemotherapy alone or chemotherapy with placebo, no neoadjuvant treatment

3. Perioperative

- Neoadjuvant and adjuvant ICI alone versus placebo or best supportive care
- Neoadjuvant and adjuvant combination of an ICI with chemotherapy versus chemotherapy alone or chemotherapy with placebo

We defined an ICI or blockade therapy as the administration of a drug or drugs targeting T-cell suppressive pathways, which could include but were not limited to: pembrolizumab and nivolumab (anti-PD-1), atezolizumab, durvalumab, and avelumab (anti-PD-L1), ipilimumab and tremelimumab (anti-CTLA-4) given as single agents or in combination.

We defined chemotherapy as the administration of a cytotoxic drug or drugs, which could include but were not limited to: carboplatin, cisplatin, paclitaxel, nab-paclitaxel, vinorelbine, pemetrexed, gemcitabine, etoposide, and docetaxel given as single agents or in combination.

We applied no restrictions on drug dosage or treatment duration. Other active antitumour treatment modalities such as radiotherapy were permitted according to the standard of care.

We excluded studies comparing differential sequencing of ICI or combination of ICI with alternate interventions (e.g. radiotherapy).

Outcome measures

Our outcomes were evaluated in older adults. If multiple age cut-offs were used, we pooled the results for each age group separately

and treated them as separate comparisons. For all the outcomes, we collected data for the following three age categories.

- 65 years and older
- 65 to 75 years old
- 75 years and older

Critical outcomes

- Overall survival (OS): the time from randomisation to death from any cause [56].
- Treatment-related adverse events (TRAEs): undesired effects of a drug or other type of treatment, such as surgery [56]. We investigated the incidence of grade 3 events (severe or medically significant but not immediately life-threatening; indication of hospitalisation or prolongation of hospitalisation; disabling; limiting self-care activities of daily living), grade 4 events (life-threatening consequences; urgent intervention indicated) and grade 5 events (death related to adverse event) based on the Common Terminology Criteria for Adverse Events (CTCAE) and Patient-Reported Outcomes CTCAE (PRO-CTCAE) [57].

Important outcomes

- Disease-free survival (DFS): the length of time, after primary treatment for a cancer ends, that the person survives without any signs or symptoms of that cancer [56], used in adjuvant trials.
- Event-free survival (EFS): the length of time, after primary treatment for a cancer ends, that the person remains free of certain complications or events that the treatment was intended to prevent or delay [56], used in perioperative and neoadjuvant trials.
- Pathological complete response (pCR): the lack of all signs of cancer in tissue samples removed during surgery or biopsy after treatment [56], used in perioperative and neoadjuvant trials.
- Major pathological response (MPR): $\leq 10\%$ viable tumour cells in resected primary tumour and lymph nodes after treatment [58, 59, 60, 61, 62, 63], used in perioperative and neoadjuvant trials.
- Health-related quality of life (HRQoL), measured via validated generic or disease-specific questionnaires, or validated items.

Survival outcomes were estimated at any time point.

Search methods for identification of studies

Electronic searches

We searched the following electronic databases to identify all eligible RCTs, with no language restrictions or publication status:

- Cochrane Central Register of Controlled Trials (CENTRAL, Issue 7, 2025), in the Cochrane Library (searched 3 July 2025);
- MEDLINE, accessed via PubMed (inception year to 3 July 2025);
- Embase (www.embase.com) via Elsevier (inception year to 3 July 2025).

We present the search strategies for MEDLINE, CENTRAL, and Embase in [Supplementary material 1](#).

We searched all databases using both controlled vocabulary (namely medical subject headings (MeSH) in MEDLINE and Emtree in Embase) and a wide range of free-text terms. We performed:

- the MEDLINE search using the Cochrane Highly Sensitive Search Strategy for identifying randomised trials in MEDLINE: sensitivity-maximising version (2008 revision); PubMed format; and
- the Embase search using the Cochrane Highly Sensitive Search Strategy used for identifying controlled trials in Embase for CENTRAL: (2023 revision); Embase.com format,

as described in the *Cochrane Handbook for Systematic Reviews of Interventions* [64].

We also conducted searches in the following clinical trial registries to identify unpublished and ongoing trials:

- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (www.clinicaltrials.gov/; searched 3 July 2025); and
- World Health Organization International Clinical Trials Registry Platform (apps.who.int/trialsearch/; searched 3 July 2025).

Searching other resources

We handsearched the references of eligible studies to identify additional studies for inclusion.

We searched the meeting abstracts of conferences from the following sources (from 2022 to 3 July 2025).

- American Society of Clinical Oncology (ASCO);
- European Society for Medical Oncology (ESMO);
- World Conference on Lung Cancer (WCLC);
- International Lung Cancer Research Association (IASLC);
- European Society for Medical Oncology Immuno-Oncology congress (ESMO IO);
- European Lung Cancer Conference (ELCC);
- American Association for Cancer Research (AACR).

We included meeting abstracts and unpublished online data only when sufficient data were provided and when a risk of bias assessment was possible.

We collected clinical study reports about ICI from the US Food and Drug Administration (www.fda.gov) and European Medicines Agency (www.ema.europa.eu) websites.

All references were scanned through Zotero, which is linked to the RetractionWatch database and automatically flags retracted publications. The search strategies were designed to identify errata and retraction notices in bibliographical databases. No eligible studies were retracted.

Data collection and analysis

Selection of studies

Two review authors (EP, EO) independently screened all titles and abstracts identified by electronic research after removal of duplicates. We used Covidence to screen titles, abstracts, and full texts [65].

The same review authors obtained the full texts for all relevant studies and independently checked the eligibility of each study against review eligibility criteria. We discussed discordant evaluations until we reached a consensus.

Data extraction and management

We developed a data extraction form. Two review authors (EP, EO) independently extracted relevant data. To reach consensus, we involved a third review author (CM) when necessary. We were not blinded to the names of study authors nor to the institutions where studies were conducted and funded. When we encountered multiple publications of the same study, we chose the first publication dealing with the primary endpoint in this review as a study identifier (study ID).

We extracted the following details from each included study.

- Source: citation, study name, and contact details.
- Study details: study design, location, setting (type and stage of disease), number of participating centres, sample size, study start date and completion date, and study follow-up.
- Characteristics of participants: inclusion and exclusion criteria, number of participants, participant and tumour characteristics (age, gender, ethnicity, smoking status, performance status, histology, molecular status, TNM stage, PD-L1 expression, TMB, geriatric characteristics (G8 screening tool, Charlson index, Activities of Daily Living (ADLs), Instrumental Activities of Daily Living (IADLs)).
- Characteristics of interventions (e.g. drugs, doses, cycle duration).
- Outcomes: primary and secondary outcomes with definitions and time points. We extracted data corresponding to the longest follow-up time available at the time of our review for each study.
- Results: number of participants allocated to each group, and for each outcome of interest, sample size, summary data for each group, estimate of effect with confidence interval and P value, and subgroup analyses, and whether analyses were performed by intention-to-treat (ITT) or per-protocol methods.
- Miscellaneous: funding source.

One review author (EP) entered the data into Review Manager (RevMan) [66], and two other review authors (EO, CM) double-checked the data for accuracy.

Risk of bias assessment in included studies

Two review authors (EP, EO) independently applied version 2 of the Cochrane risk of bias tool for randomised trials (RoB 2) per Chapter 8 of the *Cochrane Handbook for Systematic Reviews of Interventions*, to assess quality and potential biases across the included studies [67]. We discussed discordant evaluations until we reached a consensus.

We used the RoB2 Excel tool to implement RoB 2, available at www.riskofbias.info/ (current version 22 August 2019). Using RoB 2, we considered the following domains:

- bias arising from the randomisation process;
- bias due to deviations from intended interventions;
- bias due to missing outcome data;
- bias in measurement of the outcome; and
- bias in selection of the reported result.

For bias due to deviations from intended interventions, we used the effect of assignment to the interventions at baseline (regardless of whether the interventions were received as intended: the intention-

to-treat effect), as recommended in Chapter 8 of the *Cochrane Handbook for Systematic Reviews of Interventions* [67].

The RoB 2 tool provides a framework for assessing the risk of bias in a specific result from a randomised trial. Accordingly, we used the signalling questions and tool algorithms to reach a judgment of low risk, high risk, or 'some concerns' for risk of bias, at study level and for each outcome measure where possible. We provided support for our judgement of each domain with a brief description. We reached the overall risk of bias for each outcome within a study by considering all domains relevant to the outcome (i.e. both study-level entries, such as allocation sequence concealment, and outcome-specific entries, such as blinding).

We also considered other potential biases, such as bias arising from the inclusion of a subset of elderly participants from RCTs that did not stratify randomisation by age.

We provided different tables to summarise the risk of bias for each outcome, outcome measure, and time point.

Measures of treatment effect

For time-to-event outcomes (overall survival, disease-free survival, event-free survival), we used hazard ratios (HRs) to measure treatment effects. We reported each HR along with the 95% confidence interval (CI). An HR of one indicates that the hazard rate is equivalent between experimental and control groups, whilst an HR other than one indicates differences in hazard rates between the two groups. We extracted the HR from the included studies when it was available. When it was not reported in the included study, we attempted to calculate the HR using Kaplan-Meier survival curves and the dedicated methods of Parmar and Tierney [68, 69].

For the dichotomous outcomes (treatment-related adverse events, pathological complete response rate and major pathological response rate), we used risk ratios (RRs) and 95% CIs if possible.

Unit of analysis issues

The participant was the primary unit of analysis.

Dealing with missing data

When we identified missing or unclear data, we contacted the study author directly. We followed Cochrane recommendations when dealing with such issues related to data, as described in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [70] and analysed only the available data. The included studies applied appropriate strategies to deal with missing data, including censoring of patients lost to follow-up. We addressed the potential impact of missing data on the findings of the review in the [Discussion](#).

Reporting bias assessment

We searched online trial registries in order to investigate potential publication bias and to assess potential outcome reporting bias in the included studies.

We planned to generate funnel plots and to perform Egger's linear regression tests to investigate reporting biases for the outcomes considered when the number of trials included in a single meta-analysis was sufficient (at least 10 trials) [71]. However, we did not include 10 or more studies in any meta-analysis. We did not find evidence of small-study effects, and thus did not perform specific

sensitivity analyses with regression-based adjustment methods. We followed the recommendations provided in Chapter 7 of the *Cochrane Handbook for Systematic Reviews of Interventions* [72].

Synthesis methods

When sufficient clinically similar studies were available, we pooled their results in meta-analyses performed on the intention-to-treat (ITT) basis. We grouped studies for analysis according to whether they provided data disaggregated by our predefined age categories of interest. We could not perform specific meta-analyses in two of our age categories (65 to 75 years old; 75 years and older) due to lack of data, and narratively described the available data in the text. We planned to pool data according to PD-L1 tumour proportional score (TPS) or combined positive score (CPS), but the available data were insufficient to allow it.

Overall survival was evaluated without distinguishing by study design (adjuvant, neoadjuvant or perioperative) as our aim was to have a global view of the effect of the addition of an immune checkpoint inhibitor in perioperative settings, and few data were available from the included studies.

We performed meta-analyses according to recommendations in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [70]. We used a random-effects model for all outcomes. Instead of using a fixed-effect model for primary outcomes as a sensitivity analysis, we deemed it more appropriate to use a random-effects model as the studies were different in their clinical methodology. We applied the Mantel-Haenszel method for dichotomous outcomes and the inverse-variance method for continuous outcomes. We applied the DerSimonian and Laird method for the random-effects model [73].

We were able to pool all included studies in at least one meta-analysis. None of the included studies was at high risk of bias.

Investigation of heterogeneity and subgroup analysis

We followed Cochrane recommendations for the assessment of heterogeneity [70]. We visually investigated heterogeneity using forest plots generated by RevMan [66]. We assessed the statistical heterogeneity of treatment effects between pooled trials for each outcome considered by using the I^2 statistic to quantify the degree of heterogeneity [74]. We considered an $I^2 > 30\%$ as indicative of moderate heterogeneity, and an $I^2 > 75\%$ to indicate significant heterogeneity.

We did not observe significant heterogeneity and lacked sufficient data to perform the planned subgroup analyses based on clinical and geriatric characteristics.

Equity-related assessment

Health equity was not explored in this review, as it was not prespecified in the protocol.

Sensitivity analysis

We planned to investigate the robustness of the review through sensitivity analyses, by removing studies at high risk of bias and those with incomplete data, whether imputed or not.

In practice, no formal sensitivity analyses were conducted, as the number of studies in the meta-analyses did not meet the required

threshold. Additionally, no studies were deemed at high risk of bias, and appropriate methods for handling missing data, including censoring of participants lost to follow-up, were applied.

Certainty of the evidence assessment

Using GRADEpro GDT software [75], we created a summary of findings table for our comparison (perioperative immune checkpoint inhibitors ± chemotherapy versus placebo or no treatment ± chemotherapy in people ≥ 65 years old with early-stage resectable non-small cell lung cancer), in which we reported the outcomes listed below. These assessments take into account the risk of bias assessments for each set of available study data, aggregated as follows.

- Overall survival: at any time point, including data from all treatment sequences (adjuvant only, neoadjuvant only or both), aggregated as an HR with a 95% CI.
- Treatment-related adverse events (grade ≥ 3): at any time point, including data from all treatment sequences, reported as a risk ratio (RR) from one study only with a 95% CI.
- Disease-free survival: at any time point, including data from adjuvant-only studies, aggregated as an HR with a 95% CI.
- Event-free survival: at any time point, including data from neoadjuvant-only and perioperative studies, aggregated as an HR with a 95% CI.
- Pathological complete response rate: at surgery, including data from neoadjuvant-only and perioperative studies, aggregated as a RR with a 95% CI.
- Major pathological response rate: at surgery, including data from neoadjuvant-only and perioperative studies, aggregated as a RR with a 95% CI.

As recommended in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* [76], we used the five GRADE considerations of study limitations, consistency of effect, imprecision, indirectness, and publication bias to assess the certainty of a body of evidence as it relates to the studies that contribute data to the meta-analyses for the prespecified outcomes.

We used the overall RoB 2 judgement to feed into the GRADE assessment, as recommended in Chapters 8 and 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* [67, 76].

Following the recommendations in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* [76], we used the I^2 statistical test to judge inconsistency when heterogeneity was apparent in a meta-analysis. If a single study contributed data to an outcome, we did not perform a heterogeneity test, as there could not be inconsistency in this case.

Three co-authors (EP, EO and RM) independently assessed the certainty of the evidence. We discussed discordant evaluations until we reached a consensus. We justified any decisions to downgrade the certainty of the evidence using footnotes and made comments to aid the reader's understanding of the review where necessary.

Consumer involvement

Although we recognise the importance of consumer input in systematic reviews, we did not involve consumers or stakeholders because the review focused on standardised clinical efficacy and safety outcomes, where consumer perspectives were not central, and conducting consumer consultations was beyond the feasible scope given time and resource constraints. Nevertheless, all methodological steps adhered to Cochrane standards, and we sought to obtain relevant data (e.g. health-related quality of life) from published studies.

RESULTS

Description of studies

Results of the search

We retrieved 12,267 records from database searches up to 3 July 2025. No additional studies were found through searching other sources. After removing the duplicate references (3308 records), 8959 records remained. We excluded 8793 records based on title and abstract screening. We retrieved the remaining 166 records for full-text screening, corresponding to 35 unique studies. Of these, we included 11 studies (124 references; see [Supplementary material 2](#)), excluded 12 studies for various reasons (21 references; see [Supplementary material 3](#)), and identified 12 ongoing studies (21 references; see [Supplementary material 4](#)). [Figure 1](#) depicts the study flow.

Figure 1. Study flow diagram

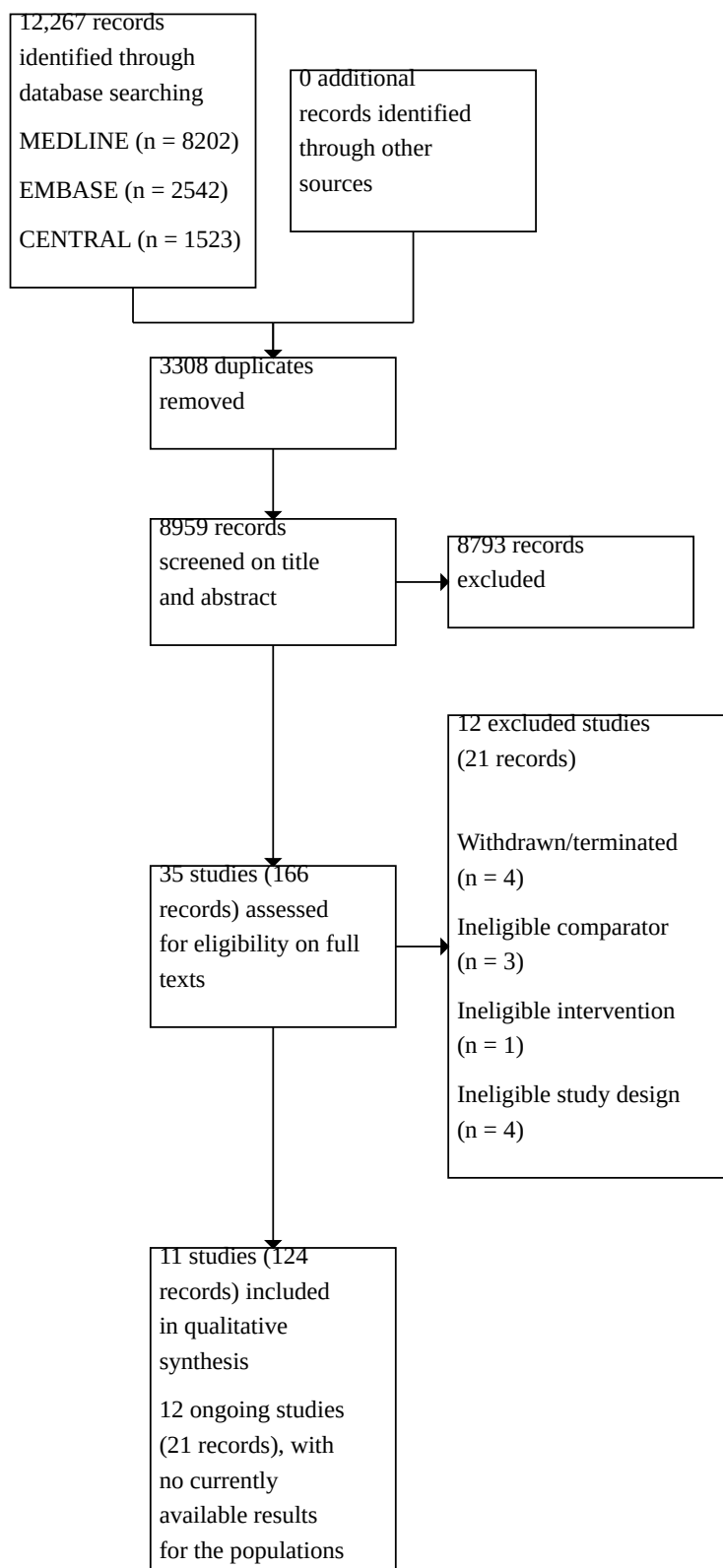
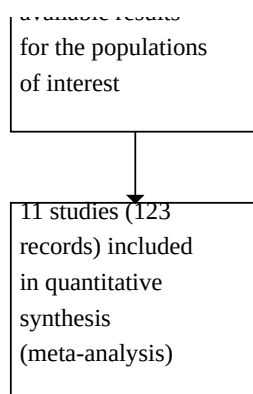


Figure 1. (Continued)



Included studies

All 11 included studies were RCTs, published in English between 2021 and 2025 (see [Table 1](#) and [Supplementary material 2](#)). They included a total of 6788 participants with resectable NSCLC, of which 3152 (46.4%) were ≥ 65 years old.

All trials were funded by pharmaceutical companies. CCTG BR.31 [77, 78, 79, 80, 81, 82] and NADIM-II [60, 83, 84, 85, 86, 87, 88] also received partial public funding – the former through support from a Canadian trial group, and the latter through European and institutional grants.

Study populations

The baseline characteristics of participants in each included study can be seen in [Supplementary material 9](#); [Supplementary material 10](#); [Supplementary material 11](#); [Supplementary material 12](#); [Supplementary material 13](#); [Supplementary material 14](#); [Supplementary material 15](#); [Supplementary material 16](#); [Supplementary material 17](#); [Supplementary material 18](#).

The inclusion criteria were similar across studies, accepting adults aged at least 18 years presenting with an early-stage resectable NSCLC (stages IB-IIIa/B from the seventh edition of TNM, stages II-IIIa/B from the eighth edition of TNM), with an Eastern Cooperative Oncology Group (ECOG) performance status score of 0 or 1. NADIM-II, NEOTORCH [89, 90, 91, 92] and TD-FOREKNOW [93, 94, 95] focused on advanced tumours (stage III). Molecular alterations were not systematically tested, but when identified, they were a criterion of exclusion from most trials (except for IMPOWER-010 [96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119]; KEYNOTE-671 [62, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130]; PEARLS/ KEYNOTE-091 [131, 132, 133, 134, 135, 136, 137, 138, 139, 140], in which it was authorised).

Overall, people aged ≥ 65 years old represented 46.4% (3152/6788) of all participants in the 11 included studies. People ≥ 75 years old represented 10.5% (87/826) of all participants in the two studies whose authors graciously shared these numbers (AEGEAN [59, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157]; NADIM-II). People over 70 years old were excluded from NEOTORCH.

Almost all studies stratified participants according to PD-L1 expression, evaluated through different assays and TPS (Tumour Proportion Score) cut-offs: AEGEAN (cut-off at 1% with SP263 assay), CCTG BR.31 (cut-offs at 1% and 25% with SP263 assay), CHECKMATE-77T [61, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168] (cut-off at 1% with 28-8 assay), IMPOWER-010 (cut-off at 5% with SP142 assay), KEYNOTE-671 (cut-off at 50% with 22C3 assay), NEOTORCH (cut-off at 1% with JS311 assay), PEARLS/ KEYNOTE-091 (cut-offs at 1% and 50% with 22C3 assay), RATIONALE-315 [63, 169, 170, 171, 172, 173] (cut-off at 1%) and TD-FOREKNOW (cut-off at 1% with SP263 assay). Most studies stratified by disease stage (AEGEAN; CCTG BR.31; CHECKMATE-77T; IMPOWER-010; KEYNOTE-671; NEOTORCH; PEARLS/ KEYNOTE-091; RATIONALE-315) and histology (CHECKMATE-77T; IMPOWER-010; KEYNOTE-671; NEOTORCH; RATIONALE-315). CCTG BR.31 and the KEYNOTE studies were also stratified by geographical region or accruing centre (KEYNOTE-671; PEARLS/ KEYNOTE-091). CCTG BR.31 had a stratification by the administration of adjuvant platinum-based chemotherapy and by the realisation of a nodal dissection. Finally, IMPOWER-010 was stratified by gender. There was no stratification in the CHECKMATE-816 [58, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194], NADIM-II and TD-FOREKNOW trials.

Most of the included and ongoing studies of our review did not enrol (or enrolled only a small number of) people from Africa and South America. One of the ongoing studies, Biswal (ELoN), evaluates low-dose Nivolumab in India, where access to healthcare is unequal.

None of them reported significant differences in baseline characteristics between study arms.

Settings

Seven included studies were international multicentre phase III trials. NEOTORCH, RATIONALE-315 and TD-FOREKNOW were phase III multicentre trials based only in China. NADIM-II was a phase II, multicentre trial restricted to Spain.

Study interventions and comparisons

Six included studies compared perioperative ICIs (anti-PD-1 or anti-PD-L1) to placebo or no treatment in association with neoadjuvant chemotherapy (AEGEAN; CHECKMATE-77T; KEYNOTE-671; NADIM-II; NEOTORCH; RATIONALE-315). CHECKMATE-816 and TD-

FOREKNOW tested the addition of neoadjuvant ICIs to neoadjuvant or perioperative chemotherapy. The three remaining studies investigated adjuvant ICIs, with or without prior adjuvant chemotherapy, in comparison to placebo or best supportive care (CCTG BR.31; IMPOWER-010; PEARLS/ KEYNOTE-091).

The interventions in the included studies are detailed in [Table 1](#). Three to four cycles of neoadjuvant regimens were administered in perioperative and neoadjuvant trials. Adjuvant regimens were usually ICI monotherapies pursued for one year, except for NEOTORCH in which a concomitant chemotherapy was associated at the first adjuvant cycle. Most of the adjuvant trials allowed prior adjuvant chemotherapy but prohibited neoadjuvant treatments, except for CCTG BR.31 which included participants previously exposed to a platinum doublet. Adjuvant radiotherapy was permitted in AEGEAN, CCTG BR.31, CHECKMATE-816 and KEYNOTE-671. All chemotherapy regimens were platinum-based doublets, tailored to histological types and participant's fitness. No cross-over was allowed.

Study durations

The median duration of follow-up was 11.7 months in AEGEAN, 60.0 months in CCTG BR.31, 25.4 months in CHECKMATE-77T, 29.5 months in CHECKMATE-816, 65.0 months in IMPOWER-010, 36.6 months in KEYNOTE-671, 26.1 months in NADIM-II, 18.3 months in NEOTORCH, 57.1 months in PEARLS/ KEYNOTE-091, 22.0 months in RATIONALE-315, and 14.1 months in TD-FOREKNOW.

Study outcomes

Primary outcomes

Disease-free survival (DFS) was the primary outcome in the three adjuvant trials (CCTG BR.31; IMPOWER-010; PEARLS/ KEYNOTE-091). Almost all perioperative and neoadjuvant trials had a coprimary endpoint with event-free survival (EFS); EFS and pathological complete response (pCR) rate for AEGEAN and CHECKMATE-816, EFS and major pathological response (MPR) rate for NEOTORCH and RATIONALE-315, EFS and overall survival (OS) for KEYNOTE-671. The unique primary outcome was EFS for CHECKMATE-77T and pCR for NADIM-II and TD-FOREKNOW.

These primary outcomes were assessed by blinded independent pathological or clinical review committees in AEGEAN, CHECKMATE-77T, CHECKMATE-816, NADIM-II, NEOTORCH and RATIONALE-315 studies. They were investigator-assessed in CCTG BR.31, IMPOWER-010, KEYNOTE-671 and PEARLS/ KEYNOTE-091 trials. Local pathologists assessed pCR in TD-FOREKNOW.

A protocol amendment for IMPOWER-010 while the trial was ongoing restricted the tested population for its primary outcome of DFS to the population with stages II-IIIa and PD-L1 $\geq 1\%$, excluding stages IB and PD-L1 $< 1\%$ tumours from the primary analysis. The primary population in CCTG BR.31 was restricted to PD-L1 $\geq 25\%$ tumours. The other studies did not select their primary population according to PD-L1 status.

None of the studies defined safety, particularly treatment-related adverse events (TRAEs), or health-related quality of life (HRQoL), as the primary outcome.

Secondary outcomes

Overall survival was a secondary outcome in all included studies (except for KEYNOTE-671, in which it was a coprimary endpoint, and TD-FOREKNOW which did not evaluate it). pCR and MPR were secondary outcomes for each neoadjuvant/perioperative study for which they were not a primary endpoint. Safety was a secondary outcome in almost all trials (AEGEAN; CCTG BR.31; CHECKMATE-77T; KEYNOTE-671; NADIM-II; NEOTORCH; PEARLS/ KEYNOTE-091; RATIONALE-315; TD-FOREKNOW). HRQoL or participant-reported outcomes (PROs) were evaluated as secondary endpoints in the four following studies: AEGEAN, CCTG BR.31, KEYNOTE-671 and RATIONALE-315. Pharmacokinetics and immunogenicity were secondary outcomes of AEGEAN and NEOTORCH. CHECKMATE-816 defined time to distant metastasis (TTDM) as an additional secondary outcome and PEARLS/ KEYNOTE-091 evaluated lung cancer-specific survival (LCSS). A number of these outcomes were evaluated in different subpopulations, notably according to PD-L1 expression, molecular alterations or stage of the disease.

Excluded studies

We excluded a total of 12 studies for the following reasons: CATHAYA [195], NALAN-01 [196], MERMAID-1 [197, 198, 199, 200], and MERMAID-2 [201, 202, 203, 204] were withdrawn due to ethical concerns and early termination following changes in the treatment landscape; ADOPT-LUNG [205, 206], INSIGHT [207], and STAR-011 [208, 209] were excluded because of ineligible study designs; CANOPY-N [210, 211], NEOCOAST-2 [212], and Huynh 2021 [213] were excluded due to the use of ineligible comparators; ESPADURVA [214] was excluded because it evaluated induction chemotherapy and chemoradiotherapy, which are not current standards of care in this setting; and the ALCHEMIST platform [215] was excluded as it does not correspond to a specific trial. Further details are provided in [Supplementary material 3](#).

Ongoing studies

We considered 12 ongoing studies for this review that met our criteria, but have not yet provided results for patients over 65 years old. Their characteristics are summarised in [Table 2](#) and detailed in [Supplementary material 4](#).

Risk of bias in included studies

We assessed the risk of bias in the included studies using the Cochrane RoB 2 tool [216]. Detailed risk of bias assessment data are available online or from authors on request. Results-level RoB 2 tables include the justification for each domain-level judgement (see [Supplementary material 5](#)). Below, we provide an overview of the overall risk of bias for each outcome.

Critical outcomes

Overall survival, at any time point

KEYNOTE-671 and NADIM-II were at low risk of bias, but there were 'some concerns' about bias for IMPOWER-010 due to possible deviations from intended interventions and selection of the reported results (it was an open-label trial and more participants withdrew consent in the atezolizumab arm; moreover there was a protocol amendment to the primary analysis population from the ITT population to the subpopulation of stages II-IIIa with PD-L1 $\geq 1\%$).

Treatment-related adverse events

NADIM-II, the only study providing data, was at low risk of bias.

Important outcomes

Disease-free survival, at any time point

CCTG BR.31 and PEARLS/ KEYNOTE-091 were at low risk of bias, while there were 'some concerns' about bias for IMPOWER-010 due to deviations from intended interventions, in measurement of the outcome, and selection of the reported results (it was an open-label trial and more participants withdrew consent in the atezolizumab arm; the outcome was assessed by unblinded investigators; and there was a protocol amendment to the primary analysis population from the ITT population to the subpopulation of stages II-IIIa with PD-L1 $\geq 1\%$).

Event-free survival, at any time point

Most studies were at low risk of bias (AEGEAN; CHECKMATE-77T; KEYNOTE-671; NADIM-II; NEOTORCH; RATIONALE-315). CHECKMATE-816 raised 'some concerns' about potential deviations from intended interventions (it was an open-label trial and 6% of participants withdrew from the control arm versus none in the intervention arm).

Pathological complete response

Four of the studies with results included in the meta-analysis of this outcome were at low risk of bias (AEGEAN; CHECKMATE-77T; NADIM-II; RATIONALE-315). CHECKMATE-816 raised 'some concerns' about potential deviations from intended interventions (it was an open-label trial and 6% of participants

withdrew from the control arm versus none in the intervention arm). TD-FOREKNOW raised 'some concerns' about outcome measurement because, in this open-label trial, the pCR was assessed locally.

Major pathological response

Four of the studies with results included in the meta-analysis of this outcome were at low risk of bias (AEGEAN; CHECKMATE-77T; NADIM-II; RATIONALE-315). CHECKMATE-816 raised 'some concerns' about potential deviations from intended interventions (it was an open-label trial and 6% of participants withdrew from the control arm versus none in the intervention arm). TD-FOREKNOW raised 'some concerns' about outcome measurement because, in this open-label trial, the MPR was assessed locally.

Synthesis of results

Summary of findings 1 and Supplementary material 6 present the review outcomes in people ≥ 65 years old treated in perioperative settings with ICI (neoadjuvant, adjuvant or both) alone or in combination with platinum-based chemotherapy for resectable NSCLC. Survival outcomes are based on median values because data were insufficient to assess these outcomes at precise time points.

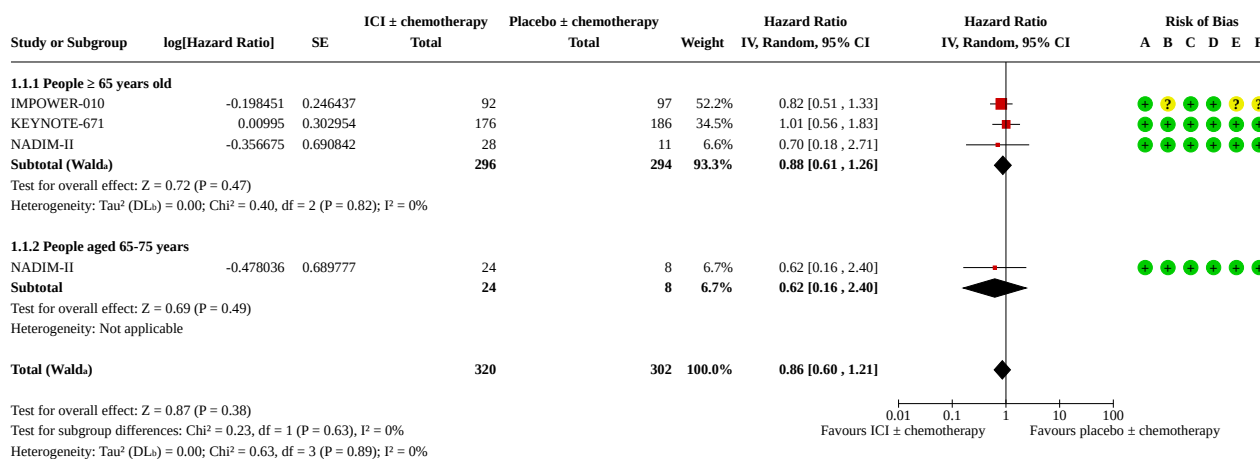
Critical outcomes

Overall survival at any time point (all studies)

Overall survival according to age

Three studies provided data for Analysis 1.1 (IMPOWER-010; KEYNOTE-671; NADIM-II) (see Figure 2).

Figure 2. Forest plot of perioperative immune checkpoint inhibitors $\pm$ chemotherapy versus placebo $\pm$ chemotherapy, outcome 1.1 Overall survival at any time according to age with single-agent ICI



Footnotes

aCI calculated by Wald-type method.

bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

For people aged ≥ 65 years old

Aggregated data from these three studies, involving 590 participants, showed that perioperative ICI probably results in little to no difference in overall survival compared to placebo or no treatment, with a hazard ratio (HR) of 0.88 (95% confidence interval (CI) 0.61 to 1.26; $P = 0.47$; $I^2 = 0\%$) with a moderate-certainty level of evidence (downgraded for risk of bias).

For people aged 65 to 75 years

Only NADIM-II provided usable information from participants between 65 and 75 years old: after approximately 39 months of follow-up, 29.2% (7/24) of participants had died in the intervention arm and 37.5% (3/8) in the control arm. In this population, perioperative ICI may result in little to no difference in overall survival compared to placebo or no treatment, with an HR of 0.62 (95% CI 0.16 to 2.40; $P = 0.49$; 32 participants).

For people aged ≥ 75 years old

No data available yet.

Overall survival according to intervention

Four studies provided data for Analysis 1.2 in [Supplementary material 6](#) (CHECKMATE-816; IMPOWER-010; KEYNOTE-671; NADIM-II).

Dual immunotherapy

Only CHECKMATE-816 evaluated the combination of two ICIs (nivolumab and ipilimumab). In participants aged ≥ 65 years, after a median follow-up of 49.2 months, the median overall survival was 56.54 months with dual ICI and was not reached in the chemotherapy arm, with an HR of 0.92 (95% CI 0.50 to 1.69; $P = 0.79$; 110 participants). Dual ICI was not compared to single-agent ICI.

Overall survival according to PD-L1 status

Three studies provided data for Analysis 1.3 in [Supplementary material 6](#) (IMPOWER-010; KEYNOTE-671; NADIM-II).

All PD-L1

KEYNOTE-671 and NADIM-II reported data for populations with all PD-L1 expression levels. Aggregated data showed an HR of 0.95 (95% CI 0.55 to 1.64; $P = 0.86$, $I^2 = 0\%$; 401 participants).

PD-L1 $\geq 1\%$ subgroup

IMPOWER-010 reported results in the PD-L1 $\geq 1\%$ population. In participants ≥ 65 years old, after a median follow-up of 65.0 months, the median overall survival was not reached in either arm, with an HR of 0.82 (95% CI 0.51 to 1.33; $P = 0.42$; 189 participants) in this population.

PD-L1 $\geq 50\%$ subgroup

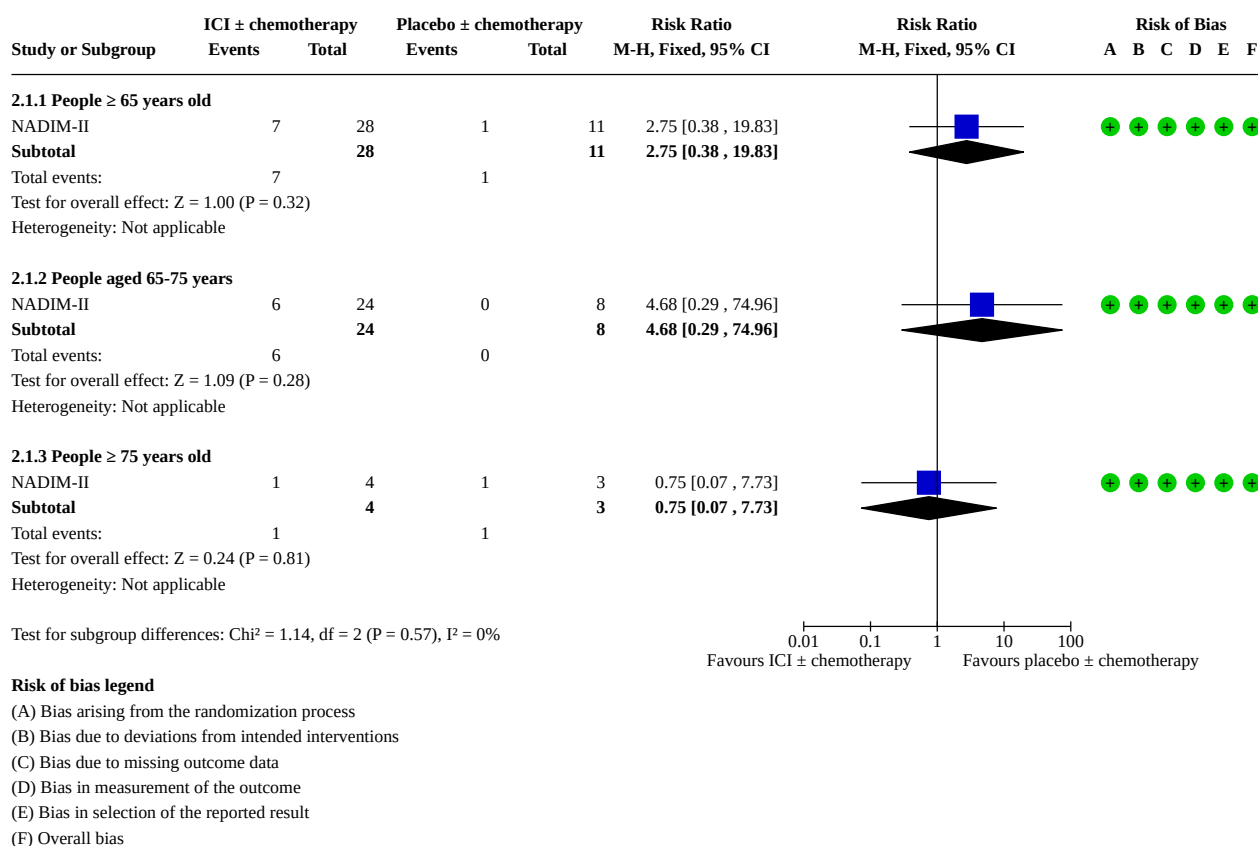
IMPOWER-010 also reported results in the PD-L1 $\geq 50\%$ population. In participants aged ≥ 65 years, after a median follow-up of 65.0 months, the median overall survival was not reached in either arm, with an HR of 0.56 (95% CI 0.26 to 1.21; $P = 0.14$; 82 participants) in this population.

Treatment-related adverse events grade ≥ 3 at any time point (all studies)

Treatment-related adverse events grade ≥ 3 according to age

Only NADIM-II provided usable information for the analysis of treatment-related adverse events (TRAEs) (Analysis 2.1). The data below were obtained after approximately 39 months of follow-up (see [Figure 3](#)).

Figure 3. Forest plot of perioperative immune checkpoint inhibitors ± chemotherapy versus placebo ± chemotherapy, outcome 2.1 Treatment-related adverse events ≥ grade 3 at any time according to age with single-agent ICI



For people aged ≥ 65 years old

Among people aged ≥ 65 years old, 25.0% (7/28) of participants presented grade ≥ 3 TRAEs in the intervention arm and 9.1% (1/11) in the control arm. The risk ratio (RR) was 2.75 (95% CI 0.38 to 19.83; $P = 0.32$; 39 participants) with a low certainty of evidence (downgraded for indirectness and imprecision), indicating that the addition of perioperative ICI to neoadjuvant chemotherapy may result in little to no difference in TRAEs ≥ grade 3 compared to chemotherapy alone.

For people aged 65 to 75 years

Among people aged between 65 and 75 years, 25.0% of participants presented grade ≥ 3 TRAEs (6/24) in the intervention arm and 0.0% (0/8) in the control arm. The RR was 4.68 (95% CI 0.29 to 74.96; $P = 0.28$; 32 participants), indicating that the addition of perioperative ICI to neoadjuvant chemotherapy may result in little to no difference in TRAEs ≥ grade 3 compared to chemotherapy alone.

For people aged ≥ 75 years old

Among people aged ≥ 75 years old, 25.0% of participants presented grade ≥ 3 TRAEs (1/4) in the intervention arm and 33.3% (1/3) in

the control arm. The RR was 0.75 (95% CI 0.07 to 7.73; $P = 0.81$; 7 participants), indicating that the addition of perioperative ICI to neoadjuvant chemotherapy may result in little to no difference in TRAEs ≥ grade 3 compared to chemotherapy alone.

Treatment-related adverse events grade ≥ 3 according to intervention

Dual immunotherapy

No data available yet.

Treatment-related adverse events grade ≥ 3 according to PD-L1 status

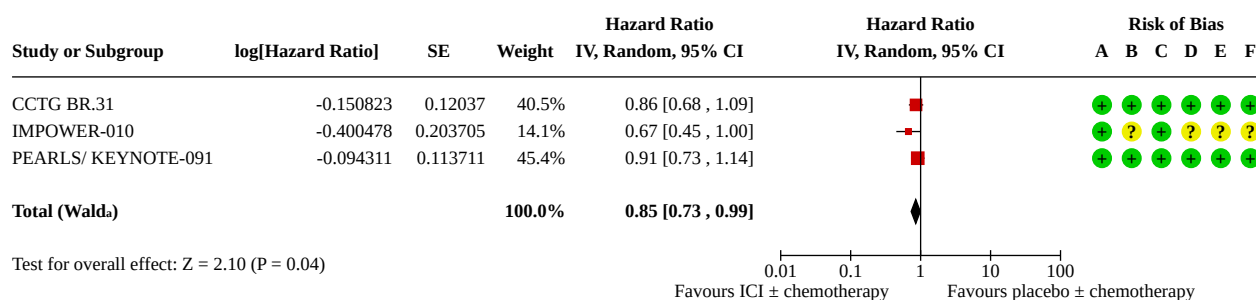
No data available yet.

Important outcomes

Disease-free survival at any time point (adjuvant studies)

Disease-free survival according to age

Three studies provided data for Analysis 3.1 (CCTG BR.31; IMPOWER-010 and PEARLS/ KEYNOTE-091) (see [Figure 4](#)).

Figure 4. Forest plot of perioperative immune checkpoint inhibitors ± chemotherapy versus placebo ± chemotherapy, outcome 3.1 Disease-free survival at any time in people ≥ 65 years old with single-agent ICI**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Bias arising from the randomization process
 (B) Bias due to deviations from intended interventions
 (C) Bias due to missing outcome data
 (D) Bias in measurement of the outcome
 (E) Bias in selection of the reported result
 (F) Overall bias

For people aged ≥ 65 years old

Aggregated data from these three studies showed that adjuvant ICI probably slightly increases disease-free survival compared to placebo or no treatment, with an HR of 0.85 (95% CI 0.73 to 0.99; P = 0.04, I² = 0%; 1403 participants) with moderate-certainty evidence (downgraded for risk of bias).

For people aged 65 to 75 years

No data available yet.

For people aged ≥ 75 years old

No data available yet.

Disease-free survival according to intervention**Dual immunotherapy**

No data available yet.

Disease-free survival according to PD-L1 status

The same three studies provided data for Analysis 3.2 in [Supplementary material 6](#).

All PD-L1

PEARLS/ KEYNOTE-091 reported results across all PD-L1 levels. In participants ≥ 65 years old, after a median follow-up of 52 months, the HR was 0.91 (95% CI 0.73 to 1.14; P = 0.41; 619 participants).

PD-L1 ≥ 1% subgroup

IMPOWER-010 reported results in the PD-L1 ≥ 1% population. In participants ≥ 65 years old, after a median follow-up of 60 months, the median disease-free survival was 65.6 months in the ICI arm and 40.6 months in the observation arm, with an HR of 0.67 (95% CI 0.45 to 1.00; P = 0.05; 189 participants) in this population.

PD-L1 ≥ 25% subgroup

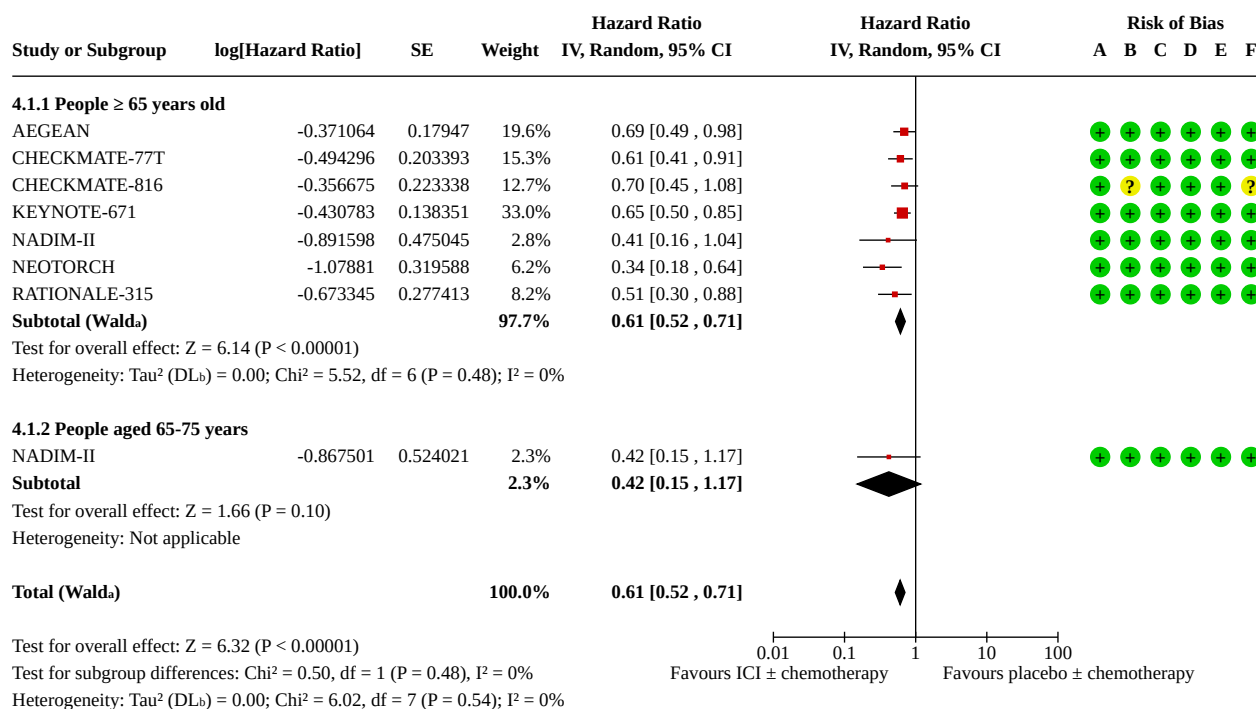
CCTG BR.31 reported results in the PD-L1 ≥ 25% population. In participants ≥ 65 years old, after a median follow-up of 60 months, 48.6% (197/405) of participants in the ICI arm had presented an event and 52.6% (100/190) of participants in the placebo arm, with an HR of 0.86 (95% CI 0.68 to 1.09; P = 0.21; 595 participants) in this population.

PD-L1 ≥ 50% subgroup

IMPOWER-010 reported results in the PD-L1 ≥ 50% population. In participants ≥ 65 years old, after a median follow-up of 60 months, the median disease-free survival was not reached in the ICI arm and was 42.9 months in the observation arm, with an HR of 0.52 (95% CI 0.27 to 1.00; P = 0.05; 82 participants) in this population.

Event-free survival at any time point (neoadjuvant and perioperative studies)**Event-free survival according to age**

Seven studies provided data for Analysis 4.1 (AEGEAN; CHECKMATE-77T; CHECKMATE-816; KEYNOTE-671; NADIM-II; NEOTORCH; RATIONALE-315) (see [Figure 5](#)).

Figure 5. Forest plot of Perioperative immune checkpoint inhibitors ± chemotherapy versus placebo ± chemotherapy, outcome 4.1 Event-free survival at any time according to age with single-agent ICI**Footnotes**^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

For people aged ≥ 65 years old

Aggregated data from seven studies showed that neoadjuvant or perioperative ICI likely increases event-free survival compared to placebo or no treatment, with an HR of 0.61 (95% CI 0.52 to 0.71; P < 0.001; I² = 0%; 1531 participants) and moderate-certainty evidence (downgraded for risk of bias).

For people aged 65 to 75 years

Only NADIM-II provided usable data for participants aged 65 to 75 years. After approximately 39 months of follow-up, 41.7% (10/24) of participants had an event in the ICI arm and 75.0% (6/8) in the control arm. The HR was 0.42 (95% CI 0.15 to 1.17; P = 0.10; 32 participants), indicating that the addition of perioperative ICI to neoadjuvant chemotherapy may result in little to no difference in event-free survival compared to chemotherapy alone.

For people aged ≥ 75 years old

No data available yet.

Event-free survival according to intervention

The same seven studies provided data on single-ICI, and one study (CHECKMATE-816) provided data on dual ICI for Analysis 4.2 in [Supplementary material 6](#).

Dual immunotherapy

Only CHECKMATE-816 evaluated the combination of two ICIs (nivolumab and ipilimumab). In participants ≥ 65 years old, after a median follow-up of 49.2 months, the median event-free survival was 54.80 months in the dual ICI arm and 24.94 months in the chemotherapy arm, with an HR of 0.84 (95% CI 0.48 to 1.47; P = 0.54; 82 participants). Dual ICI was not compared to single-agent ICI.

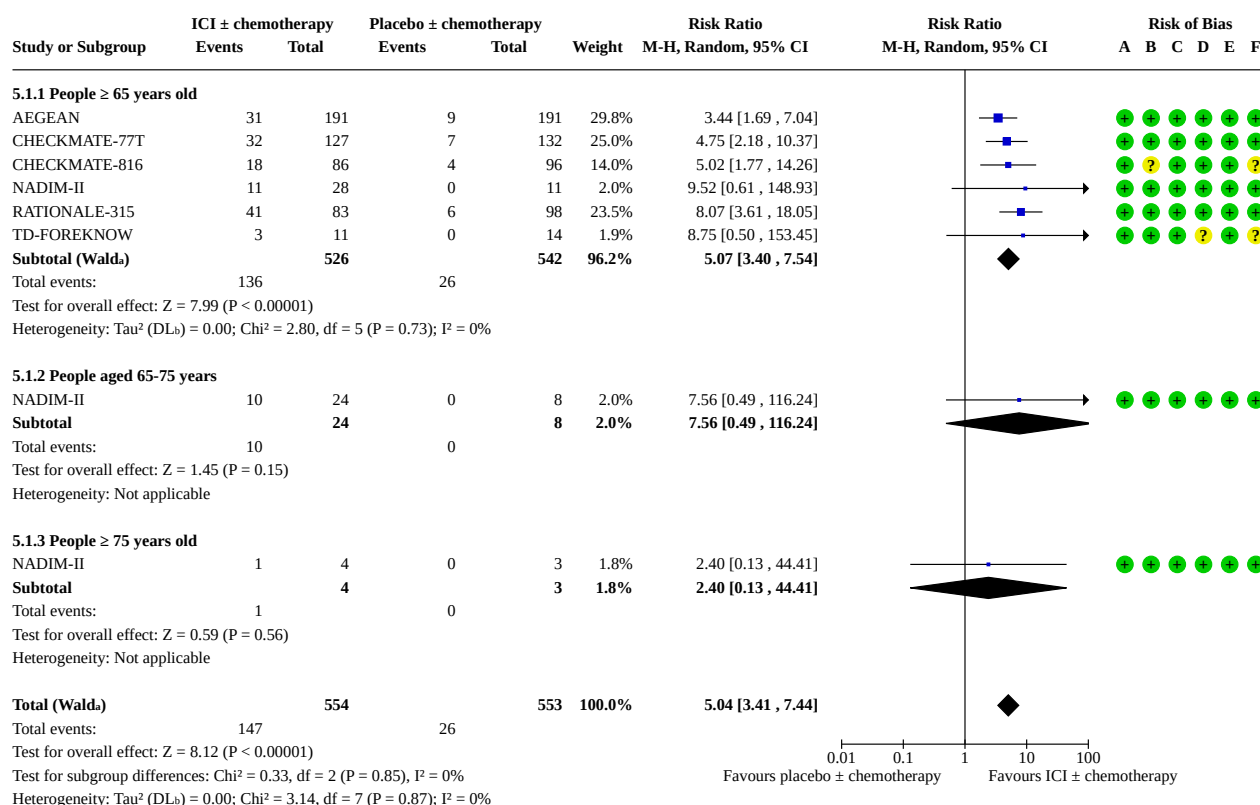
Event-free survival according to PD-L1 status

No subgroup data available yet. The seven studies reporting EFS data included all PD-L1 levels.

Pathological complete response rate at surgery (neoadjuvant and perioperative studies)**Pathological complete response rate according to age**

Six studies provided data for Analysis 5.1 (AEGEAN; CHECKMATE-77T; CHECKMATE-816; NADIM-II; RATIONALE-315; TD-FOREKNOW) (see Figure 6).

Figure 6. Forest plot of perioperative immune checkpoint inhibitors ± chemotherapy versus placebo ± chemotherapy, outcome 5.1 Pathological complete response rate according to age with single-agent ICI

**Footnotes**

^aCI calculated by Wald-type method.

^bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

For people aged ≥ 65 years old

Aggregated data from six studies showed that neoadjuvant or perioperative ICI may result in a large increase in pathological complete response rates compared to placebo or no treatment, with an RR of 5.07 (95% CI 3.40 to 7.54; $P < 0.001$, $I^2 = 0\%$; 1068 participants) and low-certainty evidence (downgraded for risk of bias and imprecision).

For people aged 65 to 75 years

Only NADIM-II provided usable data from people aged 65 to 75 years: 42% (10/24) of participants presented a pCR in the ICI arm

and 0% (0/8) in the control arm. Perioperative ICI with neoadjuvant chemotherapy may result in little to no difference in pCR compared to chemotherapy alone with an RR of 7.56 (95% CI 0.49 to 116.24; $P = 0.15$; 32 participants) in this population.

For people aged ≥ 75 years old

Only NADIM-II provided usable data in people aged ≥ 75 years: 25% (1/4) of participants presented a pCR in the ICI arm and 0% (0/3) in the control arm. Perioperative ICI with neoadjuvant chemotherapy may result in little to no difference in pCR compared

to chemotherapy alone, with an RR of 2.40 (95% CI 0.13 to 44.41; $P = 0.56$; 7 participants) in this population.

Pathological complete response rate according to intervention

The same six studies provided data on single-ICI, and one study (CHECKMATE-816) provided data on dual ICI for Analysis 5.2 in [Supplementary material 6](#).

Dual immunotherapy

Only CHECKMATE-816 evaluated the combination of two ICIs (nivolumab and ipilimumab). In people ≥ 65 years old, 25.5% (13/51; 95% CI 14.3 to 39.6) of participants presented a pCR in the dual ICI arm and 8.5% (5/59; 95% CI 2.8 to 18.7) in the chemotherapy arm,

with an RR of 3.01 (95% CI 1.15 to 7.86; $P = 0.02$; 110 participants). Dual ICI was not compared to single-agent ICI.

Pathological complete response rate according to PD-L1 status

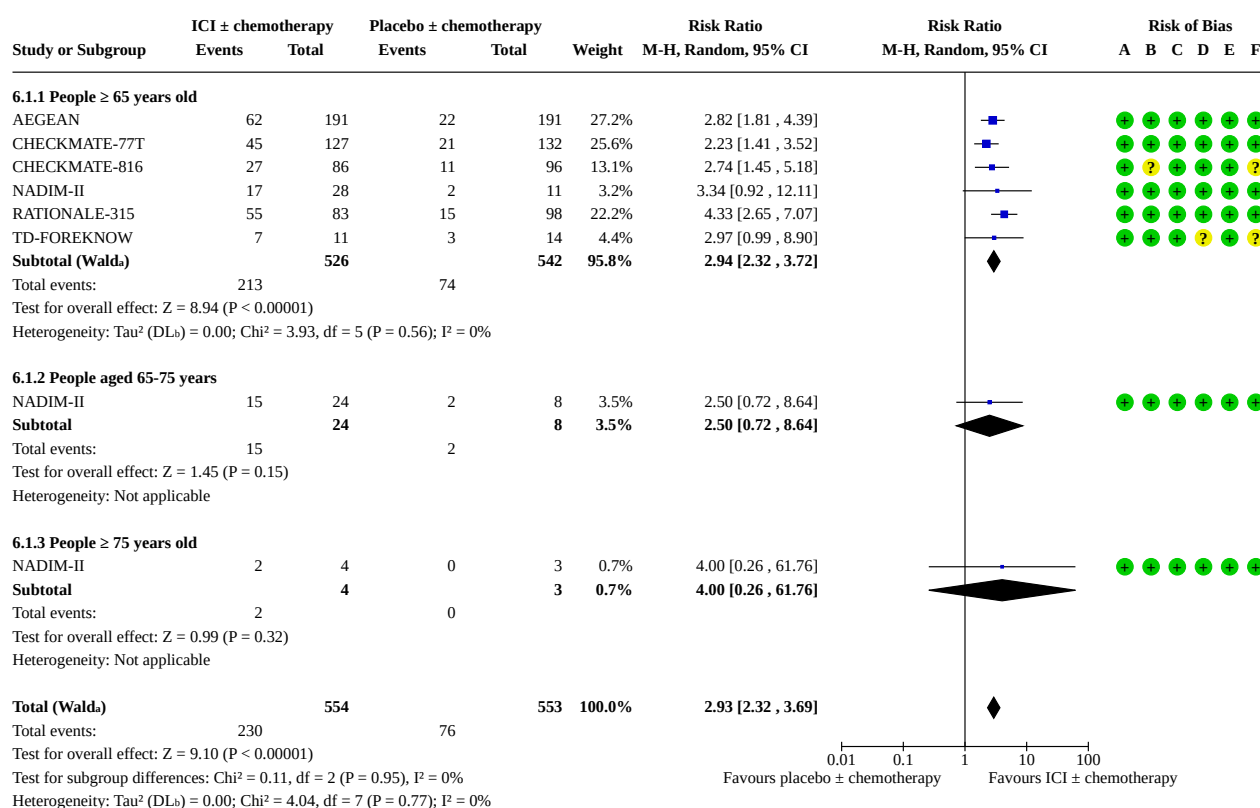
No subgroup data available yet. The six studies reporting pCR data included all PD-L1 levels.

Major pathological response rate at surgery (neoadjuvant and perioperative studies)

Major pathological response rate according to age

Six studies provided data for Analysis 6.1 (AEGEAN; CHECKMATE-77T; CHECKMATE-816; NADIM-II; RATIONALE-315; TD-FOREKNOW) (see [Figure 7](#)).

Figure 7. Forest plot of perioperative immune checkpoint inhibitors $\pm$ chemotherapy versus placebo $\pm$ chemotherapy, outcome 6.1 Major pathological response according to age with single-agent ICI



Footnotes

^aCI calculated by Wald-type method.

^b Tau^2 calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

For people aged ≥ 65 years old

Aggregated data from six studies showed that neoadjuvant or perioperative ICI probably results in a large increase of major pathological response rates compared to placebo or no treatment,

with an RR of 2.94 (95% CI 2.32 to 3.72; $P < 0.001$, $I^2 = 0\%$; 1068 participants) and moderate-certainty evidence (downgraded for risk of bias).

For people aged 65 to 75 years

Only NADIM-II provided usable data from people aged 65 to 75 years: 62% (15/24) of participants presented an MPR in the ICI arm and 25% (2/8) in the control arm. Perioperative ICI with neoadjuvant chemotherapy may result in little to no difference in MPR compared to chemotherapy alone, with a RR of 2.50 (95% CI 0.72 to 8.64; $P = 0.15$; 32 participants) in this population.

For people aged ≥ 75 years old

Only NADIM-II provided usable data in people aged ≥ 75 years: 50% of participants presented an MPR (2/4) in the ICI arm and 0% (0/3) in the control arm. Perioperative ICI with neoadjuvant chemotherapy may result in little to no difference in MPR compared to chemotherapy alone, with a RR of 4.00 (95% CI 0.26 to 61.76; $P = 0.32$; 7 participants) in this population.

Major pathological response rate according to intervention

Dual immunotherapy

No data available yet.

Major pathological response rate according to PD-L1 status

No subgroup data available yet. The six studies reporting MPR data included all PD-L1 levels.

Reporting biases

There may be a reporting bias in the adjuvant studies, since Chaff (ANVIL) [217, 218] and Peters (IMPOWER-030) [219, 220, 221, 222] have been ongoing since 2018 without releasing results. Based on our search in trial registries, we do not have additional arguments for reporting bias. Since there were fewer than 10 studies included in the meta-analyses, we did not generate funnel plots or conduct Egger's linear regression tests.

DISCUSSION

Summary of main results

We obtained data for 3152 participants from 11 studies comparing ICIs alone or in combination with a platinum-based doublet chemotherapy, in perioperative settings in people aged 65 years and older with early-stage resectable NSCLC. Most included studies were at low risk of bias, except for IMPOWER-010 (some concerns for overall survival and disease-free survival), CHECKMATE-816 and TD-FOREKNOW (some concerns for pathological complete response and major pathological response) (see [Summary of findings 1](#)).

Overall, in people aged ≥ 65 years, the perioperative use of ICIs (with or without platinum-based chemotherapy) compared to placebo or no treatment (with or without platinum-based chemotherapy):

- probably results in little to no difference in overall survival at any follow-up time point;
- may result in little to no difference in treatment-related adverse events of grade ≥ 3 ;
- probably slightly increases disease-free survival at any follow-up time point (adjuvant immunotherapy only);
- probably increases event-free survival at any follow-up time point (neoadjuvant or perioperative immunotherapy);
- may result in a large increase in pathological complete response rate (neoadjuvant or perioperative immunotherapy); and

- probably results in a large increase in major pathological response rate (neoadjuvant or perioperative immunotherapy).

Due to insufficient data, we were unable to draw conclusions about:

- health-related quality of life;
- outcomes in age subgroups other than ≥ 65 years;
- outcomes according to PD-L1 status; and
- outcomes with dual immunotherapy regimens.

Limitations of the evidence included in the review

Overall completeness and applicability of evidence

The following limitations may affect the strength of the conclusions of this review.

Data completeness in included studies

There were enough data to properly address overall survival (OS) and three of the important outcomes for people aged ≥ 65 years old: disease-free survival (DFS), event-free survival (EFS) and major pathological response rate (MPR). However, the evidence was of low certainty for treatment-related adverse events (TRAEs) and pathological complete response rate (pCR). We were unable to retrieve health-related quality of life data, or to ascertain the outcomes according to age subgroups (other than in people aged ≥ 65 years) and according to the PD-L1 tumour expression (a key biomarker). The overall survival results are not statistically significant, perhaps because of the immaturity of the available data.

Sadly, authors often fail to provide detailed data across different age groups. All but two of the included studies failed to provide evidence on participants aged between 65 and 75 years and ≥ 75 years. The elderly as a group are deeply under-represented in clinical trials, despite a gradual increase in their proportion of the overall population over recent years [223]. Based on the current available data, we cannot draw conclusions about the benefits and harms of ICIs in people aged 65 to 75 and ≥ 75 years. Implications for clinical practice are therefore limited.

We could not evaluate the effect of adding ICI to platinum-based chemotherapy according to PD-L1 status, because of the lack of data within our particular age group. Moreover, despite the inclusion of elderly people in these studies, none reported geriatric assessment, and no data on health-related quality of life were available.

Enrolment inequities

Two studies enrolled participants from China only, and another from Spain only. Most studies were international, with a vast majority of participants being from Europe, North America and Asia. The rest of the world (including South America and Africa) represented only about 10% of the enrolment. Therefore, our findings should be taken with caution in these under-represented populations.

Furthermore, the included studies recruited only fit people, i.e. OMS performance status of 0 to 1. The potential advantages of ICIs remain an open question for people in poorer general physical condition treated for resectable NSCLC.

Molecular alterations

Where investigated, participants with actionable molecular alterations (notably EGFR mutations and ALK translocations) were excluded from the studies, except for IMPOWER-010, KEYNOTE-671 and PEARLS/ KEYNOTE-091. Such molecular alterations were found in less than 10% of these studies' participants. Consequently, the results presented here cannot be extrapolated to people presenting these molecular alterations.

The IASLC (International Association for the Study of Lung Cancer) recently published recommendations from an international expert panel, emphasising the necessity of biomarker testing in resectable NSCLC, an early multidisciplinary approach and the preference for adjuvant targeted therapy for people with actionable molecular alterations [224]. Adjuvant osimertinib and alectinib are approved by the FDA and the EMA for stages IB-IIIa resected NSCLC with a sensitising EGFR or ALK alteration [225, 226, 227, 228].

Certainty of the evidence

All 11 included studies were RCTs.

We judged that two studies raised 'some concerns' due to deviations from intended interventions (CHECKMATE-816; IMPOWER-010). There were slight imbalances in withdrawal between arms in these two open-label trials (11 participants (6%) in the control arm versus none in the intervention arm of CHECKMATE-816; 20 participants (4.0%) in the control arm versus 33 participants (6.5%) in the intervention arm of IMPOWER-010).

IMPOWER-010 raised 'some concerns' in two other domains: measurement of the outcome for DFS, because the outcome was assessed by the investigators who were not blinded to the interventions; and selection of the reported results due to a protocol amendment to the primary analysis population from the ITT population to the subpopulation of participants with stages II-IIIa NSCLC with PD-L1 $\geq$ 1%.

We deemed TD-FOREKNOW to raise 'some concerns' for outcome measurement because, in this open-label study, pCR and MPR were assessed by local pathologists, who may have been influenced by knowledge of the treatment allocation.

We did not identify any risk of bias due to the randomisation process or from missing outcome data.

Using the GRADE approach, our assessment of the certainty of the evidence ranged from moderate to low (Summary of findings 1). We downgraded the certainty of the evidence for almost all outcomes (except TRAEs) due to risk of bias. We downgraded pCR and TRAEs for imprecision because of the wide confidence intervals. We further downgraded TRAEs for indirectness, since only one study provided data.

In addition, all included studies were, at least partially, funded by pharmaceutical companies, which, in some cases, intervened in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication. We thus identified potential conflicts of interest for authors in these studies.

Limitations of the review processes

We applied predefined eligibility criteria for study selection, and we followed our prespecified review methodology in performing risk of bias assessment and data analysis. We used a sensitive search strategy, with no restrictions on language and publication status. We think it is unlikely that we overlooked relevant eligible studies. Only RCTs were considered for inclusion.

We obtained relevant data on survival and pathological outcomes from most studies. We were unable to generate a funnel plot to explore publication bias, so this cannot be ruled out. The results of 12 ongoing studies could potentially affect the conclusions of this review once they are published. Two review authors independently and thoroughly analysed the risk of bias in the included studies using the Cochrane RoB 2 tool. We do not know if there are other unpublished trial reports. We prespecified all outcomes and subgroups in the review before completing the analysis.

Agreements and disagreements with other studies or reviews

To the best of our knowledge, this is the first meta-analysis that compares the effects of ICI monotherapy or combination with platinum-based chemotherapy versus platinum-based chemotherapy alone, specifically for people aged 65 years and older with resectable NSCLC. However, we describe below other recent meta-analyses whose results bear on and further illuminate the results of this review.

Banna and colleagues analysed data from eight RCTs (involving 3387 participants) that evaluated the addition of neoadjuvant or perioperative ICIs to neoadjuvant chemotherapy for stages IB-IIIb, including initially unresectable NSCLC for one study [229]. They showed similar effects in participants over 65 years old (HR 0.62, 95% CI 0.52 to 0.75) and in younger participants (HR 0.54, 95% CI 0.46 to 0.64) for two-year EFS. Results for pCR also showed similar results for participants over 65 years old (RR 6.76, 95% CI 3.57 to 12.79) as for younger participants (RR 10.68, 95% CI 2.06 to 55.35). No other endpoints were analysed.

Nuccio and colleagues included eight RCTs in their systematic review and meta-analysis, aiming to identify clinicopathological factors and to perform an indirect comparison between treatment strategies [39]. They found no significant differences in ICIs' effects between participants over or below 65 years old with, respectively: HR of 0.65 (95% CI 0.53 to 0.81) and HR of 0.58 (95% CI 0.47 to 0.72) for EFS; HR of 0.78 (95% CI 0.62 to 0.99) and HR of 0.71 (95% CI 0.57 to 0.88) for DFS; and RR of 3.81 (95% CI 1.89 to 7.71) versus RR of 10.35 (95% CI 0.95 to 112.45, with high heterogeneity, $I^2 = 67\%$) for pCR. Their data were mostly based on studies included in our meta-analysis (AEGEAN; CHECKMATE-816; IMPOWER-010; KEYNOTE-671; NADIM-II; PEARLS/ KEYNOTE-091).

Sorin and colleagues focused on the neoadjuvant setting with eight RCTs (involving a total of 3387 people) [230]. They retrieved similar results, in favour of a maintained benefit of EFS by the addition of neoadjuvant ICIs to chemotherapy across age categories with a cut-off at 65 years: HR of 0.52 (95% CI 0.41 to 0.66) for younger participants versus HR of 0.70 (95% CI 0.55 to 0.88) for participants $\geq$ 65 years old. This analysis was based solely on CHECKMATE-816 and KEYNOTE-671. They did not evaluate the other outcomes according to age.

Numerous systematic reviews and meta-analyses have been published in recent years, but they did not consider the impact of age on perioperative ICI outcomes.

Thus, our review is the only one to provide pooled data from so many trials (11 studies), in a homogeneous population (early-stage resectable NSCLC), with treatment arms that allow us to evaluate specifically the effect of intensification by the addition of ICIs to platinum-based chemotherapy treatment, in people aged ≥ 65 years old. There is a major lack of data specifically on the elderly population in the available literature.

AUTHORS' CONCLUSIONS

Implications for practice

In people aged 65 years and older with resectable non-small cell lung cancer, the addition of immune checkpoint inhibitors probably results in little to no improvement in overall survival, although the available data remain limited and largely immature. Clinical decision-making in older adults should be cautious because of important evidence gaps. Data on treatment-related adverse events are sparse, limiting the ability to fully assess the balance of benefits and harms; and no study assessed health-related quality of life.

However, the addition of immune checkpoint inhibitors in the neoadjuvant, adjuvant, or both, settings probably improves recurrence-free survival (disease-free survival and event-free survival) and major pathological response rates, and may also improve pathological complete response rates, compared to chemotherapy alone or best supportive care.

Given the heterogeneity of the elderly population, and their substantial under-representation in clinical trials, evidence remains insufficient for people aged 65 to 75 years and those aged ≥ 75 years. This limits the direct applicability of the available evidence to routine clinical practice, particularly for elderly people who are frail or have multiple comorbidities. Treatment effects observed in older adults need to be interpreted in the context of a comprehensive geriatric assessment, as chronological age alone does not capture clinical heterogeneity.

Evidence regarding dual-immunotherapy regimens and stratification by programmed death-ligand 1 (PD-L1) expression remains insufficient to support the routine use of combination immunotherapy in older adults, especially in those with multiple comorbidities. Additional data are needed to clarify whether PD-L1 status modifies treatment effects in older adults and to help identify individuals who are most likely to benefit from lower-intensity treatment approaches.

Implications for research

Given the limitations in extrapolating these findings, it seems essential to broaden inclusion criteria to better reflect the real-world people treated in routine clinical practice and to report outcomes stratified by age group. Additionally, trials specifically designed for older adults, incorporating systematic geriatric assessments and capturing relevant outcomes such as health-related quality of life, could be useful to complement existing data and support evidence-based clinical decision-making.

The Geriatric Core Dataset (G-CODE), developed through international consensus, offers a standardised and time-efficient approach that enables consistent reporting and comparison across trials, and could improve understanding of this heterogeneous population if included in future research [231]. A phase III clinical trial conducted in elderly people ≥ 70 years old, comparing standard management to treatment decisions guided by a comprehensive geriatric assessment for advanced non-small cell lung cancer, failed to demonstrate a significant benefit; however, it was associated with reduced toxicity and fewer treatment interruptions, providing supportive evidence for the integration of geriatric assessment tools into clinical trial design [232].

Finally, exploring the development of biomarkers to help identify individuals who may benefit more from specific treatment strategies, such as different drug classes (e.g. immune checkpoint inhibitors, chemotherapy, targeted therapies), treatment sequencing (perioperative, neoadjuvant, or adjuvant), or treatment intensity (monotherapy versus combinations), could contribute to improving personalised care. Greater research priority could be accorded to individuals aged 75 years and older, given their distinctive health needs, the limited available data, and the growing relevance of personalised care.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD016104](https://doi.org/10.1002/14651858.CD016104).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Risk of bias

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Differences between protocol and review

Supplementary material 9 Appendix 1 : baseline characteristics of patients in AEGEAN study

Supplementary material 10 Appendix 2 : baseline characteristics of patients in CTG BR.31 study

Supplementary material 11 Appendix 3 : baseline characteristics of patients in CHECKMATE-77T study

Supplementary material 12 Appendix 4 : baseline characteristics of patients in CHECKMATE-816 study

Supplementary material 13 Appendix 5 : baseline characteristics of patients in IMPOWER-010 study

Supplementary material 14 Appendix 6 : baseline characteristics of patients in KEYNOTE-671 study

Supplementary material 15 Appendix 7 : baseline characteristics of patients in NADIM-II study

Supplementary material 16 Appendix 8 : baseline characteristics of the patients in NEOTORCH study

Supplementary material 17 Appendix 9 : baseline characteristics of patients in PEARLS/ KEYNOTE-091 study

Supplementary material 18 Appendix 10 : baseline characteristics of patients in RATIONALE-315 study

Supplementary material 19 Appendix 11 : baseline characteristics of patients in TD-FOREKNOW study

ADDITIONAL INFORMATION

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Contributions of authors

Performed previous work that was the foundation of the review: CM, EO, RM, VW

Designed the search strategies: EP, FC

Selected studies: EO, EP

Extracted data: EO, EP

Conducted analyses: EO, EP, RM

Assessed risk of bias: EO, EP

Assessed certainty of evidence: EO, EP, RM

Wrote the draft review: EP

Critical revision: CM, EO, RM

Critical appraisal of the final review: VW

Declarations of interest

Eva Plissonneau is a Medical Oncology Resident at Besançon University Hospital. She has been conducting research (internship) at INSERM UMR 1098 /EFS (on lymphocyte exhaustion in pancreatic adenocarcinomas), funded by public sources and with no direct connection to this review. She has received support to attend courses and conferences, exclusively from academic societies and non-profit foundations: AERIO (Association pour l'Enseignement et la Recherche des Internes en Oncologie), AGECC (Agir Ensemble Contre le Cancer), ESMO (European Society for Medical Oncology) and SFC (Société Française du Cancer).

She has declared that she has no conflict of interest.

Corynne Marchal is the Managing Editor for the Cochrane Lung Cancer Group and was not involved in the editorial process for this review. She has declared that she has no conflict of interest.

Reem Malouf has declared that she has no conflict of interest.

Francois Calais is the Information Specialist for the Cochrane Lung Cancer Group and was not involved in the editorial process for this review. He has declared that he has no conflict of interest.

Virginie Westeel works as a Pulmonologist at University hospital, Besançon, France, and is the Co-ordinating Editor for the Cochrane Lung Cancer Group; she was not involved in the editorial process for this review. VW reports 'relationship/activity' leading to personal payments with AMGEN (speaker bureau, scientific committee, travel expenses and funding for research); AstraZeneca (scientific advisory board, scientific committee, speaker bureau, travel expenses), Bristol-Myers-Squibb (funding for research), Daiichi Sankyo (data and safety monitoring and funding for research), Janssen pharmaceuticals (speaker bureau and travel expenses), Merck Sharp and Dohme (speaker bureau, scientific committee, travel expenses and funding for research), Merck-Serono (funding for research), Novartis (funding for research), Ose-Immunotherapeutics (funding for research), Pfizer (speaker bureau and travel expenses), PharmaMar (funding for research), Roche (scientific committee, speaker bureau and funding for research), and, Seagen (funding for research) and Regeneron (speaker's bureau and travel expenses).

Emeline Orillard works as a Medical Oncologist at CHU Besançon, France. She has declared that she has no conflict of interest.

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Registration and protocol

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Data, code and other materials

As part of the published Cochrane Review, the following is made available for download for users of the Cochrane Library.

- Full search strategies for each database ([Supplementary material 1](#))
- Full citations of each unique report for all studies included, ongoing, or excluded on the full text screen, in the final review ([Supplementary material 2](#); [Supplementary material 3](#); [Supplementary material 4](#))
- Study characteristics for all studies included (detailed format; [Supplementary material 2](#)) and ongoing (compact format) ([Supplementary material 4](#))
- Consensus risk of bias assessments (online and [Supplementary material 5](#))
- Analyses of all outcomes included in the [Summary of findings 1](#) ([Supplementary material 6](#))
- Data extracted from included studies (online and [Supplementary material 7](#))
- And baseline characteristics of patients in each included study ([Supplementary material 11](#); [Supplementary material 12](#); [Supplementary material 13](#); [Supplementary material 14](#); [Supplementary material 15](#); [Supplementary material 16](#); [Supplementary material 17](#); [Supplementary material 18](#); [Supplementary material 19](#); [Supplementary material 9](#); [Supplementary material 10](#))

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REFERENCES

1. Bray F, Laversanne M, Sung H, Ferlay J, Siegel RL, Soerjomataram I, et al. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA : A Cancer Journal for Clinicians* 2024;**74**(3):229-63. [DOI: [10.3322/caac.21834](https://doi.org/10.3322/caac.21834)] [PMID: 38572751]
2. Barta JA, Powell CA, Wisnivesky JP. Global epidemiology of lung cancer. *Annals of Global Health* 2019;**85**(1):8. [DOI: [10.5334/aogh.2419](https://doi.org/10.5334/aogh.2419)] [PMID: 30741509]
3. Cancer stat facts — lung and bronchus cancer. <https://seer.cancer.gov/statfacts/html/lungb.html> (accessed 26 April 2024).
4. Chevallier M, Borgeaud M, Addeo A, Friedlaender A. Oncogenic driver mutations in non-small cell lung cancer: past, present and future. *World Journal of Clinical Oncology* 2021;**12**(4):217-37. [DOI: [10.5306/wjco.v12.i4.217](https://doi.org/10.5306/wjco.v12.i4.217)] [PMID: 33959476]
5. Hendriks LE, Kerr KM, Menis J, Mok TS, Nestle U, Passaro A, et al ; on behalf of the ESMO Guidelines Committee. Oncogene-addicted metastatic non-small-cell lung cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. *Annals of Oncology* 2023;**34**(4):339-57. [DOI: [10.1016/j.annonc.2022.12.013](https://doi.org/10.1016/j.annonc.2022.12.013)] [PMID: 36872130]
6. NCCN. NCCN clinical practice guidelines in oncology, non-small cell lung cancer, version 8.2024. https://www.nccn.org/professionals/physician_gls/pdf/nscl.pdf (accessed 23 August 2024).
7. Patel SP, Kurzrock R. PD-L1 Expression as a predictive biomarker in cancer immunotherapy. *Molecular Cancer Therapeutics* 2015;**14**(4):847-56. [DOI: [10.1158/1535-7163.MCT-14-0983](https://doi.org/10.1158/1535-7163.MCT-14-0983)] [PMID: 25695955]
8. Bodor JN, Bomber Y, Borghaei H. Biomarkers for immune checkpoint inhibition in non-small cell lung cancer (NSCLC). *Cancer* 2020;**126**(2):260-70. [DOI: [10.1002/cncr.32468](https://doi.org/10.1002/cncr.32468)] [PMID: 31691957]
9. Kerr KM, Bibeau F, Thunnissen E, Botling J, Ryška A, Wolf J, et al. The evolving landscape of biomarker testing for non-small cell lung cancer in Europe. *Lung Cancer* 2021;**154**:161-75. [DOI: [10.1016/j.lungcan.2021.02.026](https://doi.org/10.1016/j.lungcan.2021.02.026)] [PMID: 33690091]
10. Yu Y, Zeng D, Ou Q, Liu S, Li A, Chen Y, et al. Association of survival and immune-related biomarkers with immunotherapy in patients with non-small cell lung cancer: a meta-analysis and individual patient-level analysis. *JAMA Network Open* 2019;**2**(7):e196879. [DOI: [10.1001/jamanetworkopen.2019.6879](https://doi.org/10.1001/jamanetworkopen.2019.6879)] [PMID: 31290993]
11. Remon J, Soria J-C, Peters S; on behalf of the ESMO Guidelines Committee. Early and locally advanced non-small-cell lung cancer: an update of the ESMO Clinical Practice Guidelines focusing on diagnosis, staging and systemic and local therapy. *Annals of Oncology* 2021;**28**(4):iv1-21. [DOI: [10.1016/j.annonc.2021.08.1994](https://doi.org/10.1016/j.annonc.2021.08.1994)] [PMID: 34481037]
12. Goldstraw P, Chansky K, Crowley J, Rami-Porta R, Asamura H, Eberhardt WE, et al; International Association for the Study of Lung Cancer Staging and Prognostic Factors Committee, Advisory Boards, and Participating Institutions. The IASLC Lung Cancer Staging Project: proposals for revision of the TNM stage groupings in the forthcoming (eighth) edition of the TNM classification for lung cancer. *Journal of Thoracic Oncology* 2016;**11**(1):39-51. [DOI: [10.1016/j.jtho.2015.09.009](https://doi.org/10.1016/j.jtho.2015.09.009)] [PMID: 26762738]
13. Cortés AA, Calera Urquiza L, Cubero JH. Adjuvant chemotherapy in non-small cell lung cancer: state-of-the-art. *Translational Lung Cancer Research* 2015;**4**(2):191-7. [DOI: [10.3978/j.issn.2218-6751.2014.06.01](https://doi.org/10.3978/j.issn.2218-6751.2014.06.01)] [PMID: 25870801]
14. Burdett S, Rydzewska L, Tierney J, Auperin A, Le Pechoux C, Le Chevalier T, et al. Preoperative chemotherapy for non-small-cell lung cancer: a systematic review and meta-analysis of individual participant data. *Lancet* 2014;**383**(9928):1561-71. [DOI: [10.1016/S0140-6736\(13\)62159-5](https://doi.org/10.1016/S0140-6736(13)62159-5)] [PMID: 24576776]
15. Dunn GP, Bruce AT, Ikeda H, Old LJ, Schreiber RD. Cancer immunoediting: from immunosurveillance to tumor escape. *Nature. Immunology* 2002;**3**(11):991-8. [DOI: [10.1038/ni1102-991](https://doi.org/10.1038/ni1102-991)] [PMID: 12407406]
16. Dunn GP, Old LJ, Schreiber RD. The three Es of cancer immunoediting. *Annual Review of Immunology* 2004;**22**:329-60. [DOI: [10.1146/annurev.immunol.22.012703.104803](https://doi.org/10.1146/annurev.immunol.22.012703.104803)] [PMID: 15032581]
17. Markham A, Keam SJ. Camrelizumab: first global approval. *Drugs* 2019;**79**(12):1355-61. [DOI: [10.1007/s40265-019-01167-0](https://doi.org/10.1007/s40265-019-01167-0)] [PMID: 31313098]
18. Chen DS, Mellman I. Oncology meets immunology: the cancer-immunity cycle. *Immunity* 2013;**39**(1):1-10. [DOI: [10.1016/j.immuni.2013.07.012](https://doi.org/10.1016/j.immuni.2013.07.012)] [PMID: 23890059]
19. Xia A, Zhang Y, Xu J, Yin T, Lu X-J. T cell dysfunction in cancer immunity and immunotherapy. *Frontiers in Immunology* 2019;**10**:1719. [DOI: [10.3389/fimmu.2019.01719](https://doi.org/10.3389/fimmu.2019.01719)] [PMID: 31379886]
20. Wrona MV, Ghosh R, Coll K, Chun C, Yousefzadeh MJ. The 3 I's of immunity and aging: immunosenescence, inflammation, and immune resilience. *Frontiers in Aging* 2024;**5**:1490302. [DOI: [10.3389/fragi.2024.1490302](https://doi.org/10.3389/fragi.2024.1490302)] [PMID: 39478807]
21. National Institute of Diabetes and Digestive and Kidney Diseases. LiverTox: Clinical and Research Information on Drug-Induced Liver Injury. Bethesda (MD): NIDDK, 2022. [BOOKSHELF ID: NBK548206] [PMID: 31643531]
22. Kwok G, Yau TCC, Chiu JW, Tse E, Kwong Y-L. Pembrolizumab (Keytruda). *Human Vaccines and Immunotherapeutics* 2016;**12**(11):2777-89. [DOI: [10.1080/21645515.2016.1199310](https://doi.org/10.1080/21645515.2016.1199310)] [PMID: 27398650]
23. Zhang L, Hao B, Geng Z, Geng Q. Toripalimab: the first domestic anti-tumor PD-1 antibody in China.

Frontiers in Immunology 2022;**12**:730666. [DOI: [10.3389/fimmu.2021.730666](https://doi.org/10.3389/fimmu.2021.730666)] [PMID: 35095833]

24. Cha E, Wallin J, Kowanzetz M. PD-L1 inhibition with MPDL3280A for solid tumors. *Seminars in Oncology* 2015;**42**(3):484-7. [DOI: [10.1053/j.seminoncol.2015.02.002](https://doi.org/10.1053/j.seminoncol.2015.02.002)] [PMID: 25965367]

25. Stewart R, Morrow M, Hammond SA, Mulgrew K, Marcus D, Poon E, et al. Identification and characterization of MEDI4736, an antagonistic anti-PD-L1 monoclonal antibody. *Cancer Immunology Research* 2015;**3**(9):1052-62. [DOI: [10.1158/2326-6066.CIR-14-0191](https://doi.org/10.1158/2326-6066.CIR-14-0191)] [PMID: 25943534]

26. López-Otín C, Blasco MA, Partridge L, Serrano M, Kroemer G. The hallmarks of aging. *Cell* 2013;**153**(6):1194-217. [DOI: [10.1016/j.cell.2013.05.039](https://doi.org/10.1016/j.cell.2013.05.039)] [PMID: 23746838]

27. Aiello A, Farzaneh F, Candore G, Caruso C, Davinelli S, Gambino CM, et al. Immunosenescence and its hallmarks: How to oppose aging strategically? A review of potential options for therapeutic intervention. *Frontiers in Immunology* 2019;**10**:2247. [DOI: [10.3389/fimmu.2019.02247](https://doi.org/10.3389/fimmu.2019.02247)] [PMID: 31608061]

28. Huff WX, Kwon JH, Henriquez M, Fetcko K, Dey M. The evolving role of CD8⁺CD28⁻ immunosenescent T cells in cancer immunology. *International Journal of Molecular Science* 2019;**20**(11):2810. [DOI: [10.3390/ijms20112810](https://doi.org/10.3390/ijms20112810)] [PMID: 31181772]

29. Ferrara R, Naigeon M, Auclin E, Duchemann B, Cassard L, Jouniaux JM, et al. Circulating T-cell immunosenescence in patients with advanced non-small cell lung cancer treated with single-agent PD-1/PD-L1 inhibitors or platinum-based chemotherapy. *Clinical Cancer Research* 2021;**27**(2):492-503. [DOI: [10.1158/1078-0432.CCR-20-1420](https://doi.org/10.1158/1078-0432.CCR-20-1420)] [PMID: 32887723]

30. Dietel M, Bubendorf L, Dingemans AM, Doooms C, Elmberger G, García RC, et al. Diagnostic procedures for non-small-cell lung cancer (NSCLC): recommendations of the European Expert Group. *Thorax* 2016;**71**(2):177-84. [DOI: [10.1136/thoraxjnl-2014-206677](https://doi.org/10.1136/thoraxjnl-2014-206677)] [PMID: 26530085]

31. Zer A, Ahn MJ, Barlesi F, Bubendorf L, De Ruysscher D, Garrido P, et al. Early and locally advanced non-small-cell lung cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. *Annals of Oncology* 2025;**36**(11):1245-62. [DOI: [10.1016/j.annonc.2025.08.003](https://doi.org/10.1016/j.annonc.2025.08.003)] [PMID: 40885528]

32. Mountzios G, Remon J, Hendriks LE, García-Campelo R, Rolfo C, et al. Immune-checkpoint inhibition for resectable non-small-cell lung cancer - opportunities and challenges. *Nature Reviews. Clinical Oncology* 2023;**20**(10):664-77. [DOI: [10.1038/s41571-023-00794-7](https://doi.org/10.1038/s41571-023-00794-7)] [PMID: 37488229]

33. Pauken KE, Dougan M, Rose NR, Lichtman AH, Sharpe AH. Adverse events following cancer immunotherapy: obstacles and opportunities. *Trends in Immunology* 2019;**40**(6):511-23. [DOI: [10.1016/j.it.2019.04.002](https://doi.org/10.1016/j.it.2019.04.002)] [PMID: 31053497]

34. Boutros C, Tarhini A, Routier E, Lambotte O, Leroy Ladurie F, Carbonnel F, et al. Safety profiles of anti-CTLA-4 and anti-PD-1 antibodies alone and in combination. *Nature Reviews. Clinical*

Oncology 2016;**13**(8):473-86. [DOI: [10.1038/nrclinonc.2016.58](https://doi.org/10.1038/nrclinonc.2016.58)] [PMID: 27141885]

35. Wang DY, Salem J-E, Cohen JV, Chandra S, Menzer C, Ye F, et al. Fatal toxic effects associated with immune checkpoint inhibitors: a systematic review and meta-analysis. *JAMA Oncology* 2018;**4**(12):1721-8. [DOI: [10.1001/jamaoncol.2018.3923](https://doi.org/10.1001/jamaoncol.2018.3923)] [PMID: 30242316]

36. Xu C, Chen Y-P, Du X-J, Liu J-Q, Huang C-L, Chen L, et al. Comparative safety of immune checkpoint inhibitors in cancer: systematic review and network meta-analysis. *BMJ* 2018;**363**(8175):4226. [DOI: [10.1136/bmj.k4226](https://doi.org/10.1136/bmj.k4226)] [PMID: 30409774]

37. Martins F, Sofiya L, Sykietis GP, Lamine F, Maillard M, Fraga M, et al. Adverse effects of immune-checkpoint inhibitors: epidemiology, management and surveillance. *Nature Reviews. Clinical Oncology* 2019;**16**(9):563-80. [DOI: [10.1038/s41571-019-0218-0](https://doi.org/10.1038/s41571-019-0218-0)] [PMID: 31092901]

38. Postow MA, Sidlow R, Hellmann MD. Immune-related adverse events associated with immune checkpoint blockade. *New England Journal of Medicine* 2018;**378**(2):158-68. [DOI: [10.1056/NEJMra1703481](https://doi.org/10.1056/NEJMra1703481)] [PMID: 29320654]

39. Nuccio A, Viscardi G, Salomone F, Servetto A, Venanzi FM, Riva ST, et al. Systematic review and meta-analysis of immune checkpoint inhibitors as single agent or in combination with chemotherapy in early-stage non-small cell lung cancer: impact of clinicopathological factors and indirect comparison between treatment strategies. *European Journal of Cancer* 2023;**195**:113404. [DOI: [10.1016/j.ejca.2023.113404](https://doi.org/10.1016/j.ejca.2023.113404)] [PMID: 37948842]

40. Patel SP, Othus M, Chen Y, Wright GP Jr, Yost KJ, Hyngstrom JR, et al. Neoadjuvant-adjuvant or adjuvant-only pembrolizumab in advanced melanoma. *New England Journal of Medicine* 2023;**388**(9):813-23. [DOI: [10.1056/NEJMoa2211437](https://doi.org/10.1056/NEJMoa2211437)] [PMID: 36856617]

41. Blank CU, Lucas MW, Scolyer RA, van de Wiel BA, Menzies AM, Lopez-Yurda M, et al. Neoadjuvant nivolumab and ipilimumab in resectable stage III melanoma. *New England Journal of Medicine* 2024;**391**(18):1696-708. [DOI: [10.1056/NEJMoa2402604](https://doi.org/10.1056/NEJMoa2402604)] [PMID: 38828984]

42. Cascone T, Hamdi H, Zhang F, Poteete A, Li L, Hudgens CW, et al. Abstract 1719 Superior efficacy of neoadjuvant compared to adjuvant immune checkpoint blockade in non-small cell lung cancer. *Cancer Research* 2018;**Suppl 13**:1719. [DOI: [10.1158/1538-7445.AM2018-1719](https://doi.org/10.1158/1538-7445.AM2018-1719)]

43. Biswal N. 613TiP Evaluation of low dose nivolumab with neoadjuvant chemotherapy in non-small cell lung cancer: a phase II open label randomised clinical trial (ELON). *Annals of Oncology* 2024;**35**, **Suppl 4**:S1622. [DOI: [10.1016/j.annonc.2024.10.640](https://doi.org/10.1016/j.annonc.2024.10.640)]

44. CTRI202302050126. Evaluation of low dose immunotherapy for lung cancer [Evaluation of low dose nivolumab with neoadjuvant chemotherapy in non small cell lung cancer]. <https://ctri.nic.in/Clinicaltrials/pmaindet2.php?>

EncHid=ODExMTE=&Enc=&userName= (first posted 27 February 2023).

45. Sacher AG, Le LW, Leighl NB, Coate LE. Elderly patients with advanced NSCLC in phase III clinical trials: are the elderly excluded from practice-changing trials in advanced NSCLC? *Journal of Thoracic Oncology* 2013;**8**(3):366-8. [DOI: [10.1097/JTO.0b013e31827e2145](https://doi.org/10.1097/JTO.0b013e31827e2145)] [PMID: 23407560]

46. Mohile SG, Mohamed MR, Xu H, Culakova E, Poh Loh K, Magnuson A, et al. Evaluation of geriatric assessment and management on the toxic effects of cancer treatment (GAP70+): a cluster-randomised study. *Lancet* 2021;**398**(10314):1894-904. [DOI: [10.1016/S0140-6736\(21\)01789-X](https://doi.org/10.1016/S0140-6736(21)01789-X)] [PMID: 34741815]

47. Pallis AG, Gridelli C, Wedding U, Faivre-Finn C, Veronesi G, Jaklitsch M, et al. Management of elderly patients with NSCLC; updated expert's opinion paper: EORTC Elderly Task Force, Lung Cancer Group and International Society for Geriatric Oncology. *Annals of Oncology* 2014;**25**(7):1270-83. [DOI: [10.1093/annonc/mdl022](https://doi.org/10.1093/annonc/mdl022)] [PMID: 24638905]

48. Gomes F, Wong M, Matteo N, Battisti L, Kordbacheh T, Kiderlen M, et al. Immunotherapy in older patients with non-small cell lung cancer: Young International Society of Geriatric Oncology position paper. *British Journal of Cancer* 2020;**123**(6):874-84. [PMID: 10.1038/s41416-020-0986-4] [PMID: 32694695]

49. Marchal C, Orillard E, Calais F, Westeel V. Immunotherapy for non-small cell lung cancer in the elderly population: a generic protocol. *Cochrane Database of Systematic Reviews* 2024, Issue 7. Art. No: CD014907. [CENTRAL: CD014907] [DOI: [10.1002/14651858.CD014907.pub2](https://doi.org/10.1002/14651858.CD014907.pub2)]

50. Orillard E, Adhikari A, Malouf R, Calais F, Marchal C, Westeel V. Immune checkpoint inhibitors plus platinum-based chemotherapy compared to platinum-based chemotherapy with or without bevacizumab for first-line treatment of older people with advanced non-small cell lung cancer. *Cochrane Database of Systematic Reviews* 2024, Issue 8. Art. No: CD015495. [DOI: [10.1002/14651858.CD015495](https://doi.org/10.1002/14651858.CD015495)] [PMID: 39136258]

51. Mascaux C, Greillier L, Moreau L, Egenod T, Toffart A-C, Cadranell J, et al. First-line atezolizumab plus chemotherapy in elderly patients with advanced nscl, IFCT-1805 elderly: a randomized, multicenter, phase 3 trial. In: 2025 World Conference on Lung Cancer; 2025 Sep 6-9; Barcelona, Spain. 2025. [ORAL ABSTRACT: OA05.02]

52. Shenoy P, Harugeri A. Elderly patients' participation in clinical trials. *Perspectives in Clinical Research* 2015;**6**(4):184-9. [DOI: [10.4103/2229-3485.167099](https://doi.org/10.4103/2229-3485.167099)] [PMID: 26623388]

53. Higgins JP, Lasserson T, Thomas J, Flemyng E, Churchill R. Methodological expectations of Cochrane intervention reviews. Cochrane: London, Version August 2023. Available from <https://community.cochrane.org/mecir-manual>.

54. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71. [DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71)]

55. McKenzie JE, Brennan SE, Ryan RE, Thomson HJ, Johnston RV, Thomas J. Chapter 3: Defining the criteria for including studies and how they will be grouped for the synthesis (chapter last updated August 2023). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://cochrane.org/handbook>.

56. National Cancer Institute. NCI Dictionary of Cancer Terms. <https://www.cancer.gov/publications/dictionaries/cancer-terms> (accessed 4 June 2026).

57. Kluetz PG, Chingos DT, Basch EM, Mitchell SA. Patient-reported outcomes in cancer clinical trials: measuring symptomatic adverse events with the National Cancer Institute's patient-reported outcomes version of the Common Terminology Criteria for Adverse Events (PRO-CTCAE). In: American Society of Clinical Oncology Educational Book. Vol. 36. Alexandria, VA: American Society of Clinical Oncology (ASCO), 2016:67-73. [DOI: [10.1200/EDBK_159514](https://doi.org/10.1200/EDBK_159514)] [PMID: 27249687]

58. Forde PM, Spicer J, Lu S, Provencio M, Mitsudomi T, Awad MM, et al. Neoadjuvant nivolumab plus chemotherapy in resectable lung cancer. *New England Journal of Medicine* 2022;**386**(21):1973-85. [DOI: [10.1056/NEJMoa2202170](https://doi.org/10.1056/NEJMoa2202170)]

59. Heymach JV, Harpole D, Mitsudomi T, Taube JM, Galffy G, Hochmair M, et al. Perioperative durvalumab for resectable non-small-cell lung cancer. *New England Journal of Medicine* 2023;**389**(18):1672-84. [DOI: [10.1056/NEJMoa2304875](https://doi.org/10.1056/NEJMoa2304875)]

60. Provencio M, Nadal E, González-Larriba JL, Martínez-Martí A, Bernabé R, Bosch-Barrera J, et al. Perioperative nivolumab and chemotherapy in stage III non-small-cell lung cancer. *New England Journal of Medicine* 2023;**389**(6):504-13. [DOI: [10.1056/NEJMoa2215530](https://doi.org/10.1056/NEJMoa2215530)]

61. Cascone T, Awad MM, Spicer JD, He J, Lu S, Sepesi B, et al. Perioperative nivolumab in resectable lung cancer. *New England Journal of Medicine* 2024;**390**:1756-69. [DOI: [10.1056/NEJMoa2311926](https://doi.org/10.1056/NEJMoa2311926)] [PMID: 38749033]

62. Wakelee H, Liberman M, Kato T, Tsuboi M, Lee SH, Gao S, et al. Perioperative pembrolizumab for early-stage non-small-cell lung cancer. *New England Journal of Medicine* 2023;**389**(6):491-503. [DOI: [10.1056/NEJMoa2302983](https://doi.org/10.1056/NEJMoa2302983)]

63. Yue D, Wang W, Liu H, Chen Q, Chen C, Zhang J, et al. LBA58 Pathological response to neoadjuvant tislelizumab (TIS) plus platinum-doublet (PtDb) chemotherapy (CT) in resectable stage II-IIIa NSCLC patients (pts) in the phase III (Ph3) RATIONALE-315 trial. *Annals of Oncology* 2023;**34**(Suppl 2):S1299. [DOI: [10.1016/j.annonc.2023.10.054](https://doi.org/10.1016/j.annonc.2023.10.054)]

64. Lefebvre C, Glanville J, Briscoe S, Littlewood A, Marshall C, Metzendorf MI, et al. Chapter 4: Searching for and selecting studies. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.2 (updated February 2021). Cochrane, 2021. Available from <https://cochrane.org/handbook>.

65. Covidence. Version accessed 3 September 2024. Melbourne, Australia: Veritas Health Innovation, 2020. Available at <https://www.covidence.org>.
66. Review Manager (RevMan). Version 8.1.1. The Cochrane Collaboration, 2024. Available at <https://revman.cochrane.org>.
67. Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized trial. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from <https://cochrane.org/handbook>.
68. Parmar MK, Torri V, Stewart L. Extracting summary statistics to perform meta-analyses of the published literature for survival endpoints. *Statistics in Medicine* 1998;**17**(24):2815-34.
69. Tierney JF, Stewart LA, Ghersi D, Burdett S, Sydes MR. Practical methods for incorporating summary time-to-event data into meta-analysis. *Trials* 2007;**7**(8):16.
70. Deeks JJ, Higgins JP, Altman DG. Chapter 10: Analysing data and undertaking meta-analyses. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from <https://cochrane.org/handbook>.
71. Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;**315**(7109):629-34.
72. Boutron I, Page MJ, Higgins JP, Altman DG, Lundh A, Hróbjartsson A. Chapter 7: Considering bias and conflicts of interest among the included studies. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from <https://cochrane.org/handbook>.
73. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Controlled Clinical Trials* 1986;**7**:177-88.
74. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine* 2002;**21**:1539-58.
75. GRADEpro GDT. Hamilton (ON): McMaster University (developed by Evidence Prime), accessed 3 September 2024. Available at <https://www.gradepr.org>.
76. Schünemann HJ, Higgins JP, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from <https://cochrane.org/handbook>.
77. Goss G, Darling GE, Westeel V, Nakagawa K, Massuti Sureda B, Perrone F, et al. LBA48 CTG BR.31: a global, double-blind placebo-controlled, randomized phase III study of adjuvant durvalumab in completely resected non-small cell lung cancer (NSCLC). *Annals of Oncology* 2024;**35**(2):S1238. [DOI: [10.1016/j.annonc.2024.08.2289](https://doi.org/10.1016/j.annonc.2024.08.2289)]
78. EUCTR 2014-004946-83. Study of adjuvant MEDI4736 in completely resected lung cancer patients [A phase III prospective double blind placebo controlled randomized study of adjuvant MEDI4736 in completely resected non-small cell lung cancer]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2014-004946-83/IT> (first posted 23 December 2014). [EUCTR: 2014-004946-83]
79. ACTRN12615000323527. A phase III prospective double blind placebo controlled randomized study of adjuvant MEDI4736 in completely resected non-small cell lung cancer [A phase III prospective double blind placebo controlled randomized study on the effect of adjuvant MEDI4736 on disease free survival in patients with completely resected non-small cell lung cancer]. <https://www.anzctr.org.au/Trial/Registration/TrialReview.aspx?id=368076&isReview=true> (first posted 9 April 2015). [ACTRN: 12615000323527]
80. Mc Lachlan S-A, Mersiades A, Wright G, Millward M, Goss G, Darling G, et al. BR31: a phase 3, double blind, placebo controlled, randomised trial of adjuvant durvalumab in completely resected non-small cell lung cancer. *Asia-Pacific Journal of Clinical Oncology* 2018;**14**(Suppl 7):198-9. [DOI: [10.1111/ajco.13089](https://doi.org/10.1111/ajco.13089)]
81. Westeel V, Goss G, Darling G, Nakagawa K, Sureda BM, Perrone F, et al. LBA3: sites of relapse and subsequent therapy in the BR.31 phase III study of durvalumab vs placebo in resected stage IB-IIIA NSCLC. *Journal of Thoracic Oncology* 2025;**20**(3):S98-9. [DOI: [10.1016/S1556-0864\(25\)00330-2](https://doi.org/10.1016/S1556-0864(25)00330-2)]
82. NCT02273375. Double blind placebo controlled study of adjuvant MEDI4736 in completely resected NSCLC [A phase III prospective double blind placebo controlled randomized study of adjuvant MEDI4736 in completely resected non-small cell lung cancer]. <https://clinicaltrials.gov/study/NCT02273375> (first posted 24 October 2014). [NCT: 02273375]
83. Provencio M, Serna R, Nadal E, Glez Larriba JL, Martinez-Marti A, Bernabe R, et al. PL03.12 Progression free survival and overall survival in NADIM II study. *Journal of Clinical Oncology* 2022;**17**(9):S2-3. [DOI: [10.1016/j.jtho.2022.07.014](https://doi.org/10.1016/j.jtho.2022.07.014)]
84. NCT03838159. NADIM II: neo-adjuvant immunotherapy (NADIMII) [A randomized phase ii study of neo-adjuvant chemo/immunotherapy versus chemotherapy alone for the treatment of locally advanced and potentially resectable non-small cell lung cancer (NSCLC) patients]. <https://clinicaltrials.gov/study/NCT03838159> (first posted 12 February 2019). [NCT: 03838159]
85. Provencio-Pulla M, Nadal E, Larriba JLG, Martinez-Marti A, Bernabe R, Bosch-Barrera J, et al. Nivolumab + chemotherapy versus chemotherapy as neoadjuvant treatment for resectable stage IIIA NSCLC: primary endpoint results of pathological complete response (pCR) from phase II NADIM II trial. *Journal of Clinical Oncology* 2022;**40**(Suppl 16):8501. [DOI: [10.1200/JCO.2022.40.16_suppl.8501](https://doi.org/10.1200/JCO.2022.40.16_suppl.8501)]
86. Serna-Blasco R, Nadal E, Gonzalez-Larriba JL, Martinez-Marti A, Caro RB, Bosch-Barrera J, et al. Prognostic value of minimal residual disease in the NADIM II trial. *Journal*

of *Clinical Oncology* 2025;**20**(3):S138-9. [DOI: [10.1016/S1556-0864\(25\)00404-6](https://doi.org/10.1016/S1556-0864(25)00404-6)]

87. EUCTR 2018-004515-45-ES. Study of chemotherapy and immunotherapy for patients with lung cancer bounded in the thorax and potentially resectable [A randomized phase II study of neo-adjuvant chemo/immunotherapy versus chemotherapy alone for the treatment of locally advanced and potentially resectable non-small cell lung cancer (NSCLC) patients]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2018-004515-45/ES> (first posted 11 April 2019). [EUCTR: 2018-004515-45]

88. Serna-Blasco R, Rodriguez Festa A, Sanz-Moreno S, De Lope LR, Medel PM, Nadal E, et al. Abstract 6684: Impact of fecal microbiome on the efficacy of neoadjuvant chemo-immunotherapy and colitis adverse events: findings from the NADIM II Trial. *Cancer Research* 2024;**84**(Suppl 6):6684. [DOI: [10.1158/1538-7445.AM2024-6684](https://doi.org/10.1158/1538-7445.AM2024-6684)]

89. Lu S, Zhang W, Wu L, Wang W, Zhang P, Fang W, et al. Perioperative toripalimab plus chemotherapy for patients with resectable non-small cell lung cancer: The Neotorch randomized clinical trial. *JAMA* 2024;**331**(3):201-11. [DOI: [10.1001/jama.2023.24735](https://doi.org/10.1001/jama.2023.24735)]

90. Wang W, Wu L, Zhang W, Lu S, Fang H, Yu G, et al. 366 A randomized double-blind placebo-controlled phase III study evaluating perioperative toripalimab combined with platinum-based doublet chemotherapy in resectable stage III NSCLC. *Journal for Immunotherapy of Cancer* 2020;**8**(Suppl 3):A223-4. [DOI: [10.1136/jitc-2020-SITC2020.0366](https://doi.org/10.1136/jitc-2020-SITC2020.0366)]

91. NCT04158440. Phase III study of toripalimab versus placebo plus chemotherapy in resectable NSCLC. <https://clinicaltrials.gov/study/NCT04158440> (first posted 7 November 2019).

92. Lu S, Wu L, Zhang W, Zhang P, Wang W, Fang W, et al. Perioperative toripalimab + platinum-doublet chemotherapy versus chemotherapy in resectable stage II/III non-small cell lung cancer (NSCLC): interim event-free survival (EFS) analysis of the phase III NEOTORCH study. In: *Journal of Clinical Oncology*. Vol. 41. 2023:8501. [DOI: [10.1200/jco.2023.41.16_suppl.8501](https://doi.org/10.1200/jco.2023.41.16_suppl.8501)]

93. Lei J, Zhao J, Gong L, Ni Y, Zhou Y, Tian F, et al. Neoadjuvant camrelizumab plus platinum-based chemotherapy versus chemotherapy alone for Chinese patients with resectable stage IIIA or IIIB (T3N2) non-small cell lung cancer: The TD-FOREKNOW randomized clinical trial. *JAMA Oncology* 2023;**9**(10):1348-55. [DOI: [10.1001/jamaoncol.2023.2751](https://doi.org/10.1001/jamaoncol.2023.2751)] [PMID: 37535377]

94. Lei J, Zhao J, Yan X, Gong L, Lei G, Jiang T. 560 A randomized, controlled, multicenter phase II trial of camrelizumab combined with albumin-bound paclitaxel and cisplatin as neoadjuvant treatment in resectable stage IIIA and IIIB (T3N2) non-small-cell lung cancer. *ESMO Immuno-Oncology and Technology* 2022;**16**(Suppl 1):100161. [DOI: [10.1016/j.iotech.2022.100161](https://doi.org/10.1016/j.iotech.2022.100161)]

95. NCT04338620. Camrelizumab combined with albumin-bound paclitaxel and platinum in neoadjuvant treatment of resectable NSCLC [An explorative study on camrelizumab

combined with albumin-bound paclitaxel and platinum in neoadjuvant treatment of resectable non-small-cell lung carcinoma]. <https://clinicaltrials.gov/study/NCT04338620> (first posted 8 April 2020). [NCT: 04338620]

96. Felip E, Altorki N, Zhou C, Csösz T, Vynnychenko I, Goloborodko O, et al. Adjuvant atezolizumab after adjuvant chemotherapy in resected stage IB-IIIA non-small-cell lung cancer (IMpower010): a randomised, multicentre, open-label, phase 3 trial. *Lancet* 2021;**398**(10308):1344-57. [DOI: [10.1016/S0140-6736\(21\)02098-5](https://doi.org/10.1016/S0140-6736(21)02098-5)]

97. CTIS 2023-505981-26-00. Study to assess safety and efficacy of atezolizumab (MPDL3280A) compared to best supportive care following chemotherapy in patients with lung cancer [A phase III, open-label, randomized study to investigate the efficacy and safety of atezolizumab (anti-PD-L1 antibody) compared with best supportive care following adjuvant cisplatin-based chemotherapy in patients with completely resected stage Ib-IIIA non-small cell lung cancer]. <https://euclinicaltrials.eu/search-for-clinical-trials/?lang=en&EUCT=2023-505981-26-00> 2023. [CTIS: 2023-505981-26-00]

98. Felip E, Altorki N, Zhou C, Vallières E, Martínez-Martí A, Rittmeyer A, et al. Overall survival with adjuvant atezolizumab after chemotherapy in resected stage II-IIIA non-small-cell lung cancer (IMpower010): a randomised, multicentre, open-label, phase III trial. *Annals of Oncology* 2023;**34**(10):907-19. [DOI: [10.1016/j.annonc.2023.07.001](https://doi.org/10.1016/j.annonc.2023.07.001)]

99. Lee JM, Vallières E, Ding B, Johnson A, Bhagwakar J, Rashidi S, et al. Safety of adjuvant atezolizumab after pneumonectomy/bilobectomy in stage II-IIIA non-small cell lung cancer in the randomized phase III IMpower010 trial. *Journal of Thoracic and Cardiovascular Surgery* 2023;**166**(3):655-66.e7. [DOI: [10.1016/j.jtcvs.2023.01.012](https://doi.org/10.1016/j.jtcvs.2023.01.012)]

100. Kenmotsu H, Zhou C, Altorki N, Felip E, Vallieres E, Sugawara S, et al. PS1-5 IMpower010: results from Asian patients in a phase 3 study of adjuvant atezolizumab in resected stage IB-IIIA NSCLC. *Annals of Oncology* 2022;**33**:S464. [DOI: [10.1016/j.annonc.2022.05.063](https://doi.org/10.1016/j.annonc.2022.05.063)]

101. Wakelee HA, Altorki NK, Zhou C, Csoszi T, Vynnychenko IO, Goloborodko O, et al. IMpower010: primary results of a phase III global study of atezolizumab versus best supportive care after adjuvant chemotherapy in resected stage IB-IIIA non-small cell lung cancer (NSCLC). *Journal of Clinical Oncology (ASCO Annual Meeting)* 2021;**39**(Suppl 15):8500. [DOI: [10.1200/JCO.2021.39.15-suppl.8500](https://doi.org/10.1200/JCO.2021.39.15-suppl.8500)]

102. Felip E, Altorki N, Zhou C, Vallieres E, Vynnychenko I, Akopov A, et al. 800 Atezolizumab (atezo) versus best supportive care (BSC) in stage II-IIIA NSCLC with high PD-L1 expression: sub-analysis from the pivotal phase III IMpower010 study. *Annals of Oncology* 2022;**45**:145-6. [DOI: [10.1159/000521004](https://doi.org/10.1159/000521004)]

103. Wakelee HA, Altorki NK, Vallieres E, Zhou C, Zuo Y, Howland M, et al. A phase III trial to compare atezolizumab (atezo) versus best supportive care (BSC) following adjuvant chemotherapy in patients (pts) with completely resected

NSCLC: iMpower010. *Journal of Clinical Oncology (ASCO Annual Meeting)* 2017;**35**(Suppl 15):TPS85.

104. Felip E, Srivastava M, Reck M, Wakelee H, Altorki N, Vallieres E, et al. MA11.08 IMpower010: Exploratory analysis of tumour mutational burden and disease-free survival with adjuvant atezolizumab in NSCLC. *Journal of Thoracic Oncology* 2023;**18**(11):S139. [DOI: [10.1016/j.jtho.2023.09.193](https://doi.org/10.1016/j.jtho.2023.09.193)]

105. Wakelee H, Altorki N, Vallieres E, Zhou C, Zuo Y, Howland M, et al. P3.02c-050 IMpower010: phase III study of atezolizumab versus best supportive care after adjuvant chemotherapy in patients with completely resected NSCLC. *Journal of Thoracic Oncology* 2017;**12**(1):S1305.

106. NCT02486718. Study to assess safety and efficacy of atezolizumab (MPDL3280A) compared to best supportive care following chemotherapy in patients with lung cancer [A phase III, open-label, randomized study to investigate the efficacy and safety of atezolizumab (anti-PD-L1 antibody) compared with best supportive care following adjuvant cisplatin-based chemotherapy in patients with completely resected stage IB-IIIa non-small cell lung cancer]. <https://clinicaltrials.gov/study/NCT02486718> (first posted 1 July 2015). [NCT: 02486718]

107. Zhou C, Das Thakur M, Srivastava MK, Zou W, Xu H, Ballinger M, et al. 20 IMpower010: biomarkers of disease-free survival (DFS) in a phase III study of atezolizumab (atezo) versus best supportive care (BSC) after adjuvant chemotherapy in stage IB-IIIa NSCLC. *Annals of Oncology* 2021;**32**:S1374. [DOI: [10.1016/j.annonc.2021.10.018](https://doi.org/10.1016/j.annonc.2021.10.018)]

108. Felip E, Wakelee H, Vallieres E, Zhou C, Zuo Y, Xia F, et al. P2.04-004 IMpower010: a phase III study of atezolizumab versus best supportive care following adjuvant chemotherapy in completely resected NSCLC. *Journal of Thoracic Oncology* 2017;**12**(11):S2398-9.

109. Wakelee HA, Altorki N, Zhou C, Csösz T, Vynnychenko IO, Goloborodko A, et al. LBA8035, Poster 297 IMpower-010: disease-free survival final analysis and second overall survival interim analysis results after >5 years of follow-up of a phase III study of adjuvant atezolizumab versus best supportive care in resected stage IB-IIIa non-small cell lung cancer. In: European Society for Medical Oncology (ESMO) 2024 Annual Meeting; 2024 Sep 13-17; Barcelona, Spain. 2024.

110. Altorki N, Felip E, Zhou C, Vallieres E, Moiseyenko V, Smolin A, et al. PL02.05 IMpower010: characterization of stage IB-IIIa NSCLC patients by type and extent of therapy prior to adjuvant atezolizumab. *Journal of Thoracic Oncology* 2021;**16**(10):S845-6. [DOI: [10.1016/j.jtho.2021.08.031](https://doi.org/10.1016/j.jtho.2021.08.031)]

111. Felip E, Altorki N, Zhou C, Vallieres E, Csoszi T, Vynnychenko IO, et al. Five-year survival outcomes with atezolizumab after chemotherapy in resected stage IB-IIIa non-small cell lung cancer (IMpower010): an open-label, randomized, phase III trial. *Journal of Clinical Oncology* 2025;**43**:21. [DOI: [10.1200/JCO-24-01681](https://doi.org/10.1200/JCO-24-01681)]

112. Wakelee H, Reck M, Felip E, Altorki NK, Vallieres E, Liersch R, et al. 1211P IMpower010: ctDNA status and 5y DFS follow up in patients (pts) with resected NSCLC who received adjuvant chemotherapy (chemo) followed by atezolizumab

(atezo) or best supportive care (BSC). *Annals of Oncology* 2024;**35**(2):S779-80. [DOI: [10.1016/j.annonc.2024.08.1270](https://doi.org/10.1016/j.annonc.2024.08.1270)]

113. Wang J, Fan Y, Fang J, He J, Liu Y, Tao M, et al. IMpower010: updated overall survival and safety results from Asian patients in a Phase III study of adjuvant atezolizumab vs best supportive care in resected stage IB-IIIa NSCLC. *American Association for Cancer Research* 2024;**84**(7):CT239. [DOI: [10.1158/1538-7445.AM2024-CT239](https://doi.org/10.1158/1538-7445.AM2024-CT239)]

114. Felip E, Vallieres E, Zhou C, Wakelee H, Bondarenko I, Sakai H, et al. LBA9 IMpower010: sites of relapse and subsequent therapy from a phase III study of atezolizumab versus best supportive care after adjuvant chemotherapy in stage IB-IIIa NSCLC. *Annals of Oncology* 2021;**32**:S1319. [DOI: [10.1016/j.annonc.2021.08.2120](https://doi.org/10.1016/j.annonc.2021.08.2120)]

115. Zhou C, Altorki N, Vallieres E, Felip E, Zuo Y, Howland M, et al. 429TiP IMpower010: a phase III trial investigating atezolizumab (atezo) versus best supportive care (BSC) after adjuvant chemotherapy (chemo) in patients (pts) with completely resected NSCLC. *Annals of Oncology* 2016;**27**:ix134-5. [DOI: [10.1093/annonc/mdw592.004](https://doi.org/10.1093/annonc/mdw592.004)]

116. Kenmotsu H, Sugawara S, Watanabe Y, Saito H, Okada M, Chen-Yoshikawa TF, et al. Adjuvant atezolizumab in Japanese patients with resected stage IB-IIIa non-small cell lung cancer (IMpower010). *Cancer Science* 2022;**113**(12):4327-38. [DOI: [10.1111/cas.15564](https://doi.org/10.1111/cas.15564)]

117. Wakelee H, Altorki N, Felip E, Vallieres E, Vynnychenko IO, Akopov A, et al. PL03.09 IMpower010: overall survival interim analysis of a phase III study of atezolizumab versus best supportive care in resected NSCLC. *Journal of Thoracic Oncology* 2022;**17**(9):S2. [DOI: [10.1016/j.jtho.2022.07.013](https://doi.org/10.1016/j.jtho.2022.07.013)]

118. Reck M, Srivastava MK, Wakelee HA, Felip E, Altorki NK, Csoszi T, et al. IMpower010: exploratory analysis of disease-free survival by KRAS status in patients with stage IB-IIIa NSCLC treated with adjuvant atezolizumab versus best supportive care. *Journal of Clinical Oncology* 2023;**41**(16):8522. [DOI: [10.1200/jco.2023.41.16_suppl.8522](https://doi.org/10.1200/jco.2023.41.16_suppl.8522)]

119. EUCTR 2014-003205-15-PT. A phase III, open-label, randomized study to investigate the efficacy and safety of atezolizumab (anti-PD-L1 antibody) compared with best supportive care following lung cancer resection and chemotherapy for early-stage lung cancer [A phase III, open-label, randomized study to investigate the efficacy and safety of atezolizumab (anti-PD-L1 antibody) compared with best supportive care following adjuvant cisplatin-based chemotherapy in patients with completely resected stage IB-IIIa non-small cell lung cancer]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2014-003205-15/PT> (first posted 9 October 2015). [EUCTR: 2014-003205-15]

120. Majem M, Garassino MC, Zhao G, Mennequier B, Grohe C, Ikeda N, et al. LBA3 Perioperative pembrolizumab (pembro) plus neoadjuvant chemotherapy (chemo) in early-stage non-small-cell lung cancer (NSCLC): a 4-year update of KEYNOTE-671. *ESMO Immuno-Oncology and Technology (ESMO Immuno-Oncology Congress)* 2024;**24**:101027. [DOI: [10.1016/j.iotech.2024.101027](https://doi.org/10.1016/j.iotech.2024.101027)]

- 121.** Wakelee HA, Liberman M, Kato T, Tsuboi M, Lee S-H, He J, et al. KEYNOTE-671: randomized, double-blind, phase 3 study of pembrolizumab or placebo plus platinum-based chemotherapy followed by resection and pembrolizumab or placebo for early stage NSCLC. *Journal of Clinical Oncology* 2023;**41**(Suppl 17):LBA100. [DOI: [10.1200/JCO.2023.41.17_suppl.LBA100](https://doi.org/10.1200/JCO.2023.41.17_suppl.LBA100)]
- 122.** Spicer JD, Garassino MC, Wakelee H, Liberman M, Kato T, Tsuboi M, et al; on behalf of the KEYNOTE-671 investigators. Neoadjuvant pembrolizumab plus chemotherapy followed by adjuvant pembrolizumab compared with neoadjuvant chemotherapy alone in patients with early-stage non-small-cell lung cancer (KEYNOTE-671): a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet* 2024;**404**(10459):1240-52. [DOI: [10.1016/S0140-6736\(24\)01756-2](https://doi.org/10.1016/S0140-6736(24)01756-2)] [PMID: 39288781]
- 123.** NCT03425643. Efficacy and safety of pembrolizumab (MK-3475) With platinum doublet chemotherapy as neoadjuvant/adjuvant therapy for participants with resectable stage II, IIIA, and resectable IIIB (T3-4N⁰) non-small cell lung cancer (MK-3475-671/KEYNOTE-671) [A phase III, randomized, double-blind trial of platinum doublet chemotherapy +/- pembrolizumab (MK-3475) as neoadjuvant/adjuvant therapy for participants with resectable stage II, IIIA, and resectable IIIB (T3-4N2) non-small cell lung cancer (NSCLC) (KEYNOTE-671)]. <https://clinicaltrials.gov/study/NCT03425643> (first posted 7 February 2018). [NCT: 03425643]
- 124.** Tsuboi M, Luft A, Ursol G, Kato T, Levchenko E, Eigendorff E, et al. 1235TiP Perioperative pembrolizumab + platinum-based chemotherapy for resectable locally advanced non-small cell lung cancer: the phase III KEYNOTE-671 study. *Annals of Oncology* 2020;**31**:S801-2. [DOI: [10.1016/j.annonc.2020.08.1437](https://doi.org/10.1016/j.annonc.2020.08.1437)]
- 125.** Garassino MC, Gao S, Liberman M, Spicer JD, Chen K, Vanakesa T, et al. 1210P Neoadjuvant pembrolizumab (pembro) or placebo (pbo) plus chemotherapy and adjuvant pembro or pbo for early stage NSCLC: subgroup analyses of the phase III KEYNOTE-671 study. *Annals of Oncology* 2024;**35**:S778-9. [DOI: [10.1016/j.annonc.2024.08.1269](https://doi.org/10.1016/j.annonc.2024.08.1269)]
- 126.** Garassino MC, Wakelee HA, Spicer J, Liberman M, Kato T, Tsuboi M, et al. Health-related quality of life (HRQoL) outcomes from the randomized, double-blind phase 3 KEYNOTE-671 study of perioperative pembrolizumab for early-stage non-small-cell lung cancer (NSCLC). *Journal of Clinical Oncology (ASCO Annual Meeting)* 2024;**42**(Suppl 16):8012. [DOI: [10.1200/JCO.2024.42.16_suppl.8012](https://doi.org/10.1200/JCO.2024.42.16_suppl.8012)]
- 127.** Jones DR, Wakelee H, Spicer JD, Liberman M, Kato T, Tsuboi M, et al. Association of pathologic regression with EFS in the KEYNOTE-671 study of perioperative pembrolizumab for early-stage NSCLC. *Journal of Thoracic Oncology* 2024;**19**(10):S8. [DOI: [10.1016/j.jtho.2024.09.022](https://doi.org/10.1016/j.jtho.2024.09.022)]
- 128.** Fernando HC, Yang J, Ferraro GL, Keller SM. Randomized, double-blind phase 3 study evaluating neoadjuvant platinum-based chemotherapy with perioperative pembrolizumab or placebo in resectable stage IIB or IIIA NSCLC: KEYNOTE-671. *Journal of Clinical Oncology* 2018;**36**(Suppl 15):TPS8583. [DOI: [10.1200/JCO.2018.36.15-suppl.TPS8583](https://doi.org/10.1200/JCO.2018.36.15-suppl.TPS8583)]
- 129.** Spicer JD, Gao S, Liberman M, Kato T, Tsuboi M, Lee S-H, et al. LBA56 Overall survival in the KEYNOTE-671 study of perioperative pembrolizumab for early-stage non-small-cell lung cancer (NSCLC). *Annals of Oncology* 2023;**34**:S1297-8. [DOI: [10.1016/j.annonc.2023.10.052](https://doi.org/10.1016/j.annonc.2023.10.052)]
- 130.** EUCTR 2017-001832-21-BE. Chemo +/- pembrolizumab (MK-3475) as perioperative therapy for resectable stage IIB or IIIA NSCLC [A Phase III, randomized, double-blind trial of platinum doublet chemotherapy +/- pembrolizumab (MK-3475) as neoadjuvant/adjuvant therapy for participants with resectable stage II, IIIA, and resectable IIIB (T3-4N2) non-small cell lung cancer (NSCLC) (KEYNOTE-671)]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2017-001832-21/BE> (first posted 19 December 2017). [EUCTR: 2017-001832-21]
- 131.** O'Brien M, Paz-Ares L, Marreaud S, Dafni U, Oselin K, Havel L, et al. Pembrolizumab versus placebo as adjuvant therapy for completely resected stage IB-IIIa non-small-cell lung cancer (PEARLS/KEYNOTE-091): an interim analysis of a randomised, triple-blind, phase 3 trial. *Lancet. Oncology* 2022;**23**(10):1274-86. [DOI: [10.1016/S1470-2045\(22\)00518-6](https://doi.org/10.1016/S1470-2045(22)00518-6)]
- 132.** O'Brien MER, Paz-Ares L, Jha N, Dafni U, Oselin K, Havel L, et al. EORTC-1416-LCG/ETOP 8-15 - PEARLS/KEYNOTE-091 study of pembrolizumab versus placebo for completely resected early-stage non-small cell lung cancer (NSCLC): outcomes in subgroups related to surgery, disease burden, and adjuvant chemotherapy use. *Journal of Clinical Oncology (ASCO annual Meeting)* 2022;**40**(Suppl 16):8512. [DOI: [10.1200/JCO.2022.40.16_suppl.8512](https://doi.org/10.1200/JCO.2022.40.16_suppl.8512)]
- 133.** Peters S, Besse B, Marreaud S, Dafni U, Oselin K, Havel L, et al. 930MO PD-L1 expression and outcomes of pembrolizumab and placebo in completely resected stage IB-IIIa NSCLC: subgroup analysis of PEARLS/KEYNOTE-091. *Annals of Oncology* 2022;**33**:S971-2. [DOI: [10.1016/j.annonc.2022.07.1056](https://doi.org/10.1016/j.annonc.2022.07.1056)]
- 134.** NCT02504372. Study of pembrolizumab (MK-3475) versus placebo for participants with non-small cell lung cancer after resection with or without standard adjuvant therapy (MK-3475-091/KEYNOTE-091) [A randomized, phase 3 trial with anti-PD-1 monoclonal antibody pembrolizumab (MK-3475) versus placebo for patients with early stage NSCLC after resection and completion of standard adjuvant therapy (PEARLS)]. <https://clinicaltrials.gov/study/NCT02504372> (first posted 21 July 2015).
- 135.** Oselin K, Shim BY, Okada M, Bryl M, Bonanno L, Demirag G, et al. Pembrolizumab versus placebo for early-stage non-small-cell lung cancer after resection and adjuvant therapy: subgroup analysis of patients who received adjuvant chemotherapy in the phase 3 PEARLS/KEYNOTE-091 study. *Journal of Clinical Oncology* 2023;**41**(16):8520. [DOI: [10.1200/jco.2023.41.16_suppl.8520](https://doi.org/10.1200/jco.2023.41.16_suppl.8520)]
- 136.** Paz-Ares L, O'Brien MER, Mauer M, Dafni U, Oselin K, Havel L, et al. VP3-2022: pembrolizumab (pembro) versus placebo for early-stage non-small cell lung cancer (NSCLC) following complete resection and adjuvant chemotherapy (chemo) when indicated: randomized, triple-blind, phase III EORTC-1416-LCG/ETOP 8-15 - PEARLS/KEYNOT. *Annals of Oncology* 2022;**33**(4):451-3. [DOI: [10.1016/j.annonc.2022.02.224](https://doi.org/10.1016/j.annonc.2022.02.224)]

- 137.** Besse B, Havel L, Peters S, Marreaud SI, Jha N, Oselin K, et al. 120MO: Adjuvant pembrolizumab versus placebo for early-stage NSCLC after resection and optional chemotherapy: updated results from PEARLS/KEYNOTE-091. *ESMO Immunology and Technology* 2023;**20** Suppl:100592. [DOI: [10.1016/j.iotech.2023.100592](https://doi.org/10.1016/j.iotech.2023.100592)]
- 138.** O'Brien MER, Paz-Ares L, Jha N, Dafni U, Oselin K, Havel L, et al. EORTC-ETOP randomized, phase 3 trial with anti-PD-1 monoclonal antibody pembrolizumab versus placebo for patients with early stage non-small cell lung cancer (NSCLC) after resection and standard adjuvant chemotherapy: PEARLS (NCT02504372). *Journal of Clinical Oncology (ASCO annual Meeting)* 2016;**34**(Suppl 15):TPS8571. [DOI: [10.1200/JCO.2016.34.15_suppl.TPS8571](https://doi.org/10.1200/JCO.2016.34.15_suppl.TPS8571)]
- 139.** Paz-Ares L, Hasan B, Dafni U, Menis J, De Maio E, Oselin K, et al. A randomized, phase 3 trial with anti-PD-1 monoclonal antibody pembrolizumab (MK-3475) versus placebo for patients with early stage NSCLC after resection and completion of standard adjuvant therapy (EORTC/ETOP 1416-PEARLS). *Annals of Oncology* 2017;**28**:iii23. [DOI: [10.1093/annonc/mdx085.013](https://doi.org/10.1093/annonc/mdx085.013)]
- 140.** EUCTR 2015-000575-27-SI. Immunotherapeutic treatment with pembrolizumab (antibody) in patients with early stage non-small cell lung cancer [A randomized, phase 3 trial with anti-PD-1 monoclonal antibody pembrolizumab (MK-3475) versus placebo for patients with early stage NSCLC after resection and completion of standard adjuvant therapy (PEARLS)]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2015-000575-27/SI> (first posted 15 July 2015). [EUCTR: 2015-000575-27]
- 141.** EUCTR 2018-002997-29. A study of neoadjuvant/adjuvant durvalumab for the treatment of patients with resectable non-small cell lung cancer [A phase III, double-blind, placebo-controlled, multi-center international study of neoadjuvant/adjuvant durvalumab for the treatment of patients with resectable stages II and III non-small cell lung cancer (AEGEAN)]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2018-002997-29/HU> (first posted 12 November 2018). [EUCTR: 2018-002997-29]
- 142.** NCT03800134. A study of neoadjuvant/adjuvant durvalumab for the treatment of patients with resectable non-small cell lung cancer [A phase III, double-blind, placebo-controlled, multi-center international study of neoadjuvant/adjuvant durvalumab for the treatment of patients with resectable stages II and III non-small cell lung cancer (AEGEAN)]. <https://clinicaltrials.gov/study/NCT03800134> (first posted 11 January 2019). [NCT: 03800134]
- 143.** Heymach J, Taube J, Mitsudomi T, Harpole D, Aperghis M, Trani L, et al. P1.18-02 The AEGEAN phase 3 trial of neoadjuvant/adjuvant durvalumab in patients with resectable stage II/III NSCLC. *Journal of Thoracic Oncology* 2019;**14**(10):S625-6. [DOI: [10.1016/j.jtho.2019.08.1318](https://doi.org/10.1016/j.jtho.2019.08.1318)]
- 144.** Heymach JV, Mitsudomi T, Harpole D, Aperghis M, Jones S, Mann H, et al. Design and rationale for a phase III, double-blind, placebo-controlled study of neoadjuvant durvalumab + chemotherapy followed by adjuvant durvalumab for the treatment of patients with resectable stages II and III non-small-cell lung cancer: the AEGEAN Trial. *Clinical Lung Cancer* 2022;**23**(3):e247-51. [DOI: [10.1016/j.clcc.2021.09.010](https://doi.org/10.1016/j.clcc.2021.09.010)]
- 145.** Heymach JV, Reck M, Mitsudomi T, Taube JM, Spira A, Chaft J, et al. Outcomes with perioperative durvalumab in patients with resectable NSCLC and baseline N2 lymph node involvement (N2 R-NSCLC): an exploratory subgroup analysis of AEGEAN. In: 2024 ASCO Annual Meeting; 2024, May 31-Jun 4; Chicago (IL). 2024.
- 146.** He J, Gao S, Reck M, Harpole D, Mitsudomi T, Taube JM, et al. OA12.06 neoadjuvant durvalumab + chemotherapy followed by adjuvant durvalumab in resectable EGFR-mutated NSCLC (AEGEAN). *Journal of Thoracic Oncology* 2023;**18**(11):S72-3. [DOI: [10.1016/j.jtho.2023.09.071](https://doi.org/10.1016/j.jtho.2023.09.071)]
- 147.** Mitsudomi T, Heymach JV, Reck M, Taube JM, Gao S, Horio Y, et al. OA12.05 surgical outcomes with neoadjuvant durvalumab + chemotherapy followed by adjuvant durvalumab in resectable NSCLC (AEGEAN). *Journal of Thoracic Oncology* 2023;**18**(11):S71-2. [DOI: [10.1016/j.jtho.2023.09.070](https://doi.org/10.1016/j.jtho.2023.09.070)]
- 148.** Reck M, Gale D, Harpole D, Taube JM, Mitsudomi T, Hochmair MJ, et al. LBA59 Associations of ctDNA clearance and pathological response with neoadjuvant treatment in patients with resectable NSCLC from the phase III AEGEAN trial. *Annals of Oncology* 2023;**34**:S1300. [DOI: [10.1016/j.annonc.2023.10.055](https://doi.org/10.1016/j.annonc.2023.10.055)]
- 149.** Gale D, Zhu Z, Lai Z, Reck M, Harpole D, Taube JM, et al. Abstract CT 238 - Associations of ctDNA levels during neoadjuvant treatment with pathological response in patients with resectable NSCLC from the phase 3 AEGEAN trial. *Cancer Research* 2024;**84**(7):CT238. [DOI: [10.1158/1538-7445.AM2024-CT238](https://doi.org/10.1158/1538-7445.AM2024-CT238)]
- 150.** Heymach J, Reck M, Mitsudomi T, Taube JM, Spira AI, Chaft JE, et al. Outcomes with perioperative durvalumab (D) in pts with resectable NSCLC and baseline N2 lymph node involvement (N2 R-NSCLC): an exploratory subgroup analysis of AEGEAN. *Journal of Clinical Oncology* 2024;**42**(Suppl 16):8011. [DOI: [10.1200/JCO.2024.42.16_suppl.8011](https://doi.org/10.1200/JCO.2024.42.16_suppl.8011)]
- 151.** Heymach JV, Harpole D, Mitsudomi T, Taube JM, Gao S, Urban L, et al. Perioperative durvalumab for resectable NSCLC (R-NSCLC): updated outcomes from the phase 3 AEGEAN trial. *Journal of Thoracic Oncology* 2024;**19**(10):S38-9. [DOI: [10.1016/j.jtho.2024.09.069](https://doi.org/10.1016/j.jtho.2024.09.069)]
- 152.** Weissinger F, Heymach JV, Harpole D, Mitsudomi T, Taube JM, Galffy G, et al. Perioperative durvalumab for resectable NSCLC: updated outcomes from the phase 3 AEGEAN trial. *Journal of Clinical Oncology* 2024;**47**:200-1. [DOI: [10.1159/000540557](https://doi.org/10.1159/000540557)]
- 153.** Mitsudomi T, Heymach JV, Reck M, Taube JM, Gao S, Horio Y, et al. Surgical outcomes with neoadjuvant durvalumab plus chemotherapy followed by adjuvant durvalumab in resectable NSCLC. *Journal of Thoracic Oncology* 2025;**S1556-0864**(25):00780. [DOI: [10.1016/j.jtho.2025.06.015](https://doi.org/10.1016/j.jtho.2025.06.015)]
- 154.** Pasello G, Reck M, Harpole D, Mitsudomi T, Winder T, Zukov R, et al. Patient-reported outcomes (PROs) with perioperative durvalumab in resectable NSCLC (AEGEAN).

Journal of Thoracic Oncology 2025;**20**(3):S125-6. [DOI: [10.1016/S1556-0864\(25\)00382-X](https://doi.org/10.1016/S1556-0864(25)00382-X)]

155. Reck M, Gale D, Zhu Z, Harpole D, Taube J, Mitsudomi T, et al. LBA49 Associations of ctDNA clearance (CL) during neoadjuvant Tx with pathological response and event-free survival (EFS) in pts with resectable NSCLC (R-NSCLC): expanded analyses from AEGEAN. *Annals of Oncology* 2024;**35**:S1239. [DOI: [10.1016/j.annonc.2024.08.2290](https://doi.org/10.1016/j.annonc.2024.08.2290)]

156. Urban L, Winder T, Reck M, Zukov R, Garbaos GE, Tran TV, et al. 123P Updated safety of perioperative durvalumab for resectable NSCLC in AEGEAN. *ESMO Open* 2024;**9**:102702. [DOI: [10.1016/j.esmoop.2024.102702](https://doi.org/10.1016/j.esmoop.2024.102702)]

157. Schumann C, Heymach JV, Harpole D, Mitsudomi T, Taube JM, Galfiy G, et al. AEGEAN: a phase 3 trial of neoadjuvant durvalumab + chemotherapy followed by adjuvant durvalumab in patients with resectable NSCLC. *Oncology Research and Treatment* 2023;**46**:8-9. [DOI: [10.1159/000533576](https://doi.org/10.1159/000533576)]

158. EUCTR 2019-000262-38. A phase 3, randomized, double-blind study of chemotherapy plus nivolumab versus chemotherapy plus placebo provided before surgery, followed by nivolumab or placebo provided after surgery for participants with non-small cell lung cancer which can be completely removed by surgery [A phase 3, randomized, double-blind study of neoadjuvant chemotherapy plus nivolumab versus neoadjuvant chemotherapy plus placebo, followed by surgical resection and adjuvant treatment with nivolumab or placebo for participants with resectable stage II-IIIb non-small cell lung cancer (CheckMate 77T, CHECKpoint pathway and nivolumab clinical Trial Evaluation 77T)]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2019-000262-38/IT> (first posted 25 June 2019). [EUCTR: 2019-000262-38]

159. Cascone T, Awad MM, Spicer JD, He J, Lu S, Sepesi B, et al. LBA1 CheckMate 77T: phase III study comparing neoadjuvant nivolumab (NIVO) plus chemotherapy (chemo) versus neoadjuvant placebo plus chemo followed by surgery and adjuvant NIVO or placebo for previously untreated, resectable stage II-IIIb NSCLC. *Annals of Oncology* 2023;**34**:S1295. [DOI: [10.1016/j.annonc.2023.10.050](https://doi.org/10.1016/j.annonc.2023.10.050)]

160. Cascone T, Provencio M, Sepesi B, Lu S, Aanur N, Li S, Spicer J. Checkmate 77T: a phase III trial of neoadjuvant nivolumab (NIVO) plus chemotherapy (chemo) followed by adjuvant nivo in resectable early-stage NSCLC. *Journal of Clinical Oncology* 2020;**38**(Suppl 15):TPS9076. [DOI: [10.1200/JCO.2020.38.15-suppl.TPS9076](https://doi.org/10.1200/JCO.2020.38.15-suppl.TPS9076)]

161. NCT04025879. A study of neoadjuvant chemotherapy plus nivolumab versus neoadjuvant chemotherapy plus placebo, followed by surgical removal and adjuvant treatment with nivolumab or placebo for participants with surgically removable early stage non-small cell lung cancer [A phase 3, randomized, double-blind study of neoadjuvant chemotherapy plus nivolumab versus neoadjuvant chemotherapy plus placebo, followed by surgical resection and adjuvant treatment with nivolumab or placebo for participants with resectable stage II-IIIb non-small cell lung cancer]. <https://clinicaltrials.gov/study/NCT04025879> (first posted 19 July 2019). [NCT: 04025879]

162. Schumann C, Cascone T, Provencio M, Sepesi B, Lu S, Aanur N, et al. CheckMate 77 T: a phase 3 trial of neoadjuvant nivolumab (NIVO) + chemotherapy(chemo) followed by adjuvant NIVO in resectable early-stage non-small cell lung cancer (NSCLC). *Pneumologie* 2021;**75**(Suppl 1):S16. [DOI: [10.1055/s-0041-1723292](https://doi.org/10.1055/s-0041-1723292)]

163. Awad M, Cascone T, Spicer J, He J, Lu S, Tanaka F, et al. LBA2 Clinical outcomes with perioperative nivolumab (NIVO) in patients (PTS) with resectable NSCLC from the phase III CheckMate 77T study. *ESMO Open* 2024;**9**:102985. [DOI: [10.1016/j.esmoop.2024.102985](https://doi.org/10.1016/j.esmoop.2024.102985)]

164. Deutsch JS, Cimino-Mathews A, Thompson E, Gabrielson E, Illei P, Jedrych J, et al. Associations between percent residual viable tumor (%RVT) and efficacy with perioperative nivolumab (NIVO) for resectable NSCLC in CheckMate 77T. *Cancer Research* 2025;**85**(8):CT097. [DOI: [10.1158/1538-7445.AM2025-CT097](https://doi.org/10.1158/1538-7445.AM2025-CT097)]

165. He J, Wu L, Gao Y, Ying K, Jiang G, Lee K-Y, et al. Perioperative nivolumab (NIVO) in patients (pts) with resectable stage II-IIIb NSCLC: checkMate 77T Chinese population analysis. *Annals of Oncology* 2024;**35**:S1616. [DOI: [10.1016/j.annonc.2024.10.621](https://doi.org/10.1016/j.annonc.2024.10.621)]

166. Moro-Sibilot D, Cascone T, Awad M, Spicer J, Tanaka F, He J, et al. Outcomes by treatment (tx) cycles (cyc) with perioperative nivolumab (NIVO) in patients (pts) with resectable NSCLC from CheckMate 77 T study. *Journal of Thoracic Oncology* 2025;**20**(3):S100-2. [DOI: [10.1016/S1556-0864\(25\)00334-X](https://doi.org/10.1016/S1556-0864(25)00334-X)]

167. Provencio Pulla M, Awad M, Cascone T, Spicer JD, He J, Lu S, et al. LBA50 Perioperative nivolumab (NIVO) versus placebo (PBO) in patients (pts) with resectable NSCLC: clinical update from the phase III CheckMate 77T study. *Annals of Oncology* 2024;**35**:S1239-40.

168. Spicer J, Pulla MP, Awad M, He J, Lu S, Tanaka F, et al. Health-related quality of life (HRQoL) outcomes from CheckMate 77T: neoadjuvant nivolumab (NIVO) plus chemotherapy (chemo) followed by adjuvant NIVO in resectable NSCLC. *ESMO Open* 2024;**9**:102700. [DOI: [10.1016/j.esmoop.2024.102700](https://doi.org/10.1016/j.esmoop.2024.102700)]

169. Yue D, Wang W, Liu H, Chen Q, Chen C, Liu L, et al. Perioperative tislelizumab plus neoadjuvant chemotherapy for patients with resectable non-small-cell lung cancer (RATIONALE-315): an interim analysis of a randomised clinical trial. *Lancet. Respiratory Medicine* 2025;**13**(2):119-29. [DOI: [10.1016/S2213-2600\(24\)00269-8](https://doi.org/10.1016/S2213-2600(24)00269-8)]

170. Wang C, Wang R, Ma Y, Liang X. 94 TiP Neoadjuvant tislelizumab or placebo + platinum-based chemotherapy followed by adjuvant tislelizumab or placebo in patients with resectable non-small cell lung cancer (NSCLC): a phase III trial in progress. *Journal of Thoracic Oncology* 2021;**16**(4):S746. [DOI: [10.1016/S1556-0864\(21\)01936-5](https://doi.org/10.1016/S1556-0864(21)01936-5)]

171. NCT04379635. Comparing the efficacy and safety of a new additional treatment with tislelizumab in non-small cell lung cancer (NSCLC) [A randomized, double-blind, placebo-controlled, phase 3 study to compare the efficacy and safety of neoadjuvant treatment with tislelizumab (BGB-A317, anti-

PD-1 antibody) or placebo plus platinum-based doublet chemotherapy followed by adjuvant tislelizumab or placebo in resectable stage II or IIIa non-small cell lung cancer]. <https://clinicaltrials.gov/study/NCT04379635> (first posted 7 May 2020). [NCT: 04379635]

172. Cappuzzo F, Wang C, Wang W, Liu H, Chen Q, Yue D, et al. 1213P Neoadjuvant tislelizumab (TIS) plus chemotherapy (CT) with adjuvant TIS vs. neoadjuvant placebo (PBO) plus CT with adjuvant PBO in resectable non-small cell lung cancer (NSCLC): patient-reported outcomes (PRO) in the RATIONALE-315 trial. *Annals of Oncology* 2024;**35**(Suppl 2):S781-2.

173. Yue D, Tan L, Xu S, Mao N, Hu J, Zhang L, et al. 1080 Surgical outcomes from RATIONALE-315: randomized, double-blind, phase III study of perioperative tislelizumab with neoadjuvant chemotherapy in resectable NSCLC. *ESMO Open* 2024;**9**(Suppl 3):102687. [DOI: [10.1016/j.esmoop.2024.102687](https://doi.org/10.1016/j.esmoop.2024.102687)]

174. Wang C, Chen K-N, Chen Q, Wu L, Wang Q, Li X, et al. Neoadjuvant nivolumab (NIVO) + chemotherapy (chemo) versus chemo in Chinese patients (pts) with resectable NSCLC in CheckMate 816. *Cancer Research* 2023;**83**(Suppl 8):CT081. [DOI: [10.1158/1538-7445.AM2023-CT081](https://doi.org/10.1158/1538-7445.AM2023-CT081)]

175. Forde PM, Spicer J, Girard N, Provencio M, Lu S, Wang C, et al. 840 Neoadjuvant nivolumab (N) + platinum-doublet chemotherapy (C) for resectable NSCLC: 3-y update from CheckMate 816. *Journal of Thoracic Oncology* 2023;**18**(4):S89-90. [DOI: [10.1016/S1556-0864\(23\)00338-6](https://doi.org/10.1016/S1556-0864(23)00338-6)]

176. Deutsch JS, Cimino-Mathews A, Thompson E, Provencio M, Forde PM, Spicer J, et al. LBA50 Analysis of pathological features and efficacy outcomes with neoadjuvant nivolumab (N) plus platinum-doublet chemotherapy (C) for resectable non-small cell lung cancer (NSCLC) in CheckMate 816. *Annals of Oncology* 2023;**11**:A1739. [DOI: [10.1136/jitc-2023-SITC2023.222-N](https://doi.org/10.1136/jitc-2023-SITC2023.222-N)]

177. Forde PM, Spicer J, Lu S, Provencio M, Mitsudomi T, Awad MM, et al. Nivolumab (NIVO) + platinum-doublet chemotherapy (chemo) versus chemo as neoadjuvant treatment (tx) for resectable (IB-IIIa) non-small cell lung cancer (NSCLC) in the phase 3 CheckMate 816 trial. *Cancer Research* 2021;**81**(Suppl 13):CT003. [DOI: [10.1158/1538-7445.AM2021-CT003](https://doi.org/10.1158/1538-7445.AM2021-CT003)]

178. Spicer J, Forde PM, Provencio M, Lu S, Wang C, Mitsudomi T, et al. Clinical outcomes with neoadjuvant nivolumab (N) + chemotherapy (C) vs C by definitive surgery in patients (pts) with resectable NSCLC: 3-y results from the phase 3 CheckMate 816 trial. *Journal of Clinical Oncology* 2023;**41**(16):8521. [DOI: [10.1200/jco.2023.41.16_suppl.8521](https://doi.org/10.1200/jco.2023.41.16_suppl.8521)]

179. NCT02998528. A neoadjuvant study of nivolumab plus ipilimumab or nivolumab plus chemotherapy versus chemotherapy alone in early stage non-small cell lung cancer (NSCLC) (CheckMate 816) [Randomized, openlabel, phase 3 trial of nivolumab plus ipilimumab or nivolumab plus platinum doublet chemotherapy versus platinum doublet chemotherapy in early stage NSCLC]. <https://clinicaltrials.gov/study/NCT02998528> (first posted 20 December 2016).

180. Mitsudomi T, Ito H, Okada M, Sugawara S, Shio Y, Tomii K, et al. Neoadjuvant nivolumab plus chemotherapy in resectable non-small-cell lung cancer in Japanese patients from CheckMate 816. *Cancer Science* 2023;**115**(2):540-54. [DOI: [10.1111/cas.16030](https://doi.org/10.1111/cas.16030)]

181. Spicer J, Wang C, Tanaka F, Chen K-N, Ito H, Liberman M, et al. Event-free survival (EFS) with neoadjuvant nivolumab (NIVO) + platinum-doublet chemotherapy (chemo) versus chemo by surgical outcomes in resectable nonsmall cell lung cancer (NSCLC) from CheckMate 816. *Lung Cancer* 2023;**178**:S26-7. [DOI: [10.1016/S0169-5002\(23\)00487-7](https://doi.org/10.1016/S0169-5002(23)00487-7)]

182. Forde PM, Chaft JE, Felip E, Broderick S, Girard N, Awad MM, et al. Checkmate 816: a phase 3, randomized, open-label trial of nivolumab plus ipilimumab vs platinum-doublet chemotherapy as neoadjuvant treatment for early-stage NSCLC. *Journal of Clinical Oncology* 2017;**35**(Suppl 15):TPS8577. [DOI: [10.1200/JCO.2017.35.15_suppl.TPS8577](https://doi.org/10.1200/JCO.2017.35.15_suppl.TPS8577)]

183. Girard N, Spicer J, Provencio M, Lu S, Broderick S, Awad MM, et al. Abstract CT012: Nivolumab (NIVO) + platinum-doublet chemotherapy (chemo) vs chemo as neoadjuvant treatment for resectable (IB-IIIa) non-small cell lung cancer (NSCLC): event-free survival (EFS) results from the phase 3 CheckMate 816 trial. *Cancer Research (AACR Annual Meeting)* 2022;**82**(Suppl 12):CT012. [DOI: [10.1158/1538-7445.AM2022-CT012](https://doi.org/10.1158/1538-7445.AM2022-CT012)]

184. Provencio Pulla M, Forde PM, Spicer JD, Wang C, Lu S, Mitsudomi T, et al. LBA57 Neoadjuvant nivolumab (N) + chemotherapy (C) in the phase III CheckMate 816 study: 3-y results by tumor PD-L1 expression. *Annals of Oncology* 2023;**34**:S1298-9. [DOI: [10.1016/j.annonc.2023.10.053](https://doi.org/10.1016/j.annonc.2023.10.053)]

185. Spicer JD, Girard N, Provencio Pulla M, Wang C, Mitsudomi T, Awad MM, et al. LBA 8010 Neoadjuvant nivolumab plus chemotherapy versus chemotherapy in patients with resectable NSCLC: 4-year update from Checkmate 816. In: 2024 ASCO Annual Meeting; 2024; May 31-Jun 4; Chicago (IL). 2024.

186. Felip E, Wang C, Ciuleanu T-E, Saylors G, Tanaka F, Chen K-N, et al. 932MO Nivolumab (NIVO) plus platinum-doublet chemotherapy (chemo) versus chemo as neoadjuvant treatment for resectable non-small cell lung cancer (NSCLC): health-related quality of life (HRQoL) outcomes from CheckMate 816. *Annals of Oncology* 2022;**33**:S973-4. [DOI: [10.1016/j.annonc.2022.07.1058](https://doi.org/10.1016/j.annonc.2022.07.1058)]

187. Awad MM, Forde PM, Girard N, Spicer J, Wang C, Lu S, et al. Neoadjuvant nivolumab plus ipilimumab versus chemotherapy in resectable lung cancer. *Journal of Clinical Oncology* 2025;**43**(12):1453-62. [DOI: [10.1200/JCO-24-02239](https://doi.org/10.1200/JCO-24-02239)]

188. Damonte A, García Morato J, Courreges S, Omati Porte A, Perfetti A, Levit P, et al. Surgical outcomes post neoadjuvant immunotherapy in lung cancer: CheckMate 816 in Argentine subpopulation [Resultados quirúrgicos post neoadyuvancia con inmunoterapia en cáncer de pulmón: checkmate 816 en subpoblación Argentina]. *Medicina* 2025;**85**(1):47-55. [PMID: 39900048]

189. Forde PM, Spicer JD, Provencio M, Mitsudomi T, Awad MM, Wang C, et al. Overall survival with neoadjuvant nivolumab plus

chemotherapy in lung cancer. *New England Journal of Medicine* 2025;**393**:742-52. [DOI: [10.1056/NEJMoa2502931](https://doi.org/10.1056/NEJMoa2502931)]

190. Provencio-Pulla M, Spicer J, Taube JM, Martin C, Spigel DR, Wang C, et al. Neoadjuvant nivolumab (NIVO) + platinum-doublet chemotherapy (chemo) versus chemo for resectable (IB-IIIA) non-small cell lung cancer (NSCLC): association of pathological regression with event-free survival (EFS) in CheckMate 816. *Journal of Clinical Oncology* 2022;**40**(Suppl 17):LBA8511. [DOI: [10.1200/JCO.2022.40.17_suppl.LBA8511](https://doi.org/10.1200/JCO.2022.40.17_suppl.LBA8511)]

191. Awad MM, Forde PM, Girard N, Spicer JD, Wang C, Lu S, et al. 1261O Neoadjuvant nivolumab (N) + ipilimumab (I) vs chemotherapy (C) in the phase III CheckMate 816 trial. *Annals of Oncology* 2023;**34**:S731. [DOI: [10.1016/j.annonc.2023.09.739](https://doi.org/10.1016/j.annonc.2023.09.739)]

192. Felip E, Brahmer J, Broderick S, Swanson S, Awad M, Mitsudomi T, et al. P2.16-03 CheckMate 816: A phase 3 trial of neoadjuvant nivolumab plus ipilimumab or chemotherapy versus chemotherapy in early-stage NSCLC. *Journal of Thoracic Oncology* 2018;**13**(10):S831-2. [DOI: [10.1016/j.jtho.2018.08.1478](https://doi.org/10.1016/j.jtho.2018.08.1478)]

193. Spicer J, Wang C, Tanaka F, Saylor GB, Chen K-N, Liberman M, et al. Surgical outcomes from the phase 3 CheckMate 816 trial: nivolumab (NIVO) + platinum-doublet chemotherapy (chemo) versus chemo alone as neoadjuvant treatment for patients with resectable non small cell lung cancer (NSCLC). *Journal of Clinical Oncology* 2021;**39**(Suppl 15):8503. [DOI: [10.1200/JCO.2021.39.15_suppl.8503](https://doi.org/10.1200/JCO.2021.39.15_suppl.8503)]

194. EUCTR 2016-003536-21-ES. A neoadjuvant study of nivolumab and ipilimumab compared to chemotherapy doublet in early stage non-small cell lung cancer (NSCLC) [Randomized, open-label, phase 3 trial of nivolumab and ipilimumab versus platinum-doublet chemotherapy in early stage NSCLC]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2016-003536-21/ES> (first posted 27 December 2016). [EUCTR: 2016-003536-21]

195. NCT04611776. A study evaluating the efficacy and safety of adjuvant platinum-doublet chemotherapy, with or without atezolizumab, in patients who are ctDNA positive after complete surgical resection of stage IB to select IIIB non-small cell lung cancer (CATHAYA) [A phase II, randomized, double-blind, placebo-controlled study evaluating the efficacy and safety of adjuvant platinum-doublet chemotherapy, with or without atezolizumab, in patients who are ctDNA positive after complete surgical resection of stage IB to select IIIB non-small cell lung cancer]. <https://clinicaltrials.gov/study/NCT04611776> (first posted 2 November 2020).

196. NCT04422392. Neoadjuvant PD-1 antibody plus chemotherapy in resectable stage IIIA-N2 non-small-cell lung cancer [Neoadjuvant PD-1 antibody plus chemotherapy in resectable stage IIIA-N2 non-small-cell lung cancer: a randomized phase II study]. <https://clinicaltrials.gov/study/NCT04422392> (first posted 9 June 2020).

197. Peters S, Spigel D, Ahn M, Tsuboi M, Chافت J, Harpole D, et al. P03.03 MERMAID-1: a phase III study of adjuvant durvalumab plus chemotherapy in resected NSCLC patients with MRD+ post-surgery. *Journal of Thoracic Oncology* 2021;**16**(3):S258-9. [DOI: [10.1016/j.jtho.2021.01.376](https://doi.org/10.1016/j.jtho.2021.01.376)]

198. Reinmuth N, Peters S. MERMAID-1: a phase III study of adjuvant durvalumab plus chemotherapy in resected NSCLC patients with MRD+ post-surgery. *Oncology Research and Treatment* 2021;**44**(Suppl 2):252. [DOI: [10.1159/000518417](https://doi.org/10.1159/000518417)]

199. NCT04385368. Phase III study to determine the efficacy of durvalumab in combination with chemotherapy in completely resected stage II-III non-small cell lung cancer (NSCLC) (MERMAID-1) [A phase III, randomized, multicenter, double-blind, placebo-controlled study to determine the efficacy of adjuvant durvalumab in combination with platinum-based chemotherapy in completely resected stage II-III NSCLC (MERMAID-1)]. <https://clinicaltrials.gov/study/NCT04385368> (first posted 12 May 2020).

200. EUCTR2020-000556-35-HU. Phase III study to determine the efficacy of Durvalumab in combination with chemotherapy in completely resected stage II-III non-small cell lung cancer [A phase III, randomized, multicenter, double-blind, placebo-controlled study to determine the efficacy of adjuvant durvalumab in combination with platinum-based chemotherapy in completely resected stage II-III NSCLC (MeRmaiD-1)]. EUCTR (European Clinical Trials Registry) (first posted 20 May 2020). [EUCTR: 2020-000556-35]

201. Reinmuth N, Spigel D, Peters S, Ahn MJ, Tsuboi M, Chافت J, et al. MERMAID-2: phase 3 study of durvalumab in patients with resected, stage II-III NSCLC who become MRD+ after curative-intent therapy. *Oncology Research and Treatment* 2021;**44**(Suppl 2):252-3. [DOI: [10.1159/000518417](https://doi.org/10.1159/000518417)]

202. EUCTR2020-000612-30-HU. Phase III study to evaluate the efficacy and safety of durvalumab compared to placebo in patients with NSCLC who have minimal residual disease after surgery +/- additional therapy before or after surgery [A phase III, randomized, multicenter, double-blind, placebo-controlled study of durvalumab for the treatment of stage II-III NSCLC patients with minimal residual disease following surgery and curative intent therapy (MERMAID-2)]. EUCTR (European Clinical Trials Registry) (first posted 2 December 2020). [EUCTR: 2020-000612-30]

203. Spigel DR, Peters S, Ahn M-J, Tsuboi M, Chافت J, Harpole D, et al. 93 TiP MERMAID-2: phase III study of durvalumab in patients with resected, stage II-III NSCLC who become MRD+ after curative-intent therapy. *Journal of Thoracic Oncology* 2021;**16**(4):S745-6. [DOI: [10.1016/S1556-0864\(21\)01935-3](https://doi.org/10.1016/S1556-0864(21)01935-3)]

204. NCT04642469. Phase III study to determine efficacy of durvalumab in stage II-III non-small cell lung cancer (NSCLC) after curative intent therapy(MERMAID-2) [A phase III, randomized, multicenter, double-blind, placebo-controlled study of durvalumab for the treatment of stage II-III NSCLC patients with minimal residual disease following surgery and curative intent therapy]. <https://clinicaltrials.gov/study/NCT04642469> (first posted 24 November 2020).

205. Schmid S, Dimopoulou G, van Schil P, König D, Mauti L, Itchins M, et al. ETOP 25-23 ADOPT-lung: an international, multicentre, open-label randomised phase III trial to evaluate the benefit of adding adjuvant durvalumab after neoadjuvant chemotherapy plus durvalumab in patients with stage IIB-IIIB

(N2) resectable NSCLC. *Lung Cancer* 2025;**206**:108635. [DOI: [10.1016/j.lungcan.2025.108635](https://doi.org/10.1016/j.lungcan.2025.108635)] [PMID: 40582281]

206. NCT06284317. A study to evaluate the benefit of adding durvalumab after chemotherapy, durvalumab and surgery in patients with early-stage, operable, non-small cell lung cancer (ADOPT-lung) [An International, multicentre, open-label randomised phase III trial to evaluate the benefit of adding adjuvant durvalumab after neoadjuvant chemotherapy plus durvalumab in patients with stage IIB-IIIb (N2) resectable NSCLC]. <https://clinicaltrials.gov/study/NCT06284317> (first posted 28 February 2024).

207. NCT06498635. Immunotherapy after surgery for people who have no remaining cancer cells after standard treatment for early-stage non-small cell lung cancer, INSIGHT trial [Randomized phase III trial INcorporating pathologic complete reSponse in participants with early staGe non small cell lung cancer to optimize immunotHerapy in the adjuvant setting (INSIGHT)]. <https://clinicaltrials.gov/study/NCT06498635> (first posted 12 July 2024).

208. NCT06620822. Efficacy of PD-1 inhibitor combination therapy in non-small cell lung cancer patients who have not achieved major pathologic response after neoadjuvant immunotherapy: a multicenter, phase II clinical trial. <https://clinicaltrials.gov/study/NCT06620822> (first posted 1 October 2024).

209. ChiCTR2400091455. Efficacy of PD-1 inhibitor combination therapy in non-small cell lung cancer patients who have not achieved major pathologic response after neoadjuvant immunotherapy [Efficacy of PD-1 inhibitor combination therapy in non-small cell lung cancer patients who have not achieved major pathologic response after neoadjuvant immunotherapy: a multicenter, phase II clinical trial]. <https://www.chictr.org.cn/showprojEN.html?proj=247064> (first posted 29 October 2024).

210. Mok TSK, Pujol J-L, Tsuboi M, Lee J, Kim E, Leonov O, et al. LBA4 CANOPY-N: A phase II study of canakinumab (CAN) or pembrolizumab (PEM), alone or in combination, as neoadjuvant therapy in patients (pts) with resectable stage Ib-IIIa non-small cell lung cancer (NSCLC). *Annals of Oncology* 2022;**33**:S1547-48. [DOI: [10.1016/j.annonc.2022.10.322](https://doi.org/10.1016/j.annonc.2022.10.322)]

211. Lee J, Mok T, Garrido P, Kim ES, Arslan C, Tsuboi M, et al. Canakinumab or pembrolizumab as monotherapy or in combination as neoadjuvant therapy in patients with surgically resected non-small cell lung cancer: CANOPY-N Trial. *American Journal of Respiratory and Critical Care Medicine* 2021;**203**:A4803. [DOI: https://doi.org/10.1164/ajrccm-conference.2021.203.1_MeetingAbstracts.A4803]

212. EUCTR2021-003369-37-ES. Study of neoadjuvant and adjuvant treatment in patients with resectable non-small cell lung cancer [A phase II open-label, multicentre, randomised study of neoadjuvant and adjuvant treatment in patients with resectable, early-stage (II to IIIA) non-small cell lung cancer (NeoCOAST-2)]. <https://www.clinicaltrialsregister.eu/ctr-search/trial/2021-003369-37/ES> (first posted 2 October 2022).

213. Huynh C, Rayes R, Gaudreau P, Hirsh V, Ofiara L, Owen S, et al. P79.05 Phase II randomized trial of neoadjuvant

pembrolizumab +/- chemotherapy for operable stage IA3-IIA non-small cell lung cancer. *Journal of Thoracic Oncology* 2021;**16**(3):S647-8. [DOI: [10.1016/j.jtho.2021.01.1184](https://doi.org/10.1016/j.jtho.2021.01.1184)]

214. NCT04202809. Phase-II trial of induction chemotherapy and chemoradiotherapy plus/minus durvalumab and consolidation immunotherapy in patients with resectable stage III NSCLC (ESPADURVA) [Prospective phase-II trial of induction chemotherapy and chemoradiotherapy plus/minus the PD-L1 antibody durvalumab followed by surgery or definitive chemoradiation boost and consolidation durvalumab in resectable stage III NSCLC]. <https://clinicaltrials.gov/study/NCT04202809> (first posted 18 December 2019).

215. Sands J, Mandrekar SJ, Oxnard GR, Kozono DE, Hillman SL, Dahlberg SE, et al. ALCHEMIST: adjuvant targeted therapy or immunotherapy for high-risk resected NSCLC. *Journal of Clinical Oncology* 2020;**38**(Suppl 15):TPS9077. [DOI: [10.1200/JCO.2020.38.15-suppl.TPS9077](https://doi.org/10.1200/JCO.2020.38.15-suppl.TPS9077)]

216. Sterne JA, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;**366**:l4898.

217. Chaft JE, Dahlberg SE, Khullar OV, Edelman MJ, Simone CB, Heymach J, et al. EA5142 adjuvant nivolumab in resected lung cancers (ANVIL). *Journal of Clinical Oncology* 2018;**36**(Suppl 15):TPS8581. [DOI: [10.1200/JCO.2018.36.15-suppl.TPS8581](https://doi.org/10.1200/JCO.2018.36.15-suppl.TPS8581)]

218. NCT02595944. Nivolumab after surgery and chemotherapy in treating patients with stage IB-IIIa non-small cell lung cancer (An ALCHEMIST treatment trial) (ANVIL) [Adjuvant nivolumab in resected lung cancers (ANVIL) - a randomized phase III study of nivolumab after surgical resection and adjuvant chemotherapy in non-small cell lung cancers]. <https://clinicaltrials.gov/study/NCT02595944> (first posted 2015).

219. Peters S, Kim AW, Solomon B, Gandara DR, Dziadziuszko R, Brunelli A, et al. IMpower030: phase III study evaluating neoadjuvant treatment of resectable stage II-IIIb non-small cell lung cancer (NSCLC) with atezolizumab (atezo) + chemotherapy. *Annals of Oncology* 2019;**30**:30. [DOI: [10.1093/annonc/mdz064.014](https://doi.org/10.1093/annonc/mdz064.014)]

220. Rizvi N, Gandara D, Solomon B, Kim A, Brunelli A, Sun S, et al. P2.17-27 IMpower030: phase III study evaluating neoadjuvant treatment of resectable stage II-IIIb NSCLC with atezolizumab + chemotherapy. *Journal of Thoracic Oncology* 2018;**13**(10):S863. [DOI: [10.1016/j.jtho.2018.08.1553](https://doi.org/10.1016/j.jtho.2018.08.1553)]

221. NCT03456063. A study of neoadjuvant atezolizumab plus chemotherapy versus placebo plus chemotherapy in patients with resectable stage II, IIIA, or select IIIB non-small cell lung cancer (IMpower030) [A phase III, double-blinded, multicenter, randomized study evaluating the efficacy and safety of neoadjuvant treatment with atezolizumab or placebo in combination with platinum-based chemotherapy in patients with resectable stage II, IIIA, or select IIIB non-small cell lung cancer]. <https://clinicaltrials.gov/study/NCT03456063> (first posted 7 March 2018).

222. EURCT2017-002857-12-DE. A study evaluating the efficacy and safety of neoadjuvant treatment with atezolizumab or

placebo in combination with platinum-based chemotherapy in patients with resectable stage II, IIIA, or select IIIB non-small cell lung cancer [A phase III, double-blinded, multicenter, randomized study evaluating the efficacy and safety of neoadjuvant treatment with atezolizumab or placebo in combination with platinum-based chemotherapy in patients with resectable stage II, IIIA, or select IIIB non-small cell lung cancer]. EUCTR (European Clinical Trials Registry) (first posted 19 February 2018). [EUCTR: 2017-002857-12]

223. Bluethmann SM, Mariotto AB, Rowland JH. Anticipating the "Silver Tsunami": prevalence trajectories and comorbidity burden among older cancer survivors in the United States. *Cancer Epidemiology, Biomarkers & Prevention (AACR)* 2016;**25**(7):1029-36. [DOI: [10.1158/1055-9965.EPI-16-0133](https://doi.org/10.1158/1055-9965.EPI-16-0133)] [PMID: 27371756]

224. Spicer JD, Cascone T, Wynes MW, Ahn M-J, Dacic S, Felip E, et al. Neoadjuvant and adjuvant treatments for early-stage resectable NSCLC: consensus recommendations from the international association for the study of lung cancer. *Journal of Thoracic Oncology* 2024;**19**(10):1373-414. [DOI: [10.1016/j.jtho.2024.06.010](https://doi.org/10.1016/j.jtho.2024.06.010)] [PMID: 38901648]

225. European Medicines Agency (EMA). Alecensa. <https://www.ema.europa.eu/en/medicines/human/EPAR/alecensa> (accessed 15 February 2026).

226. European Medicines Agency (EMA). Tagrisso. <https://www.ema.europa.eu/en/medicines/human/EPAR/tagrisso> (accessed 15 February 2026).

227. US Food & Drug Administration (FDA). FDA approves alectinib as adjuvant treatment for ALK-positive non-small cell lung cancer; 18 April 2024. <https://www.fda.gov/drugs/>

[resources-information-approved-drugs/fda-approves-alectinib-adjuvant-treatment-alk-positive-non-small-cell-lung-cancer](https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-alectinib-adjuvant-treatment-alk-positive-non-small-cell-lung-cancer) (accessed 15 February 2026).

228. US Food & Drug Administration (FDA). FDA approves osimertinib as adjuvant therapy for non-small cell lung cancer with EGFR mutations; 18 December 2020. <https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-osimertinib-adjuvant-therapy-non-small-cell-lung-cancer-egfr-mutations> (accessed 15 February 2026).

229. Banna GL, Hassan MA, Signori A, Giunta EF, Maniam A, Anpalakhan S, et al. Neoadjuvant chemo-immunotherapy for early-stage non-small cell lung cancer: a systematic review and meta-analysis. *JAMA Network Open* 2024;**7**(4):e246837. [DOI: [10.1001/jamanetworkopen.2024.6837](https://doi.org/10.1001/jamanetworkopen.2024.6837)]

230. Sorin M, Prosty C, Ghaleb L, Nie K, Katergi K, Shahzad MH, et al. Neoadjuvant chemoimmunotherapy for NSCLC: a systematic review and meta-analysis. *JAMA Oncology* 2024;**10**(5):621-33. [DOI: [10.1001/jamaoncol.2024.0057](https://doi.org/10.1001/jamaoncol.2024.0057)]

231. Paillaud E, Soubeyran P, Caillet P, Cudennec T, Brain E, Terret C; for the G-CODE collaborators. Multidisciplinary development of the Geriatric Core Dataset for clinical research in older patients with cancer: a French initiative with international survey. *European Journal of Cancer* 2018;**103**:61-8. [DOI: [10.1016/j.ejca.2018.07.137](https://doi.org/10.1016/j.ejca.2018.07.137)] [PMID: 30212804]

232. Corre R, Greillier L, Le Caër H, Audigier-Valette C, Baize N, Bérard H, et al. Use of a comprehensive geriatric assessment for the management of elderly patients with advanced non-small-cell lung cancer: the phase III randomized ESOGIA-GFPC-GECP 08-02 study. *Journal of Clinical Oncology* 2016;**34**(13):1476-83. [DOI: [10.1200/JCO.2015.63.5839](https://doi.org/10.1200/JCO.2015.63.5839)] [PMID: 26884557]

ADDITIONAL TABLES

Table 1. Design and interventions of included studies

Study	Population	Arm	Neoadjuvant tested regimen	Neoadjuvant chemotherapy	Adjuvant chemotherapy	Adjuvant radiotherapy	Adjuvant tested regimen	Relevant outcomes
Perioperative trials								
AEGEAN	Resectable stage II to IIIB NSCLC 802 participants International	Intervention	4 cycles every 3 weeks Durvalumab (1500 mg)	4 cycles every 3 weeks For squamous carcinoma: carboplatin + paclitaxel or cisplatin + gemcitabine or carboplatin + gemcitabine	No	Allowed if indicated according to local guidance	12 cycles every 4 weeks Durvalumab (1500 mg)	Primary outcomes: EFS and pCR Others: MPR, OS, safety
		Control	4 cycles every 3 weeks Placebo	For non-squamous carcinoma: cisplatin + pemetrexed or carboplatin + pemetrexed			12 cycles every 4 weeks Placebo	
CHECK-MATE-77T	Resectable stage II to IIIB NSCLC 461 participants International	Intervention	4 cycles every 3 weeks Nivolumab (360 mg)	4 cycles every 3 weeks For squamous carcinoma: cisplatin + docetaxel or carboplatin + paclitaxel	No	Allowed	12 cycles every 4 weeks Nivolumab (480 mg)	Primary outcome: EFS Others: MPR, OS, pCR, safety
		Control	4 cycles every 3 weeks Placebo	For non-squamous carcinoma: cisplatin + pemetrexed or carboplatin + pemetrexed or carboplatin + paclitaxel			12 cycles every 4 weeks Placebo	
KEYNOTE-671	Resectable stage II to IIIB NSCLC 797 participants International	Intervention	4 cycles every 3 weeks Pembrolizumab (200 mg)	4 cycles every 3 weeks For squamous carcinoma: cisplatin + gemcitabine For non-squamous carcinoma: cisplatin + pemetrexed	No	Allowed if R1/R2 disease, extra-capsular nodal extension, or no surgery	13 cycles every 3 weeks Pembrolizumab (200 mg)	Primary outcomes: EFS and OS Others: HRQoL, MPR, pCR, safety
		Control	4 cycles every 3 weeks Placebo				13 cycles every 3 weeks Placebo	
NADIM-II	Resectable stage IIIA or IIIB NSCLC 86 participants	Intervention	3 cycles every 3 weeks	3 cycles every 3 weeks Carboplatin + paclitaxel	No	NA	If R0 6 cycles every 4 weeks	Primary outcome: pCR

Table 1. Design and interventions of included studies (Continued)

Spain only		Nivolumab (360 mg)					Nivolumab (480mg)	Others: MPR, OS, safety
		Control	No placebo				Observation	
NEOTORCH	Resectable stage II to IIIB NSCLC	Intervention	3 cycles every 3 weeks	3 cycles every 3 weeks	1 cycle	Allowed	13 cycles every 3 weeks	Primary out- comes: EFS and MPR
	404 participants		Toripalimab (240 mg)	For squamous carcinoma: cis- platin or carboplatin + docetax- el or paclitaxel	Same reg- imen as in the neoad- juvant phase		Toripalimab (240 mg)	Others: OS, pCR, safety
	China only	Control	3 cycles every 3 weeks	For non-squamous carcinoma: cisplatin or carboplatin + peme- trexed			13 cycles every 3 weeks	
			Placebo				Placebo	
RATIONALE-315	Resectable stage II to IIIA NSCLC	Intervention	3–4 cycles every 3 weeks	3–4 cycles every 3 weeks	No	NA	8 cycles every 6 weeks	Primary out- comes: EFS and MPR
	453 participants		Tislelizumab (200 mg)	For squamous carcinoma: cis- platin or carboplatin + paclitax- el			Tislelizumab (400 mg)	Others: HRQoL, safe- ty, OS
	China only	Control	3–4 cycles every 3 weeks	For non-squamous carcinoma: cisplatin or carboplatin + peme- trexed			8 cycles every 6 weeks	
			Placebo				Placebo	
Neoadjuvant trials								
CHECK- MATE-816	Resectable stage IB to IIIA NSCLC	Intervention	3 cycles every 3 weeks	3 cycles every 3 weeks	4 cycles every 3 weeks	Allowed if indicated	No	Primary out- comes: EFS, pCR
	358 participants		Nivolumab (360 mg)	For any histology: carboplatin + paclitaxel or cisplatin + vinorel- bine or cisplatin + docetaxel	Same reg- imen as in the neoad- juvant phase			Others: MPR, OS, safety
	International	Control	No placebo	For squamous carcinoma: cis- platin + gemcitabine				
				For non-squamous carcinoma: cisplatin + pemetrexed				
TD-FOR- KNOW	Resectable stage IIIA to IIIB NSCLC	Intervention	3 cycles every 3 weeks	3 cycles every 3 weeks	No	NA	No	Primary out- come: pCR
	94 participants							

Table 1. Design and interventions of included studies (Continued)

China only		Camrelizum-ab (200 mg)		Nab-paclitaxel + platinum (cisplatin, carboplatin or nedaplatin)		Others: EFS, MPR, safety			
		Control	No placebo						
Adjuvant trials									
CCTG BR.31	Resected stage IB to IIIA NSCLC	Intervention	No	No	Adjuvant platinum-doublet allowed	For N2 disease only	12 cycles every 4 weeks of Durvalumab (20 mg/kg)	Primary outcome: DFS	
	PD-L1 ≥ 25%	Control					12 cycles every 4 weeks of Placebo	Others: OS, safety, HRQoL	
	1219 participants								
	International								
IMPOWER-010	Resected stage IB to IIIA NSCLC	Intervention	No	No	4 cycles every 3 weeks	No	16 cycles every 3 weeks	Primary outcome: DFS	
	PD-L1 ≥ 1%	Control	Cisplatin + vinorelbine or docetaxel or gemcitabine or pemetrexed (only for non-squamous carcinoma) Before atezolizumab			Atezolizumab (1200 mg)	Other: OS		
	1005 participants					Best supportive care			
	International								
PEARLS/KEYNOTE-091	Resected stage IB to IIIA NSCLC	Intervention	No	No	4 cycles	No	18 cycles every 3 weeks	Primary outcome: DFS	
	1177 participants	Control	Not mandatory but to be considered for stage IB and strongly recommended for stages II-III A			Pembrolizumab (200 mg)	Others: OS and safety		
	International					18 cycles every 3 weeks	Placebo		

Before pem-
brolizumab

Table 1. Design and interventions of included studies *(Continued)*

Chemotherapy dosage: carboplatin (AUC 5–6), cisplatin (75 mg/m²), docetaxel (60–75 mg/m²), gemcitabine (1000–1250 mg/m²), paclitaxel (175–200 mg/m²), pemetrexed (500 mg/m²), vinorelbine (25–30 mg/m²). Gemcitabine and vinorelbine could be administered on day 1 and day 8.

Abbreviations

DFS: disease-free survival; EFS: event-free survival; HRQoL: health-related quality of life; Gy: gray; MPR: major pathological response; NA: information not available; NSCLC: non-small cell lung cancer; N2: N2-level nodal involvement; OS: overall survival; pCR: pathological complete response; R0: complete resection; R1: microscopically incomplete resection; R2: macroscopically incomplete resection

Table 2. Design of ongoing studies

Study	Population	Arm	Neoadjuvant regimen	Adjuvant regimen	Relevant outcomes
Perioperative trial					
IMPOWER-030	Resectable stage II-IIIB NSCLC 453 participants International	Intervention	Atezolizumab + chemotherapy	Atezolizumab	Primary outcome: EFS
		Control	Placebo + chemotherapy	Observation	Others: HRQoL, MPR, pCR, OS, safety
Neoadjuvant trials					
NCT06083454 (Ascorbate)	Resectable stage I NSCLC 36 expected participants USA only	Intervention 1	Durvalumab	No	Secondary outcomes: EFS, MPR, pCR, OS
		Intervention 2	Durvalumab + ascorbate		
		Control	Surgery alone		
CTDNEO	Resectable stage II NSCLC Negative ctDNA 48 expected participants China only	Intervention	Toripril + platinum-doublet	No	Primary outcome: pCR
		Control	Platinum-doublet		Others: EFS, MPR, OS, safety
ELoN	Resectable stage IIB-IIIA NSCLC 50 expected participants India only	Intervention	Nivolumab + carboplatin + paclitaxel	Allowed as per clinician discretion	Primary outcome: MPR
		Control	Carboplatin + paclitaxel		Other: safety
XWh-NS-CLC-NEO-IIT-SHR1210	Resectable stage IIB-IIIA NSCLC 38 expected participants China only	Intervention	Camrelizumab + carboplatin + nab-paclitaxel	No	Primary outcome: MPR
		Control	Carboplatin + nab-paclitaxel		Other: pCR
Adjuvant trials					
ACCIO	Resected stage II-III NSCLC 1210 expected participants USA only	Intervention 1	No	Chemotherapy followed by pembrolizumab ± radiotherapy	Primary outcomes: DFS and OS
		Intervention 2		Concomitant chemotherapy + pembrolizumab (no radiotherapy)	Others: HRQoL, safety

Table 2. Design of ongoing studies (Continued)

		Control		Chemotherapy ± radiotherapy	
ANVIL	Resected stage IB-IIIA NSCLC	Intervention	No	Standard adjuvant treatment followed by nivolumab	Primary outcomes: DFS and OS
	903 expected participants				Other: safety
	USA only	Control		Standard adjuvant treatment followed by observation	
ARCH	Resected stage II-IIIA NSCLC	Intervention	No	Cemiplimab	Primary outcome: DFS
	PD-L1 ≥ 1%	Control		Observation	Others: OS, safety
	390 expected participants				
	International				
BT-CRC-LUN18-153	Resected stage I NSCLC	Intervention	No	Pembrolizumab	Primary outcome: DFS
	244 expected participants	Control		Observation	Others: OS, safety
	USA only				
DUBHE-L-304	Resected stage II-IIIB NSCLC	Intervention	No	QL1706 + chemotherapy	Primary outcome: DFS
	632 expected participants				Others: OS, safety
	China only	Control		Placebo + chemotherapy	
NADIM-adjuvant	Resected stage IB-IIIA NSCLC	Intervention	No	Nivolumab + carboplatin + paclitaxel	Primary outcome: DFS
	210 participants				Others: OS, safety
	Spain only	Control		Carboplatin + paclitaxel	
NCT07120282 (adjuvant tislelizumab)	Resected stage I NSCLC	Intervention	No	Tislelizumab ± platinum-doublet	Primary outcome: DFS
	High-risk feature				Others: OS, safety
	108 expected participants	Control		No treatment or platinum-doublet	
	China only				

Abbreviations

ctDNA: circulating tumour DNA; DFS: disease-free survival; EFS: event-free survival; HRQoL: health-related quality of life; MPR: major pathological response; NSCLC: non-small cell lung cancer; PD-L1: programmed-death ligand 1; OS: overall survival; pCR: pathological complete response.